My Project

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Chapter 4

文件索引

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/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/**baseLidarObjectDetector.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/**classicLidarDetector.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/**functionLidarDetector.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/**imu.pb.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/**lidarTools.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/**patchworkpp.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/**prediction.pb.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/**defineColor.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/**geometry_m.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/**globalPlanning_m.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/**interpolate_m.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/**localPlanning_m.hpp** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/**planning.hpp** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/geometry_m.cpp
Geometry 376
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/globalPlanning_m.cpp
GlobalPlanning 377
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/interpolate_m.cpp
Interpolate speed along the path 379
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/localPlanning_m.cpp
Local planning 380
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/planningFigure_m.cpp
Planning 385
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/**control.pb.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/**controllerPID.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/**defineColor.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/**imu.pb.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/**localization_MapAnalysis.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/**localization_MapAnalysis_m.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml.hpp 386
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml_print.hpp 392
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml_utils.hpp 395
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/**steeringController.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/**ztgeographycoordinatetransform.h** . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/**ztgeographycoordinatetransform_m.h** . . . . . ??

Chapter 5

命名空文档

5.1 geometry 命名空参考

是一个算欧式距离的namespace。

- struct crossInput
- struct crossOutput
- struct crossParam
- struct inOrOutPolygonInput
- struct inOrOutPolygonOutput
- struct inOrOutPolygonParam
- struct isInSimpleInput
- struct isInSimpleOutput
- struct isInSimpleParam
- struct isLineIntersectionInput
- struct isLineIntersectionOutput
- struct isLineIntersectionParam
- struct isOnlineInput
- struct isOnlineOutput
- struct isOnlineParam
- struct isPointInPolygonInput
- struct isPointInPolygonOutput
- struct isPointInPolygonParam
- struct isPolygonsIntersectionInput
- struct isPolygonsIntersectionOutput
- struct isPolygonsIntersectionParam
- struct point_t
- struct pointToLineInput
- struct pointToLineOutput
- struct pointToLineParam
- struct pointToPolygonInput
- struct pointToPolygonOutput
- struct pointToPolygonParam

函数

- void pointToPolygon (const pointToPolygonParam &param, const pointToPolygonInput &input, pointToPolygonOutput &output)

  算点到凸多形的距离
- void cross (const crossParam &param, const crossInput &input, crossOutput &output)

  点乘
- void pointToLine (const pointToLineParam &param, const pointToLineInput &input, pointToLineOutput &output)

  算点到段的距离
- void isOnline (const isOnlineParam &param, const isOnlineInput &input, isOnlineOutput &output)

  判断点是否在段上
- void isInSimple (const isInSimpleParam &param, const isInSimpleInput &input, isInSimpleOutput &output)

  判断点是否在横平直的多形内部
- void isLineIntersection (const isLineIntersectionParam &param, const isLineIntersectionInput &input, isLineIntersectionOutput &output)

  判断两段是否相交
- void isPolygonsIntersection (const isPolygonsIntersectionParam &param, const isPolygonsIntersectionInput &input, isPolygonsIntersectionOutput &output)

  判断两多形是否相交
- void isPointInPolygon (const isPointInPolygonParam &param, const isPointInPolygonInput &input, isPointInPolygonOutput &output)

  判断点是否在多形内部
- void inOrOutPolygon (const inOrOutPolygonParam &param, const inOrOutPolygonInput &input, inOrOutPolygonOutput &output)

  判断点是否在多形内部

5.1.1 描述

是一个算欧式距离的namespace。

5.1.2 函数明

5.1.2.1 cross()

```
void geometry::cross (
            const crossParam & param,
            const crossInput & input,
            crossOutput & output )
```

点乘

参数

[IN]	param 无
[IN]	input 向量原点，
[OUT]	output 点到凸多形的距离 @cn_name: 点乘 @granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

5.1.2.2 inOrOutPolygon()

```
void geometry::inOrOutPolygon (
            const inOrOutPolygonParam & param,
            const inOrOutPolygonInput & input,
            inOrOutPolygonOutput & output )
```

判断点是否在多形内部

参数

[IN]	param 无
[IN]	input 多形点数、多形点X，多形点Y，点X，点Y
[OUT]	output 判断果 @cn_name: 判断点是否在多形内部 @granularity: atomic //函数件的粒度，atomic：基件（原子），composite>：合件（在前端可以向下一展）

5.1.2.3 isInSimple()

```
void geometry::isInSimple (
            const isInSimpleParam & param,
            const isInSimpleInput & input,
            isInSimpleOutput & output )
```

判断点是否在横平直的多形内部

参数

[IN]	param 无
[IN]	input 多形点，多形点数，点
[OUT]	output 判断果 @cn_name: 判断点是否在横平直的多形内部 @granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

5.1.2.4 isLineIntersection()

```
void geometry::isLineIntersection (
            const isLineIntersectionParam & param,
            const isLineIntersectionInput & input,
            isLineIntersectionOutput & output )
```

判断两段是否相交

参数

[IN]	param 无
[IN]	input 两段起点、点
[OUT]	output 判断果 @cn_name: 判断两段是否相交 @granularity: atomic //函数件的粒度， atomic ： 基件（原子），composite> ： 合件（在前端可以向下一展）

5.1.2.5 isOnline()

```
void geometry::isOnline (
            const isOnlineParam & param,
            const isOnlineInput & input,
            isOnlineOutput & output )
```

判断点是否在段上

参数

[IN]	param 无
[IN]	input 段起点，段点，点
[OUT]	output 判断果 @cn_name: 判断点是否在段上 @granularity: atomic //函数件的粒度， atomic ： 基件（原子），composite> ： 合件（在前端可以向下一展）

5.1.2.6 isPointInPolygon()

```
void geometry::isPointInPolygon (
            const isPointInPolygonParam & param,
            const isPointInPolygonInput & input,
            isPointInPolygonOutput & output )
```

判断点是否在多形内部

参数

[IN]	param 无
[IN]	input 多形点、点
[OUT]	output 判断果 @cn_name: 判断点是否在多形内部 @granularity: composite //函数件的粒度，atomic ： 基件（原子），composite> ： 合件（在前端可以向下一展）

5.1.2.7 isPolygonsIntersection()

```
void geometry::isPolygonsIntersection (
            const isPolygonsIntersectionParam & param,
            const isPolygonsIntersectionInput & input,
            isPolygonsIntersectionOutput & output )
```

判断两多形是否相交

参数

[IN]	param 无
[IN]	input 两多形点
[OUT]	output 判断果 @cn_name: 判断两多形是否相交 @granularity: composite //函数件的粒度，atomic：基件（原子），composite>：合件（在前端可以向下一展）

5.1.2.8 pointToLine()

```
void geometry::pointToLine (
            const pointToLineParam & param,
            const pointToLineInput & input,
            pointToLineOutput & output )
```

算点到段的距离

参数

[IN]	param 无
[IN]	input 段起点，段点，点
[OUT]	output 点到段的距离 @cn_name: 算点到段的距离 @granularity: atomic //函数件的粒度，atomic：基件（原子），composite>：合件（在前端可以向下一展）

5.1.2.9 pointToPolygon()

```
void geometry::pointToPolygon (
            const pointToPolygonParam & param,
            const pointToPolygonInput & input,
            pointToPolygonOutput & output )
```

算点到凸多形的距离

参数

[IN]	param 无
[IN]	input 多形点向量数，点
[OUT]	output 点到凸多形的距离 @cn_name: 算点到凸多形的距离 @granularity: composite //函数件的粒度，atomic：基件（原子），composite>：合件（在前端可以向下一展）

5.2 GolbalPlanning 命名空参考

是一个名GolbalPlanning的namespace。

- class Astar
- struct AstarCalcFInput
- struct AstarCalcFOutput
- struct AstarCalcFParam
- struct AstarCalcGInput
- struct AstarCalcGOutput
- struct AstarCalcGParam
- struct AstarCalcHInput
- struct AstarCalcHOutput
- struct AstarCalcHParam
- struct AstarFindLaneInput
- struct AstarFindLaneOutput
- struct AstarFindLaneParam
- struct AstarFindLaneRInput
- struct AstarFindLaneROutput
- struct AstarFindLaneRParam
- struct AstarFindPathInput
- struct AstarFindPathOutput
- struct AstarFindPathParam
- struct AstarGetLeastFInput
- struct AstarGetLeastFOutput
- struct AstarGetLeastFParam
- struct AstarGetNextRoadInput
- struct AstarGetNextRoadOutput
- struct AstarGetNextRoadParam
- struct AstarGetPathInput
- struct AstarGetPathOutput
- struct AstarGetPathParam
- struct AstarInitLinkInput
- struct AstarInitLinkOutput
- struct AstarInitLinkParam
- struct AstarInitRoadInput
- struct AstarInitRoadOutput
- struct AstarInitRoadParam
- struct AstarMapToAstarInput
- struct AstarMapToAstarOutput
- struct AstarMapToAstarParam
- struct AstarModuleSelfCheckInput
- struct AstarModuleSelfCheckOutput
- struct AstarModuleSelfCheckParam
- struct AstarModuleSelfCheckPrintInput
- struct AstarModuleSelfCheckPrintOutput
- struct AstarModuleSelfCheckPrintParam
- struct AstarRecursiveFindLaneInput
- struct AstarRecursiveFindLaneOutput
- struct AstarRecursiveFindLaneParam
- struct AstarResetInput
- struct AstarResetOutput
- struct AstarResetParam
- class cmp
- class node
- struct road

函数

- void AstarMapToAstar (const AstarMapToAstarParam &param, const AstarMapToAstarInput &input, AstarMapToAstarOutput &output)

  使用*Map*构建*Astar*
- void AstarGetPath (const AstarGetPathParam &param, const AstarGetPathInput &input, AstarGetPathOutput &output)

  使用*Astar*算法取路径，包括理、后理等部分
- void AstarInitRoad (const AstarInitRoadParam &param, const AstarInitRoadInput &input, AstarInitRoadOutput &output)

  *Astar*道路初始化
- void AstarInitLink (const AstarInitLinkParam &param, const AstarInitLinkInput &input, AstarInitLinkOutput &output)

  *Astar*道路接系初始化
- void AstarReset (const AstarResetParam &param, const AstarResetInput &input, AstarResetOutput &output)

  在保留地的情况下，初始化*Astar*参数
- void AstarModuleSelfCheck (const AstarModuleSelfCheckParam &param, const AstarModuleSelfCheckInput &input, AstarModuleSelfCheckOutput &output)

  *Astar*道路*Id*
- void AstarModuleSelfCheckPrint (const AstarModuleSelfCheckPrintParam &param, const AstarModuleSelfCheckPrintInput &input, AstarModuleSelfCheckPrintOutput &output)

  *Astar*道路*Id*打印
- void AstarFindLane (const AstarFindLaneParam &param, const AstarFindLaneInput &input, AstarFindLaneOutput &output)

  根据*Astar*道路全局划果，得道全局划果
- void AstarFindLaneR (const AstarFindLaneRParam &param, const AstarFindLaneRInput &input, AstarFindLaneROutput &output)

  根据*Astar*道路全局划果，使用的方法得道全局划果
- void AstarRecursiveFindLane (const AstarRecursiveFindLaneParam &param, const AstarRecursiveFindLaneInput &input, AstarRecursiveFindLaneOutput &output)

  *Astar*得道全局划果主体函数
- void AstarFindPath (const AstarFindPathParam &param, const AstarFindPathInput &input, AstarFindPathOutput &output)

  *Astar*算法的主要部分
- void AstarCalcG (const AstarCalcGParam &param, const AstarCalcGInput &input, AstarCalcGOutput &output)

  *Astar*算法 算起点到当前点的已知度
- void AstarCalcH (const AstarCalcHParam &param, const AstarCalcHInput &input, AstarCalcHOutput &output)

  *Astar*算法 算当前点到点的估度
- void AstarCalcF (const AstarCalcFParam &param, const AstarCalcFInput &input, AstarCalcFOutput &output)

  *Astar*算法 算当前点估最路径度
- void AstarGetLeastF (const AstarGetLeastFParam &param, const AstarGetLeastFInput &input, AstarGetLeastFOutput &output)

  *Astar*算法 找到估最路径度最小的点
- void AstarGetNextRoad (const AstarGetNextRoadParam &param, const AstarGetNextRoadInput &input, AstarGetNextRoadOutput &output)

  *Astar*算法 找到下一条路

量

- AstarMapToAstarParam **param_map2astar** {m, as}
- AstarMapToAstarInput **input_map2astar** {}
- AstarMapToAstarOutput **output_map2astar** {as}
- AstarGetPathParam **param_get_path** {as}
- AstarGetPathInput **input_get_path** {origin, destination, originPointId, destinationPointId}
- AstarGetPathOutput **output_get_path** {path, as}
- AstarInitRoadParam **param_init_road** {as}
- AstarInitRoadInput **input_init_road** {number, xStart, yStart, xEnd, yEnd, length}
- AstarInitRoadOutput **output_init_road** {as}
- AstarInitLinkParam **param_init_link** {as}
- AstarInitLinkInput **input_init_link** {from, to}
- AstarInitLinkOutput **output_init_link** {as}
- AstarResetParam **param_reset** {as}
- AstarResetInput **input_reset** {}
- AstarResetOutput **output_reset** {as}
- AstarModuleSelfCheckParam **param_module_selfcheck** {as}
- AstarModuleSelfCheckInput **input_module_selfcheck** {path}
- AstarModuleSelfCheckOutput **output_module_selfcheck** {}
- AstarModuleSelfCheckPrintParam **param_module_selfcheck_print** {as}
- AstarModuleSelfCheckPrintInput **input_module_selfcheck_print** {pathLanes}
- AstarModuleSelfCheckPrintOutput **output_module_selfcheck_print** {}
- AstarFindLaneParam **param_find_lane** {as}
- AstarFindLaneInput **input_find_lane** {m, path}
- AstarFindLaneOutput **output_find_lane** {as}
- AstarFindLaneRParam **param_find_lane_r** {as}
- AstarFindLaneRInput **input_find_lane_r** {roads, path, pathLanes, lane_id}
- AstarFindLaneROutput **output_find_lane_r** {as}
- AstarRecursiveFindLaneParam **param_recursive_find_lane** {as}
- AstarRecursiveFindLaneInput **input_recursive_find_lane** {roads, path, optional_lanes, road_id, lane_id}
- AstarRecursiveFindLaneOutput **output_recursive_find_lane** {as}

5.2.1 描述

是一个名GolbalPlanning的namespace。

5.2.2 函数明

5.2.2.1 AstarCalcF()

```
void GolbalPlanning::AstarCalcF (
            const AstarCalcFParam & param,
            const AstarCalcFInput & input,
            AstarCalcFOutput & output )
```

Astar算法 算当前点估最路径度

参数

[IN]	param Astar
[IN]	input
[OUT]	output F

@cn_name: Astar算法的主要部分

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.2.2.2 AstarCalcG()

```
void GolbalPlanning::AstarCalcG (
            const AstarCalcGParam & param,
            const AstarCalcGInput & input,
            AstarCalcGOutput & output )
```

Astar算法 算起点到当前点的已知度

参数

[IN]	param Astar
[IN]	input 出点 当前点
[OUT]	output G

@cn_name: Astar算法的主要部分

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.2.2.3 AstarCalcH()

```
void GolbalPlanning::AstarCalcH (
            const AstarCalcHParam & param,
            const AstarCalcHInput & input,
            AstarCalcHOutput & output )
```

Astar算法 算当前点到点的估度

参数

[IN]	param Astar
[IN]	input 当前点 点
[OUT]	output H

@cn_name: Astar算法的主要部分

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.2.2.4 AstarFindLane()

```
void GolbalPlanning::AstarFindLane (
            const AstarFindLaneParam & param,
            const AstarFindLaneInput & input,
            AstarFindLaneOutput & output )
```

根据Astar道路全局划果，得道全局划果

参数

[IN]	param Astar
[IN]	input 地、道路全局划果
[OUT]	output 更新的Astar

@cn_name: Astar得全局划道果

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.2.2.5 AstarFindLaneR()

```
void GolbalPlanning::AstarFindLaneR (
            const AstarFindLaneRParam & param,
            const AstarFindLaneRInput & input,
            AstarFindLaneROutput & output )
```

根据Astar道路全局划果，使用的方法得道全局划果

参数

[IN]	param Astar
[IN]	input 道路向量数、道路全局划果、存果的向量数、点道Id
[OUT]	output 更新的Astar

@cn_name: Astar地得全局划道果

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.2.2.6 AstarFindPath()

```
void GolbalPlanning::AstarFindPath (
            const AstarFindPathParam & param,
            const AstarFindPathInput & input,
            AstarFindPathOutput & output )
```

Astar算法的主要部分

参数

[IN]	param Astar
[IN]	input 出点 目的地
[OUT]	output 合理的道路

@cn_name: Astar算法的主要部分

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.2.2.7 AstarGetLeastF()

```
void GolbalPlanning::AstarGetLeastF (
            const AstarGetLeastFParam & param,
            const AstarGetLeastFInput & input,
            AstarGetLeastFOutput & output )
```

Astar算法 找到估最路径度最小的点

参数

[IN]	param Astar
[IN]	input
[OUT]	output openList下

@cn_name: Astar算法的主要部分

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.2.2.8 AstarGetNextRoad()

```
void GolbalPlanning::AstarGetNextRoad (
            const AstarGetNextRoadParam & param,
            const AstarGetNextRoadInput & input,
            AstarGetNextRoadOutput & output )
```

Astar算法 找到下一条路

参数

[IN]	param Astar
[IN]	input road id
[OUT]	output 下一条路id

@cn_name: Astar算法的主要部分

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.2.2.9 AstarGetPath()

```
void GolbalPlanning::AstarGetPath (
            const AstarGetPathParam & param,
            const AstarGetPathInput & input,
            AstarGetPathOutput & output )
```

使用Astar算法取路径，包括理、后理等部分

参数

[IN]	param Astar
[IN]	input 起点、点、起点Id、点Id
[OUT]	output 更新的Astar，起点到点的路径

@cn_name: Astar取全局路径

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.2.2.10 AstarInitLink()

```
void GolbalPlanning::AstarInitLink (
            const AstarInitLinkParam & param,
```

```
            const AstarInitLinkInput & input,
            AstarInitLinkOutput & output )
```

Astar道路接系初始化

参数

[IN]	param Astar
[IN]	input 起始道路Id，到达道路Id
[OUT]	output 更新的Astar

@cn_name: Astar添加道路接系

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.2.2.11 AstarInitRoad()

```
void GolbalPlanning::AstarInitRoad (
            const AstarInitRoadParam & param,
            const AstarInitRoadInput & input,
            AstarInitRoadOutput & output )
```

Astar道路初始化

参数

[IN]	param Astar
[IN]	input 道路Id，起点XY，点XY，度
[OUT]	output 更新的Astar

@cn_name: Astar添加道路

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.2.2.12 AstarMapToAstar()

```
void GolbalPlanning::AstarMapToAstar (
            const AstarMapToAstarParam & param,
            const AstarMapToAstarInput & input,
            AstarMapToAstarOutput & output )
```

使用Map构建Astar

参数

[IN]	param 地，初始化的Astar
[IN]	input 无
[OUT]	output 更新的Astar

@cn_name: 使用地构建Astar

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.2.2.13 AstarModuleSelfCheck()

```
void GolbalPlanning::AstarModuleSelfCheck (
            const AstarModuleSelfCheckParam & param,
            const AstarModuleSelfCheckInput & input,
            AstarModuleSelfCheckOutput & output )
```

Astar道路Id

参数

[IN]	param Astar
[IN]	input 无
[OUT]	output 无

@cn_name: Astar道路Id

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.2.2.14 AstarModuleSelfCheckPrint()

```
void GolbalPlanning::AstarModuleSelfCheckPrint (
            const AstarModuleSelfCheckPrintParam & param,
            const AstarModuleSelfCheckPrintInput & input,
            AstarModuleSelfCheckPrintOutput & output )
```

Astar道路Id打印

参数

[IN]	param Astar
[IN]	input 无
[OUT]	output 无

@cn_name: Astar道路Id打印

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.2.2.15 AstarRecursiveFindLane()

```
void GolbalPlanning::AstarRecursiveFindLane (
            const AstarRecursiveFindLaneParam & param,
            const AstarRecursiveFindLaneInput & input,
            AstarRecursiveFindLaneOutput & output )
```

Astar得道全局划果主体函数

参数

[IN]	param Astar
[IN]	input 道路、道路全局划果、可道、点道路Id、点道Id
[OUT]	output 更新的Astar

@cn_name: Astar得道全局划果主体函数

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.2.2.16 AstarReset()

```
void GolbalPlanning::AstarReset (
            const AstarResetParam & param,
            const AstarResetInput & input,
            AstarResetOutput & output )
```

在保留地的情况下，初始化Astar参数

参数

[IN]	param Astar
[IN]	input 无
[OUT]	output 更新的Astar

@cn_name: Astar参数初始化

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.3 imageProcess 命名空参考

imageProcess namespace。

函数

- bool loadImage (const char *filename, cv::Mat &clr_image)

  Loads an image from file into a cv::Mat.
- bool saveImage (cv::Mat input, const char *filename)

  Saves an image to file.
- void imShow (cv::Mat input, const char *window_name)

  Displays an image in a window.
- bool imGray (cv::Mat input, cv::Mat &output)

  Converts a color image to grayscale.
- bool imLaplaceTrans (cv::Mat input, cv::Mat &output, bool is_norm)

  Applies Laplacian edge detection.
- bool imEqualizeHistogram (cv::Mat input, cv::Mat &output)

  Performs histogram equalization on grayscale image.
- bool imGammaCorrection (cv::Mat input, cv::Mat &output, double gamma)

  Applies gamma correction to image.
- bool imInvertTransform (cv::Mat input, cv::Mat &output)

  Inverts image colors.
- bool imLogTransform (cv::Mat input, cv::Mat &output, double c)

  Applies logarithmic transform to image.
- bool imMeanFilter (cv::Mat input, cv::Mat &output, int kernel_size_x, int kernel_size_y)

  Applies mean filter to image.
- bool imMedianFilter (cv::Mat input, cv::Mat &output, int kernel_size)

  Applies median filter to image.
- void imGaussianFilter (cv::Mat input, cv::Mat &output, int kernel_size, double sigma)

  Applies Gaussian filtering to an image.
- void imSharpeningFilter (cv::Mat input, cv::Mat &output, double sigma, double amount, int threshold)

  Applies sharpening filter to image using unsharp masking.
- bool imBilateralFilter (cv::Mat input, cv::Mat &output, int diameter, double sigma_color, double sigma_space)

  Applies bilateral filter to image.
- void imDrawRectangle (cv::Mat input, cv::Rect rect, cv::Scalar color, int width)

  Draws a rectangle on an image.
- void imDrawText (cv::Mat input, char *str, cv::Point point, int fontFace, float size, cv::Scalar color)

  Draws text on an image.
- void imDrawCircle (cv::Mat input, cv::Point center, int radius, cv::Scalar color, int thickness, int lineType, int shift)

  Draws a circle on an image.
- void imLaplacian (cv::Mat input, cv::Mat &output, int kernel_size, double scale, double delta)

  Applies Laplacian operator to image.
- void imRoberts (cv::Mat input, cv::Mat &output)

  Applies Roberts edge detection.
- void imPrewitt (cv::Mat input, cv::Mat &output)

Applies Prewitt edge detection.

- void imMorphologyDilation (cv::Mat input, cv::Mat &output, int kernel_size)

  Applies morphological dilation.
- void imMorphologyErosion (cv::Mat input, cv::Mat &output, int kernel_size)

  Applies morphological erosion.
- void imMorphologyOpen (cv::Mat input, cv::Mat &output, int kernel_size)

  Applies morphological opening.
- void imMorphologyClose (cv::Mat input, cv::Mat &output, int kernel_size)

  Applies morphological closing.
- void imNearestInterpolation (cv::Mat input, cv::Mat &output, double sx, double sy)

  Performs nearest neighbor interpolation.
- void imBiLineInterpolate (cv::Mat input, cv::Mat &output, double scale_x, double scale_y)

  Performs bilinear interpolation.
- void imCubicInterpolation (cv::Mat input, cv::Mat &output, double scale_x, double scale_y)

  Performs bicubic interpolation.
- void imGetNearestNeighbor (cv::Mat input, uchar &result, double pos_x, double pos_y)

  Gets nearest neighbor pixel value.
- void imGetBilinearInterpolation (cv::Mat input, uchar &result, double pos_x, double pos_y)

  Gets bilinearly interpolated pixel value.

5.3.1 描述

imageProcess namespace。

5.3.2 函数明

5.3.2.1 imBilateralFilter()

```
bool imageProcess::imBilateralFilter (
            cv::Mat input,
            cv::Mat & output,
            int diameter,
            double sigma_color,
            double sigma_space )
```

Applies bilateral filter to image.

参数

input	Input image
output	Output filtered image
diameter	Diameter of pixel neighborhood
sigma_color	Filter sigma in color space
sigma_space	Filter sigma in coordinate space

返回

true if successful, false if failed

5.3.2.2 imBiLineInterpolate()

```
void imageProcess::imBiLineInterpolate (
            cv::Mat input,
            cv::Mat & output,
            double scale_x,
            double scale_y )
```

Performs bilinear interpolation.

参数

input	Input image
output	Output interpolated image
scale↩ _x	Scale factor in x direction
scale↩ _y	Scale factor in y direction

5.3.2.3 imCubicInterpolation()

```
void imageProcess::imCubicInterpolation (
            cv::Mat input,
            cv::Mat & output,
            double scale_x,
            double scale_y )
```

Performs bicubic interpolation.

参数

input	Input image
output	Output interpolated image
scale↩ _x	Scale factor in x direction
scale↩ _y	Scale factor in y direction

5.3.2.4 imDrawCircle()

```
void imageProcess::imDrawCircle (
            cv::Mat input,
            cv::Point center,
            int radius,
            cv::Scalar color,
            int thickness,
            int lineType,
            int shift )
```

Draws a circle on an image.

参数

input	Image to draw on
center	Circle center point
radius	Circle radius
color	Circle color
thickness	Line thickness
lineType	Line type
shift	Number of fractional bits

5.3.2.5 imDrawRectangle()

```
void imageProcess::imDrawRectangle (
            cv::Mat input,
            cv::Rect rect,
            cv::Scalar color,
            int width )
```

Draws a rectangle on an image.

参数

input	Image to draw on
rect	Rectangle coordinates
color	Rectangle color
width	Rectangle line width

5.3.2.6 imDrawText()

```
void imageProcess::imDrawText (
            cv::Mat input,
            char * str,
```

```
            cv::Point point,
            int fontFace,
            float size,
            cv::Scalar color )
```

Draws text on an image.

参数

input	Image to draw on
str	Text string to draw
point	Text position
fontFace	Font type
size	Font size
color	Text color

5.3.2.7 imEqualizeHistogram()

```
bool imageProcess::imEqualizeHistogram (
            cv::Mat input,
            cv::Mat & output )
```

Performs histogram equalization on grayscale image.

参数

input	Input image
output	Output equalized image

返回

true if successful, false if failed

5.3.2.8 imGammaCorrection()

```
bool imageProcess::imGammaCorrection (
            cv::Mat input,
            cv::Mat & output,
            double gamma )
```

Applies gamma correction to image.

参数

input	Input image
output	Output gamma corrected image
gamma	Gamma correction value

返回

true if successful, false if failed

5.3.2.9 imGaussianFilter()

```
void imageProcess::imGaussianFilter (
            cv::Mat input,
            cv::Mat & output,
            int kernel_size,
            double sigma )
```

Applies Gaussian filtering to an image.

参数

input	Input source image
output	Output filtered image
kernel_size	Size of Gaussian kernel (must be odd)
sigma	Standard deviation of Gaussian kernel

5.3.2.10 imGetBilinearInterpolation()

```
void imageProcess::imGetBilinearInterpolation (
            cv::Mat input,
            uchar & result,
            double pos_x,
            double pos_y )
```

Gets bilinearly interpolated pixel value.

参数

input	Input image
result	Output pixel value
pos↩_x	X coordinate
pos↩_y	Y coordinate

5.3.2.11 imGetNearestNeighbor()

```
void imageProcess::imGetNearestNeighbor (
            cv::Mat input,
```

```
            uchar & result,
            double pos_x,
            double pos_y )
```

Gets nearest neighbor pixel value.

参数

input	Input image
result	Output pixel value
pos↩ _x	X coordinate
pos↩ _y	Y coordinate

5.3.2.12 imGray()

```
bool imageProcess::imGray (
            cv::Mat input,
            cv::Mat & output )
```

Converts a color image to grayscale.

参数

input	Input color image
output	Output grayscale image

返回

true if successful, false if failed

5.3.2.13 imInvertTransform()

```
bool imageProcess::imInvertTransform (
            cv::Mat input,
            cv::Mat & output )
```

Inverts image colors.

参数

input	Input image
output	Output inverted image

返回

true if successful, false if failed

5.3.2.14 imLaplaceTrans()

```
bool imageProcess::imLaplaceTrans (
            cv::Mat input,
            cv::Mat & output,
            bool is_norm )
```

Applies Laplacian edge detection.

参数

input	Input image
output	Output edge detected image
is_norm	Whether to normalize output to 0-255 range

返回

true if successful, false if failed

5.3.2.15 imLaplacian()

```
void imageProcess::imLaplacian (
            cv::Mat input,
            cv::Mat & output,
            int kernel_size,
            double scale,
            double delta )
```

Applies Laplacian operator to image.

参数

input	Input image
output	Output edge detected image
kernel_size	Size of Laplacian kernel
scale	Optional scale factor
delta	Optional delta value

5.3.2.16 imLogTransform()

```
bool imageProcess::imLogTransform (
            cv::Mat input,
            cv::Mat & output,
            double c )
```

Applies logarithmic transform to image.

参数

input	Input image
output	Output transformed image
c	Scaling constant

返回

true if successful, false if failed

5.3.2.17 imMeanFilter()

```
bool imageProcess::imMeanFilter (
            cv::Mat input,
            cv::Mat & output,
            int kernel_size_x,
            int kernel_size_y )
```

Applies mean filter to image.

参数

input	Input image
output	Output filtered image
kernel_size_x	Kernel width
kernel_size_y	Kernel height

返回

true if successful, false if failed

5.3.2.18 imMedianFilter()

```
bool imageProcess::imMedianFilter (
            cv::Mat input,
```

```
            cv::Mat & output,
            int kernel_size )
```

Applies median filter to image.

参数

input	Input image
output	Output filtered image
kernel_size	Size of median filter kernel

返回

true if successful, false if failed

5.3.2.19 imMorphologyClose()

```
void imageProcess::imMorphologyClose (
            cv::Mat input,
            cv::Mat & output,
            int kernel_size )
```

Applies morphological closing.

参数

input	Input image
output	Output closed image
kernel_size	Size of structuring element

5.3.2.20 imMorphologyDilation()

```
void imageProcess::imMorphologyDilation (
            cv::Mat input,
            cv::Mat & output,
            int kernel_size )
```

Applies morphological dilation.

参数

input	Input image
output	Output dilated image
kernel_size	Size of structuring element

5.3.2.21 imMorphologyErosion()

```
void imageProcess::imMorphologyErosion (
            cv::Mat input,
            cv::Mat & output,
            int kernel_size )
```

Applies morphological erosion.

参数

input	Input image
output	Output eroded image
kernel_size	Size of structuring element

5.3.2.22 imMorphologyOpen()

```
void imageProcess::imMorphologyOpen (
            cv::Mat input,
            cv::Mat & output,
            int kernel_size )
```

Applies morphological opening.

参数

input	Input image
output	Output opened image
kernel_size	Size of structuring element

5.3.2.23 imNearestInterpolation()

```
void imageProcess::imNearestInterpolation (
            cv::Mat input,
            cv::Mat & output,
            double sx,
            double sy )
```

Performs nearest neighbor interpolation.

参数

input	Input image
output	Output interpolated image
sx	Scale factor in x direction
sy	Scale factor in y direction

5.3.2.24 imPrewitt()

```
void imageProcess::imPrewitt (
            cv::Mat input,
            cv::Mat & output )
```

Applies Prewitt edge detection.

参数

input	Input image
output	Output edge detected image

5.3.2.25 imRoberts()

```
void imageProcess::imRoberts (
            cv::Mat input,
            cv::Mat & output )
```

Applies Roberts edge detection.

参数

input	Input image
output	Output edge detected image

5.3.2.26 imSharpeningFilter()

```
void imageProcess::imSharpeningFilter (
            cv::Mat input,
            cv::Mat & output,
            double sigma,
            double amount,
            int threshold )
```

Applies sharpening filter to image using unsharp masking.

参数

input	Input image
output	Output sharpened image
sigma	Gaussian blur sigma (default 1.0)
amount	Sharpening strength (default 1.5)
threshold	Minimum difference for sharpening (default 0)

5.3.2.27 imShow()

```
void imageProcess::imShow (
            cv::Mat input,
            const char * window_name )
```

Displays an image in a window.

参数

input	Image to display
window_name	Name of the window

5.3.2.28 loadImage()

```
bool imageProcess::loadImage (
            const char * filename,
            cv::Mat & clr_image )
```

Loads an image from file into a cv::Mat.

参数

filename	Path to the image file to load
clr_image	Output color image matrix

返回

true if successful, false if failed to load

5.3.2.29 saveImage()

```
bool imageProcess::saveImage (
            cv::Mat input,
            const char * filename )
```

Saves an image to file.

参数

input	Image matrix to save
filename	Output filename to save to

返回

true if successful, false if failed

5.4 interp_m 命名空参考

是一个名interp_m的namespace。

- struct interpConstAccInput
- struct interpConstAccOutput
- struct interpConstAccParam
- struct interpLinearInput
- struct interpLinearOutput
- struct interpLinearParam

函数

- void linear (const interpLinearParam &param, const interpLinearInput &input, interpLinearOutput &output)
  根据初、和数，行性插
- void constant_acceleration (const interpConstAccParam &param, const interpConstAccInput &input, interpConstAccOutput &output)
  在等隔的情况下，行匀加速度插

5.4.1 描述

是一个名interp_m的namespace。

5.4.2 函数明

5.4.2.1 constant_acceleration()

```
void interp_m::constant_acceleration (
            const interpConstAccParam & param,
            const interpConstAccInput & input,
            interpConstAccOutput & output )
```

在等隔的情况下，行匀加速度插

参数

[IN]	param 无
[IN]	input 初、、数、加速度
[O↩ UT]	output 存插的向量数

@cn_name: 匀加速度插

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.4.2.2 linear()

```
void interp_m::linear (
            const interpLinearParam & param,
            const interpLinearInput & input,
            interpLinearOutput & output )
```

根据初、和数，行性插

参数

[IN]	param 无
[IN]	input 初、、数
[O↩ UT]	output 存插的向量数

@cn_name: 性插

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.5 lidarTools 命名空参考

lidarTools namespace。

- struct ClusterDistance
- struct ClusterResult

函数

- void loadPCD (std::string filename, pcl::PointCloud< pcl::PointXYZI >::Ptr output)

  Load point cloud from PCD file.
- void pcdIntensityThresholdLowerFilter (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl::PointXYZI >::Ptr output, int intensity_thres)

  Filter points based on intensity threshold.
- void pcdVoxelFilter (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl::PointXYZI >::Ptr output, const double size_x, const double size_y, const double size_z)

  Apply voxel grid filter to downsample point cloud.
- void pcdRadiusFilter (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl::PointXYZI >::Ptr output, const double radius, const int min_neighbors)

  Apply radius outlier removal filter.
- void pcdStatisticFilter (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl::PointXYZI >↩ ::Ptr output, const double mean_K, const double std)

  Apply statistical outlier removal filter.
- float gaussian (float x, float sigma)

  Calculate Gaussian weight based on distance.
- void pcdGaussianFilter (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl::PointXY↩ ZI >::Ptr output, const double sigma, const double search_radius)

  Apply Gaussian filter to smooth point cloud.
- void updateDistances (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud, int point_idx, std::vector< float > &min↩ _distances, pcl::KdTreeFLANN< PointXYZI >::Ptr kdtree)

  Update minimum distances between points and clusters.
- int findFarthestPoint (std::vector< float > &distances)

  Find point with maximum distance in distance vector.
- void pcdFarthestPointSample (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl::Point↩ XYZI >::Ptr output, int num_samples, bool random_seed)

  Perform farthest point sampling on point cloud.
- void pcdRandomSample (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl::PointXYZI >::Ptr output, const int num_samples)

  Perform random sampling on point cloud.
- void initializeCentroids (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud, pcl::PointCloud< pcl::PointXYZI >::Ptr centroids)

  Initialize centroids for k-means clustering.
- void assignPointsToClusters (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud, pcl::PointCloud< pcl::PointXYZI >::Ptr centroids, std::vector< int > &labels)

  Assign points to nearest cluster centers.
- void updateCentroids (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud, pcl::PointCloud< pcl::PointXYZI >::Ptr centroids, std::vector< int > &labels, int k)

  Update cluster centroids based on point assignments.
- float calculateDistance (pcl::PointXYZI &p1, pcl::PointXYZI &p2)

  Calculate Euclidean distance between two points.
- float calculateTotalError (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud, pcl::PointCloud< pcl::PointXYZI >↩ ::Ptr centroids, std::vector< int > &labels)

  Calculate total clustering error.
- void pcdKmeansCluster (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, ClusterResult &result, int k, const int max_iterations_, const double convergence_threshold)

  Perform k-means clustering on point cloud.
- void pcdDBScanCluster (pcl::PointCloud< pcl::PointXYZI >::Ptr input, std::vector< pcl::PointIndices > &cluster_indices, double epsilon, int min_points)

  Perform DBSCAN clustering on point cloud.
- double computeCloudResolution (const pcl::PointCloud< pcl::PointXYZI >::Ptr &cloud)

Compute resolution of point cloud.

- void ISSKeyPointExtract (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl::PointXYZI >::Ptr keypoints, double salient_radius, double non_max_radius, double normal_radius, double threshold21, double threshold32, double min_neighbors, int threads)

  Extract ISS keypoints from point cloud.
- void SIFTKeyPointExtract (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl::PointWith↩Scale >::Ptr keypoints, float min_scale_, int nr_octaves, int nr_scales_per_octave, float min_contrast)

  Extract SIFT keypoints from point cloud.
- void harrisKeyPointKeyPointExtract (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl↩::PointXYZI >::Ptr keypoints, float threshold, float radius, bool non_maxima, int method)

  Extract Harris keypoints from point cloud.
- void PFHExtractFeature (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, double radius)

  Extracts Point Feature Histogram (PFH) features from a point cloud.
- void ICPPcMatch (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl::PointXYZI >::Ptr target_cloud, Eigen::Matrix< float, 4, 4 > &transformation_matrix)

  Performs Iterative Closest Point (ICP) matching between two point clouds.
- void GICPPcMatch (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl::PointXYZI >::Ptr target_cloud, Eigen::Matrix< float, 4, 4 > &transformation_matrix)

  Performs Generalized Iterative Closest Point (GICP) matching between two point clouds.
- void NTDPcMatch (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl::PointXYZI >::Ptr target_cloud, Eigen::Matrix< float, 4, 4 > &transformation_matrix, double thre, double step, double res)

  Performs Normal Distributions Transform (NDT) matching between two point clouds.
- bool get_filelist_from_dir (std::string _path, std::vector< std::string > &_files)

  Gets list of files from a directory.
- void convertPCDtoBin (std::string &in_file, std::string &out_file)

  Converts a PCD file to binary format.
- void saveToBin (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud, std::string &out_file)

  Saves point cloud to binary format.
- void pcd2Bin (string &pcd_dir, string &bin_dir)

  Converts all PCD files in a directory to binary format.
- bool convertBin2PCD (const std::string &input_path, const std::string &output_path)

  Converts a binary point cloud file to PCD format.
- std::vector< pcl::PointXYZI > readBinaryFile (const std::string &filename)

  Reads points from a binary file.
- void bin2Pcd (string &bin_dir, string &pcd_dir)

  Converts all binary files in a directory to PCD format.
- void pcdClip (pcl::PointCloud< pcl::PointXYZI >::Ptr input, pcl::PointCloud< pcl::PointXYZI >::Ptr output, double min_x, double max_x, double min_y, double max_y, double min_z, double max_z)

  Clips a point cloud to specified boundaries.
- void pcdRotateWithMat (pcl::PointCloud< pcl::PointXYZI >::Ptr input, pcl::PointCloud< pcl::PointXYZI >::Ptr output, Eigen::Matrix3f rotationMatrix)

  Rotates a point cloud using a rotation matrix.
- void pcdRotateWithEuler (pcl::PointCloud< pcl::PointXYZI >::Ptr input, pcl::PointCloud< pcl::PointXYZI >↩::Ptr output, double roll, double pitch, double yaw)

  Rotates a point cloud using Euler angles.
- void pcdRotateWithQuaternions (pcl::PointCloud< pcl::PointXYZI >::Ptr input, pcl::PointCloud< pcl::Point↩XYZI >::Ptr output, Eigen::Quaterniond q)

  Rotates a point cloud using quaternions.
- void pcdTrans (pcl::PointCloud< pcl::PointXYZI >::Ptr input, pcl::PointCloud< pcl::PointXYZI >::Ptr output, double trans_x, double trans_y, double trans_z)

  Translates a point cloud.
- void pcdComputeCentroid (const pcl::PointCloud< pcl::PointXYZI >::Ptr input, Eigen::Vector3f &centroid_)

Computes the centroid of a point cloud.

- void pcdComputeCovarianceMatrix (const pcl::PointCloud< pcl::PointXYZI >::Ptr input, Eigen::Vector3f &centroid_, Eigen::Matrix3f &covariance_matrix_)

  Computes the covariance matrix of a point cloud.

- void pcdComputeEigenVectors (Eigen::Matrix3f &covariance_matrix_, Eigen::Matrix3f &eigenvectors_, Eigen::Vector3f &eigenvalues_)

  Computes eigenvectors and eigenvalues from a covariance matrix.

- void pcdComputePCA (pcl::PointCloud< pcl::PointXYZI >::Ptr input, Eigen::Matrix3f &eigenvectors_, Eigen::Vector3f &eigenvalues_)

  Performs Principal Component Analysis (PCA) on a point cloud.

- void pcdToKDTree (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud, pcl::KdTreeFLANN< pcl::PointXYZI >::Ptr kdtree)

  Creates a KD-tree structure from a point cloud for efficient nearest neighbor searches.

- void pcdGetNearestNeighbor (pcl::PointCloud< pcl::PointXYZI >::Ptr input, pcl::KdTreeFLANN< pcl::PointXYZI >::Ptr kdtree, pcl::PointXYZI &result, pcl::PointXYZI &point, double radius)

  Finds the nearest neighbor point to a query point within a given radius.

- void pcdEuclideanClustering (pcl::PointCloud< pcl::PointXYZI >::Ptr input, std::vector< pcl::PointIndices > &cluster_indices, double radius, int min_cluster_size, int max_cluster_size)

  Performs Euclidean clustering on a point cloud.

- void pcdSpectralClustering (pcl::PointCloud< pcl::PointXYZI >::Ptr input, std::vector< pcl::PointIndices > &cluster_indices, int n_clusters, double sigma, int k_neighbors)

  Performs spectral clustering on a point cloud.

- void pcdAffinityPropagationCluster (pcl::PointCloud< pcl::PointXYZI >::Ptr input, std::vector< pcl::PointIndices > &cluster_indices, double damping, int max_iter)

  Performs Affinity Propagation clustering on a point cloud.

- void pcdMeanShiftCluster (pcl::PointCloud< pcl::PointXYZI >::Ptr input, std::vector< pcl::PointIndices > &cluster_indices, float bandwidth, int max_iter)

  Performs mean-shift clustering on a point cloud.

- void pcdWardCluster (pcl::PointCloud< pcl::PointXYZI >::Ptr input, std::vector< pcl::PointIndices > &cluster_indices, float distance_threshold, int min_cluster_size, int max_cluster_size)

  Performs Ward hierarchical clustering on a point cloud.

- void pcdOPTICSCluster (pcl::PointCloud< pcl::PointXYZI >::Ptr input, std::vector< pcl::PointIndices > &cluster_indices, double eps, int min_pts, double xi)

  OPTICS (Ordering Points To Identify the Clustering Structure) clustering algorithm.

- void pcdBIRCHCluster (pcl::PointCloud< pcl::PointXYZI >::Ptr input, std::vector< pcl::PointIndices > &cluster_indices, int branching_factor, double threshold, int max_leaf_size)

  Performs BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies) clustering on point cloud data.

- void **harrisKeyPointKeyPointExtract** (pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud, pcl::PointCloud< pcl::PointWithScale >::Ptr keypoints, float threshold=1e-6, float radius=0.01, bool non_maxima=true, int method=1)

5.5.1 描述

lidarTools namespace。

5.5.2 函数明

5.5.2.1 assignPointsToClusters()

```
void lidarTools::assignPointsToClusters (
            pcl::PointCloud< pcl::PointXYZI >::Ptr cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr centroids,
            std::vector< int > & labels )
```

Assign points to nearest cluster centers.

参数

in	*cloud*	Input point cloud
in	*centroids*	Cluster centroid points
out	*labels*	Point cluster assignments

5.5.2.2 bin2Pcd()

```
void lidarTools::bin2Pcd (
            string & bin_dir,
            string & pcd_dir )
```

Converts all binary files in a directory to PCD format.

参数

bin_dir	Input binary directory path
pcd_dir	Output PCD directory path

5.5.2.3 calculateDistance()

```
float lidarTools::calculateDistance (
            pcl::PointXYZI & p1,
            pcl::PointXYZI & p2 )
```

Calculate Euclidean distance between two points.

参数

in	*p1*	First point
in	*p2*	Second point

返回

Squared Euclidean distance between points

5.5.2.4 calculateTotalError()

```
float lidarTools::calculateTotalError (
            pcl::PointCloud< pcl::PointXYZI >::Ptr cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr centroids,
            std::vector< int > & labels )
```

Calculate total clustering error.

参数

in	*cloud*	Input point cloud
in	*centroids*	Cluster centroid points
in	*labels*	Point cluster assignments

返回

Total sum of squared distances to cluster centers

5.5.2.5 computeCloudResolution()

```
double lidarTools::computeCloudResolution (
            const pcl::PointCloud< pcl::PointXYZI >::Ptr & cloud )
```

Compute resolution of point cloud.

参数

in	*cloud*	Input point cloud

返回

Average distance between points

5.5.2.6 convertBin2PCD()

```
bool lidarTools::convertBin2PCD (
            const std::string & input_path,
            const std::string & output_path )
```

Converts a binary point cloud file to PCD format.

参数

input_path	Input binary file path
output_path	Output PCD file path

返回

true if conversion successful

5.5.2.7 convertPCDtoBin()

```
void lidarTools::convertPCDtoBin (
            std::string & in_file,
            std::string & out_file )
```

Converts a PCD file to binary format.

参数

in_file	Input PCD file path
out_file	Output binary file path

5.5.2.8 findFarthestPoint()

```
int lidarTools::findFarthestPoint (
            std::vector< float > & distances )
```

Find point with maximum distance in distance vector.

参数

in	*distances*	Vector of distances

返回

Index of point with maximum distance

5.5.2.9 gaussian()

```
float lidarTools::gaussian (
            float x,
            float sigma )
```

Calculate Gaussian weight based on distance.

参数

in	*x*	Distance value
in	*sigma*	Standard deviation parameter

返回

Gaussian weight value

5.5.2.10 get_filelist_from_dir()

```
bool lidarTools::get_filelist_from_dir (
            std::string _path,
            std::vector< std::string > & _files )
```

Gets list of files from a directory.

参数

	_path	Directory path
out	*_files*	Vector to store filenames

返回

true if successful

5.5.2.11 GICPPcMatch()

```
void lidarTools::GICPPcMatch (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr target_cloud,
            Eigen::Matrix< float, 4, 4 > & transformation_matrix )
```

Performs Generalized Iterative Closest Point (GICP) matching between two point clouds.

参数

	in_cloud	Source point cloud
	target_cloud	Target point cloud
out	*transformation_matrix*	The resulting transformation matrix

5.5.2.12 harrisKeyPointKeyPointExtract()

```
void lidarTools::harrisKeyPointKeyPointExtract (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr keypoints,
            float threshold,
            float radius,
            bool non_maxima,
            int method )
```

Extract Harris keypoints from point cloud.

参数

in	*in_cloud*	Input point cloud
out	*keypoints*	Output keypoint cloud
in	*threshold*	Response threshold
in	*radius*	Neighborhood size
in	*non_maxima*	Enable/disable non-maxima suppression
in	*method*	Response calculation method

5.5.2.13 ICPPcMatch()

```
void lidarTools::ICPPcMatch (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr target_cloud,
            Eigen::Matrix< float, 4, 4 > & transformation_matrix )
```

Performs Iterative Closest Point (ICP) matching between two point clouds.

参数

	in_cloud	Source point cloud
	target_cloud	Target point cloud
out	*transformation_matrix*	The resulting transformation matrix

5.5.2.14 initializeCentroids()

```
void lidarTools::initializeCentroids (
            pcl::PointCloud< pcl::PointXYZI >::Ptr cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr centroids )
```

Initialize centroids for k-means clustering.

参数

in	*cloud*	Input point cloud
out	*centroids*	Output centroid points

5.5.2.15 ISSKeyPointExtract()

```
void lidarTools::ISSKeyPointExtract (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr keypoints,
            double salient_radius,
            double non_max_radius,
            double normal_radius,
            double threshold21,
            double threshold32,
            double min_neighbors,
            int threads )
```

Extract ISS keypoints from point cloud.

参数

in	*in_cloud*	Input point cloud
out	*keypoints*	Output keypoint cloud
in	*salient_radius*	Radius for computing covariance matrix
in	*non_max_radius*	Radius for non-maximum suppression
in	*normal_radius*	Radius for normal estimation
in	*threshold21*	First eigenvalue threshold
in	*threshold32*	Second eigenvalue threshold
in	*min_neighbors*	Minimum number of neighbors
in	*threads*	Number of threads to use

5.5.2.16 loadPCD()

```
void lidarTools::loadPCD (
            std::string filename,
            pcl::PointCloud< pcl::PointXYZI >::Ptr output )
```

Load point cloud from PCD file.

参数

in	*filename*	Path to PCD file to load
out	*output*	Output point cloud containing loaded points

5.5.2.17 NTDPcMatch()

```
void lidarTools::NTDPcMatch (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr target_cloud,
            Eigen::Matrix< float, 4, 4 > & transformation_matrix,
            double thre,
            double step,
            double res )
```

Performs Normal Distributions Transform (NDT) matching between two point clouds.

参数

	in_cloud	Source point cloud
	target_cloud	Target point cloud
out	*transformation_matrix*	The resulting transformation matrix
	thre	Transformation epsilon threshold
	step	Step size for optimization
	res	Resolution of the voxel grid

5.5.2.18 pcd2Bin()

```
void lidarTools::pcd2Bin (
            string & pcd_dir,
            string & bin_dir )
```

Converts all PCD files in a directory to binary format.

参数

pcd_dir	Input PCD directory path
bin_dir	Output binary directory path

5.5.2.19 pcdAffinityPropagationCluster()

```
void lidarTools::pcdAffinityPropagationCluster (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            std::vector< pcl::PointIndices > & cluster_indices,
            double damping,
            int max_iter )
```

Performs Affinity Propagation clustering on a point cloud.

Implements the Affinity Propagation algorithm which:

1. Computes pairwise similarities between points
2. Iteratively updates responsibility and availability matrices
3. Identifies exemplars (cluster centers) through message passing
4. Assigns points to their nearest exemplar to form clusters

参数

in	*input*	Input point cloud to be clustered
out	*cluster_indices*	Vector containing point indices for each cluster
in	*damping*	Damping factor (0-1) to avoid numerical oscillations
in	*max_iter*	Maximum number of iterations for convergence

注解

The algorithm works by:

1. Building similarity matrix based on negative squared distances
2. Iteratively updating responsibility and availability matrices
3. Applying damping to avoid oscillations
4. Identifying exemplars where responsibility + availability > 0
5. Assigning points to nearest exemplar to form final clusters

5.5.2.20 pcdBIRCHCluster()

```
void lidarTools::pcdBIRCHCluster (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            std::vector< pcl::PointIndices > & cluster_indices,
            int branching_factor,
            double threshold,
            int max_leaf_size )
```

Performs BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies) clustering on point cloud data.

参数

input	Input point cloud to be clustered
cluster_indices	Output vector containing the indices of points for each cluster
branching_factor	Maximum number of child nodes for each non-leaf node (default: 50)
threshold	Maximum radius of a sub-cluster (default: 0.5)
max_leaf_size	Maximum number of points in a leaf node (default: 100)

5.5.2.21 pcdClip()

```
void lidarTools::pcdClip (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            pcl::PointCloud< pcl::PointXYZI >::Ptr output,
            double min_x,
            double max_x,
            double min_y,
            double max_y,
            double min_z,
            double max_z )
```

Clips a point cloud to specified boundaries.

参数

input	Input point cloud
output	Output point cloud after clipping
min↩ _x	Minimum X boundary
max↩ _x	Maximum X boundary
min↩ _y	Minimum Y boundary
max↩ _y	Maximum Y boundary
min↩ _z	Minimum Z boundary
max↩ _z	Maximum Z boundary

5.5.2.22 pcdComputeCentroid()

```
void lidarTools::pcdComputeCentroid (
            const pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            Eigen::Vector3f & centroid_ )
```

Computes the centroid of a point cloud.

参数

	input	Input point cloud
out	*centroid↩ _*	Computed centroid vector

5.5.2.23 pcdComputeCovarianceMatrix()

```
void lidarTools::pcdComputeCovarianceMatrix (
            const pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            Eigen::Vector3f & centroid_,
            Eigen::Matrix3f & covariance_matrix_ )
```

Computes the covariance matrix of a point cloud.

参数

	input	Input point cloud
	*centroid_*	Centroid of the point cloud
out	*covariance_matrix_*	Computed covariance matrix

5.5.2.24 pcdComputeEigenVectors()

```
void lidarTools::pcdComputeEigenVectors (
            Eigen::Matrix3f & covariance_matrix_,
            Eigen::Matrix3f & eigenvectors_,
            Eigen::Vector3f & eigenvalues_ )
```

Computes eigenvectors and eigenvalues from a covariance matrix.

参数

	*covariance_matrix_*	Input covariance matrix
out	*eigenvectors_*	Computed eigenvectors
out	*eigenvalues_*	Computed eigenvalues

5.5.2.25 pcdComputePCA()

```
void lidarTools::pcdComputePCA (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            Eigen::Matrix3f & eigenvectors_,
            Eigen::Vector3f & eigenvalues_ )
```

Performs Principal Component Analysis (PCA) on a point cloud.

参数

	input	Input point cloud
out	*eigenvectors_*	Computed eigenvectors
out	*eigenvalues_*	Computed eigenvalues

5.5.2.26 pcdDBScanCluster()

```
void lidarTools::pcdDBScanCluster (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            std::vector< pcl::PointIndices > & cluster_indices,
            double epsilon,
            int min_points )
```

Perform DBSCAN clustering on point cloud.

参数

in	*input*	Input point cloud
out	*cluster_indices*	Vector of point indices for each cluster
in	*epsilon*	Search radius for finding neighbors
in	*min_points*	Minimum number of points to form a cluster

5.5.2.27 pcdEuclideanClustering()

```
void lidarTools::pcdEuclideanClustering (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            std::vector< pcl::PointIndices > & cluster_indices,
            double radius,
            int min_cluster_size,
            int max_cluster_size )
```

Performs Euclidean clustering on a point cloud.

Groups points into clusters based on Euclidean distance between points

参数

in	*input*	Input point cloud to be clustered
out	*cluster_indices*	Vector of point indices for each cluster found
in	*radius*	Maximum distance between points in a cluster
in	*min_cluster_size*	Minimum number of points required to form a cluster
in	*max_cluster_size*	Maximum number of points allowed in a cluster

5.5.2.28 pcdFarthestPointSample()

```
void lidarTools::pcdFarthestPointSample (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
```

```
            pcl::PointCloud< pcl::PointXYZI >::Ptr output,
            int num_samples,
            bool random_seed )
```

Perform farthest point sampling on point cloud.

参数

in	*in_cloud*	Input point cloud
out	*output*	Output sampled point cloud
in	*num_samples*	Number of points to sample
in	*random_seed*	Whether to use random initial point

5.5.2.29 pcdGaussianFilter()

```
void lidarTools::pcdGaussianFilter (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr output,
            const double sigma,
            const double search_radius )
```

Apply Gaussian filter to smooth point cloud.

参数

in	*in_cloud*	Input point cloud
out	*output*	Output smoothed point cloud
in	*sigma*	Controls smoothing strength
in	*search_radius*	Controls neighborhood size for smoothing

5.5.2.30 pcdGetNearestNeighbor()

```
void lidarTools::pcdGetNearestNeighbor (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            pcl::KdTreeFLANN< pcl::PointXYZI >::Ptr kdtree,
            pcl::PointXYZI & result,
            pcl::PointXYZI & point,
            double radius )
```

Finds the nearest neighbor point to a query point within a given radius.

Uses KD-tree to efficiently search for the closest point to the query point

参数

in	*input*	Input point cloud containing all points

参数

in	*kdtree*	KD-tree structure built from the input cloud
out	*result*	The nearest neighbor point found
in	*point*	Query point to find neighbors for
in	*radius*	Search radius limit

5.5.2.31 pcdIntensityThresholdLowerFilter()

```
void lidarTools::pcdIntensityThresholdLowerFilter (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr output,
            int intensity_thres )
```

Filter points based on intensity threshold.

参数

in	*in_cloud*	Input point cloud
out	*output*	Output point cloud containing filtered points
in	*intensity_thres*	Intensity threshold value - points with intensity above this are kept

5.5.2.32 pcdKmeansCluster()

```
void lidarTools::pcdKmeansCluster (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            ClusterResult & result,
            int k,
            const int max_iterations_,
            const double convergence_threshold )
```

Perform k-means clustering on point cloud.

参数

in	*in_cloud*	Input point cloud
out	*result*	Clustering result containing labels and centroids
in	*k*	Number of clusters
in	*max_iterations_*	Maximum number of iterations
in	*convergence_threshold*	Error threshold for convergence

5.5.2.33 pcdMeanShiftCluster()

```
void lidarTools::pcdMeanShiftCluster (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            std::vector< pcl::PointIndices > & cluster_indices,
            float bandwidth,
            int max_iter )
```

Performs mean-shift clustering on a point cloud.

Uses mean-shift algorithm to cluster points by iteratively shifting points towards local density maxima

参数

input	Input point cloud to be clustered
cluster_indices	Output vector containing indices for each cluster
bandwidth	Kernel bandwidth parameter that determines the size of the window used to compute mean shift
max_iter	Maximum number of iterations for mean shift convergence

注解

The algorithm works by:

1. For each point, calculating weighted mean of nearby points within bandwidth
2. Shifting the point towards the weighted mean
3. Repeating until convergence or max iterations reached
4. Points that converge to the same mode are grouped into clusters

5.5.2.34 pcdOPTICSCluster()

```
void lidarTools::pcdOPTICSCluster (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            std::vector< pcl::PointIndices > & cluster_indices,
            double eps,
            int min_pts,
            double xi )
```

OPTICS (Ordering Points To Identify the Clustering Structure) clustering algorithm.

Performs density-based clustering using the OPTICS algorithm, which creates an augmented ordering of the data points that represents the density-based clustering structure of the data

参数

in	*input*	Input point cloud to be clustered
out	*cluster_indices*	Vector of point indices for each cluster found
in	*eps*	Maximum distance between two points for them to be considered neighbors (default: 1.0)
in	*min_pts*	Minimum number of points required to form a dense region (default: 5)
in	*xi*	Steepness threshold for detecting cluster boundaries (default: 0.05)

5.5.2.35 pcdRadiusFilter()

```
void lidarTools::pcdRadiusFilter (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr output,
            const double radius,
            const int min_neighbors )
```

Apply radius outlier removal filter.

参数

in	*in_cloud*	Input point cloud
out	*output*	Output filtered point cloud
in	*radius*	Search radius for finding neighbors
in	*min_neighbors*	Minimum number of neighbors required to keep point

5.5.2.36 pcdRandomSample()

```
void lidarTools::pcdRandomSample (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr output,
            const int num_samples )
```

Perform random sampling on point cloud.

参数

in	*in_cloud*	Input point cloud
out	*output*	Output sampled point cloud
in	*num_samples*	Number of points to sample

5.5.2.37 pcdRotateWithEuler()

```
void lidarTools::pcdRotateWithEuler (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            pcl::PointCloud< pcl::PointXYZI >::Ptr output,
            double roll,
            double pitch,
            double yaw )
```

Rotates a point cloud using Euler angles.

参数

input	Input point cloud
output	Output rotated point cloud
roll	Roll angle in radians
pitch	Pitch angle in radians
yaw	Yaw angle in radians

5.5.2.38 pcdRotateWithMat()

```
void lidarTools::pcdRotateWithMat (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            pcl::PointCloud< pcl::PointXYZI >::Ptr output,
            Eigen::Matrix3f rotationMatrix )
```

Rotates a point cloud using a rotation matrix.

参数

input	Input point cloud
output	Output rotated point cloud
rotationMatrix	3x3 rotation matrix

5.5.2.39 pcdRotateWithQuaternions()

```
void lidarTools::pcdRotateWithQuaternions (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            pcl::PointCloud< pcl::PointXYZI >::Ptr output,
            Eigen::Quaterniond q )
```

Rotates a point cloud using quaternions.

参数

input	Input point cloud
output	Output rotated point cloud
q	Quaternion for rotation

5.5.2.40 pcdSpectralClustering()

```
void lidarTools::pcdSpectralClustering (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
```

```
            std::vector< pcl::PointIndices > & cluster_indices,
            int n_clusters,
            double sigma,
            int k_neighbors )
```

Performs spectral clustering on a point cloud.

参数

input	Input point cloud to be clustered
cluster_indices	Output vector containing the indices of points in each cluster
n_clusters	Number of clusters to generate
sigma	Gaussian kernel bandwidth parameter
k_neighbors	Number of nearest neighbors for similarity graph construction

5.5.2.41 pcdStatisticFilter()

```
void lidarTools::pcdStatisticFilter (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr output,
            const double mean_K,
            const double std )
```

Apply statistical outlier removal filter.

参数

in	*in_cloud*	Input point cloud
out	*output*	Output filtered point cloud
in	*mean↩ _K*	Number of nearest neighbors to use for mean distance estimation
in	*std*	Standard deviation multiplier threshold

5.5.2.42 pcdToKDTree()

```
void lidarTools::pcdToKDTree (
            pcl::PointCloud< pcl::PointXYZI >::Ptr cloud,
            pcl::KdTreeFLANN< pcl::PointXYZI >::Ptr kdtree )
```

Creates a KD-tree structure from a point cloud for efficient nearest neighbor searches.

Builds a KD-tree data structure from the input point cloud to enable fast spatial queries

参数

in	*cloud*	Input point cloud to build the KD-tree from
out	*kdtree*	Output KD-tree structure

5.5.2.43 pcdTrans()

```
void lidarTools::pcdTrans (
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            pcl::PointCloud< pcl::PointXYZI >::Ptr output,
            double trans_x,
            double trans_y,
            double trans_z )
```

Translates a point cloud.

参数

input	Input point cloud
output	Output translated point cloud
trans↩ _x	Translation in X direction
trans↩ _y	Translation in Y direction
trans↩ _z	Translation in Z direction

5.5.2.44 pcdVoxelFilter()

```
void lidarTools::pcdVoxelFilter (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr output,
            const double size_x,
            const double size_y,
            const double size_z )
```

Apply voxel grid filter to downsample point cloud.

参数

in	*in_cloud*	Input point cloud
out	*output*	Output downsampled point cloud
in	*size_x*	Voxel size in x dimension
in	*size_y*	Voxel size in y dimension
in	*size_z*	Voxel size in z dimension

5.5.2.45 pcdWardCluster()

```
void lidarTools::pcdWardCluster (
```

```
            pcl::PointCloud< pcl::PointXYZI >::Ptr input,
            std::vector< pcl::PointIndices > & cluster_indices,
            float distance_threshold,
            int min_cluster_size,
            int max_cluster_size )
```

Performs Ward hierarchical clustering on a point cloud.

参数

input	Input point cloud to be clustered
cluster_indices	Output vector containing the indices of points in each cluster
distance_threshold	Maximum distance between clusters to be merged
min_cluster_size	Minimum number of points a cluster must contain
max_cluster_size	Maximum number of points a cluster can contain

5.5.2.46 PFHExtractFeature()

```
void lidarTools::PFHExtractFeature (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            double radius )
```

Extracts Point Feature Histogram (PFH) features from a point cloud.

参数

in_cloud	Input point cloud
radius	Search radius for feature computation

5.5.2.47 readBinaryFile()

```
std::vector< pcl::PointXYZI > lidarTools::readBinaryFile (
            const std::string & filename )
```

Reads points from a binary file.

参数

filename	Input binary file path

返回

Vector of points read from file

5.5.2.48 saveToBin()

```
void lidarTools::saveToBin (
            pcl::PointCloud< pcl::PointXYZI >::Ptr cloud,
            std::string & out_file )
```

Saves point cloud to binary format.

参数

cloud	Input point cloud
out_file	Output binary file path

5.5.2.49 SIFTKeyPointExtract()

```
void lidarTools::SIFTKeyPointExtract (
            pcl::PointCloud< pcl::PointXYZI >::Ptr in_cloud,
            pcl::PointCloud< pcl::PointWithScale >::Ptr keypoints,
            float min_scale_,
            int nr_octaves,
            int nr_scales_per_octave,
            float min_contrast )
```

Extract SIFT keypoints from point cloud.

参数

in	*in_cloud*	Input point cloud
out	*keypoints*	Output keypoint cloud
in	*min_scale_*	Minimum scale for keypoint detection
in	*nr_octaves*	Number of octaves
in	*nr_scales_per_octave*	Number of scales per octave
in	*min_contrast*	Minimum contrast threshold

5.5.2.50 updateCentroids()

```
void lidarTools::updateCentroids (
            pcl::PointCloud< pcl::PointXYZI >::Ptr cloud,
            pcl::PointCloud< pcl::PointXYZI >::Ptr centroids,
            std::vector< int > & labels,
            int k )
```

Update cluster centroids based on point assignments.

参数

in	*cloud*	Input point cloud
in,out	*centroids*	Cluster centroid points to update
in	*labels*	Point cluster assignments
in	*k*	Number of clusters

5.5.2.51 updateDistances()

```
void lidarTools::updateDistances (
            pcl::PointCloud< pcl::PointXYZI >::Ptr cloud,
            int point_idx,
            std::vector< float > & min_distances,
            pcl::KdTreeFLANN< PointXYZI >::Ptr kdtree )
```

Update minimum distances between points and clusters.

参数

in	*cloud*	Input point cloud
in	*point_idx*	Index of current point
in,out	*min_distances*	Vector of minimum distances to update
in	*kdtree*	KD-tree for neighbor search

5.6 LocalPlanning 命名空参考

是一个名LocalPlanning的namespace。

- struct ableToPassYellowLightInput
- struct ableToPassYellowLightOutput
- struct ableToPassYellowLightParam
- struct angleRegulateInput
- struct angleRegulateOutput
- struct angleRegulateParam
- struct assessTrajectoryInput
- struct assessTrajectoryOutput
- struct assessTrajectoryParam
- struct boundaryCollisionCheckInput
- struct boundaryCollisionCheckOutput
- struct boundaryCollisionCheckParam
- struct calculateCurvatureInput
- struct calculateCurvatureOutput
- struct calculateCurvatureParam

- struct changeCurrentPointInput
- struct changeCurrentPointOutput
- struct changeCurrentPointParam
- struct changeLocalToGlobalInput
- struct changeLocalToGlobalOutput
- struct changeLocalToGlobalParam
- struct clearPathListInput
- struct clearPathListOutput
- struct clearPathListParam
- struct coordTran2DInput
- struct coordTran2DOutput
- struct coordTran2DParam
- struct courseAngleReviseInput
- struct courseAngleReviseOutput
- struct courseAngleReviseParam
- struct cruiseControllerInput
- struct cruiseControllerOutput
- struct cruiseControllerParam
- struct DecisionData
- struct findMaxInput
- struct findMaxOutput
- struct findMaxParam
- struct findMinInput
- struct findMinOutput
- struct findMinParam
- struct fixRoadLaneIndexInput
- struct fixRoadLaneIndexOutput
- struct fixRoadLaneIndexParam
- struct frenet2CartesianInput
- struct frenet2CartesianOutput
- struct frenet2CartesianParam
- struct gaussConvertInput
- struct gaussConvertOutput
- struct gaussConvertParam
- struct generateBezierPathInFrenetInput
- struct generateBezierPathInFrenetOutput
- struct generateBezierPathInFrenetParam
- struct generateBezierPathListInFrenetInput
- struct generateBezierPathListInFrenetOutput
- struct generateBezierPathListInFrenetParam
- struct generateSegmentedBezierPathListInFrenetInput
- struct generateSegmentedBezierPathListInFrenetOutput
- struct generateSegmentedBezierPathListInFrenetParam
- struct generateSpeedListInput
- struct generateSpeedListOutput
- struct generateSpeedListParam
- struct generateTrajectoryListInput
- struct generateTrajectoryListOutput
- struct generateTrajectoryListParam
- struct getAngleDiffInput
- struct getAngleDiffOutput
- struct getAngleDiffParam
- struct getAngleInput
- struct getAngleOutput
- struct getAngleParam

- struct getCurrentPositionInput
- struct getCurrentPositionOutput
- struct getCurrentPositionParam
- struct getDistanceInput
- struct getDistanceOutput
- struct getDistanceParam
- struct getFrenetLocationPointInput
- struct getFrenetLocationPointOutput
- struct getFrenetLocationPointParam
- struct getMinDistanceOfPointInput
- struct getMinDistanceOfPointOutput
- struct getMinDistanceOfPointParam
- struct getNextRoadLaneIdInput
- struct getNextRoadLaneIdOutput
- struct getNextRoadLaneIdParam
- struct getOptimalTrajectoryIndexInput
- struct getOptimalTrajectoryIndexOutput
- struct getOptimalTrajectoryIndexParam
- struct getPlanningDistanceInput
- struct getPlanningDistanceOutput
- struct getPlanningDistanceParam
- struct getPlanningPointInput
- struct getPlanningPointOutput
- struct getPlanningPointParam
- struct getReferenceLineInput
- struct getReferenceLineOutput
- struct getReferenceLineParam
- struct getTrajectoryPlanningResultInput
- struct getTrajectoryPlanningResultOutput
- struct getTrajectoryPlanningResultParam
- struct hasPrecedingVehicleInput
- struct hasPrecedingVehicleOutput
- struct hasPrecedingVehicleParam
- struct inAreaInput
- struct inAreaOutput
- struct inAreaParam
- struct initLocalPlanningInput
- struct initLocalPlanningOutput
- struct initLocalPlanningParam
- struct innerCollisionCheckInput
- struct innerCollisionCheckOutput
- struct innerCollisionCheckParam
- struct limitPlanningDistanceInput
- struct limitPlanningDistanceOutput
- struct limitPlanningDistanceParam
- struct loadStopPointsFromYamlInput
- struct loadStopPointsFromYamlOutput
- struct loadStopPointsFromYamlParam
- struct loadStopPointsInput
- struct loadStopPointsOutput
- struct loadStopPointsParam
- struct PlanningPoint
- struct PlanningTrajectory
- struct pointCollisionCheckInput
- struct pointCollisionCheckOutput

- struct pointCollisionCheckParam
- struct pointDistanceInput
- struct pointDistanceOutput
- struct pointDistanceParam
- struct pointOnCubicBezierInput
- struct pointOnCubicBezierOutput
- struct pointOnCubicBezierParam
- struct pointToLineDistanceInput
- struct pointToLineDistanceOutput
- struct pointToLineDistanceParam
- struct processTrafficLightInput
- struct processTrafficLightOutput
- struct processTrafficLightParam
- struct ReferenceLine
- struct restartInput
- struct restartOutput
- struct restartParam
- struct stopInput
- struct stopOutput
- struct stopParam
- struct stopPointJudgeInput
- struct stopPointJudgeOutput
- struct stopPointJudgeParam
- struct trajectoryCollisionCheckInput
- struct trajectoryCollisionCheckOutput
- struct trajectoryCollisionCheckParam

函数

- void gaussConvert (const gaussConvertParam &param, const gaussConvertInput &input, gaussConvertOutput &output)

  度全局坐
- void coordTran2D (const coordTran2DParam &param, const coordTran2DInput &input, coordTran2DOutput &output)

  全局坐局部坐
- void inArea (const inAreaParam &param, const inAreaInput &input, inAreaOutput &output)

  判断点是否在区域内
- void pointDistance (const pointDistanceParam &param, const pointDistanceInput &input, pointDistanceOutput &output)

  算两点之的距离
- void courseAngleRevise (const courseAngleReviseParam &param, const courseAngleReviseInput &input, courseAngleReviseOutput &output)

  逆旋坐系*90*度，向*0*度成向北*0*度
- void loadStopPoints (const loadStopPointsParam &param, const loadStopPointsInput &input, loadStopPointsOutput &output)

  从文件中加停止点
- void loadStopPointsFromYaml (const loadStopPointsFromYamlParam &param, const loadStopPointsFromYamlInput &input, loadStopPointsFromYamlOutput &output)

  从文件中加停止点
- void getNextRoadLaneId (const getNextRoadLaneIdParam &param, const getNextRoadLaneIdInput &input, getNextRoadLaneIdOutput &output)

  从全局划中，取后道和道路*Id*

- void getPlanningDistance (const getPlanningDistanceParam &param, const getPlanningDistanceInput &input, getPlanningDistanceOutput &output)

  根据速度、曲率取划距离
- void getPlanningPoint (const getPlanningPointParam &param, const getPlanningPointInput &input, getPlanningPointOutput &output)

  根据信息取划瞄点
- void limitPlanningDistance (const limitPlanningDistanceParam &param, const limitPlanningDistanceInput &input, limitPlanningDistanceOutput &output)

  限制划距离，防止突
- void getAngle (const getAngleParam &param, const getAngleInput &input, getAngleOutput &output)

  算坐系中两点角度
- void getAngleDiff (const getAngleDiffParam &param, const getAngleDiffInput &input, getAngleDiffOutput &output)

  算两角度之差，并正化
- void **localPlanning** (const pc::Imu &imu, double velocity, const RoadMap &map, DecisionData &decision↩Data, const prediction::ObjectList &predictionMsg, const std::vector< std::tuple< int32_t, int32_t >> &routingList, const std::vector< GaussRoadPoint > stopPoints, const infopack::TrafficLight &trafficLight, std::vector< infopack::ObjectsProto > objectsCmd)
- void restart (const restartParam &param, const restartInput &input, restartOutput &output)

  停后，判断能否重新后
- void stop (const stopParam &param, const stopInput &input, stopOutput &output)

  停修改速度曲
- void stopPointJudge (const stopPointJudgeParam &param, const stopPointJudgeInput &input, stopPointJudgeOutput &output)

  判断是否在停区域
- void pointOnCubicBezier (const pointOnCubicBezierParam &param, const pointOnCubicBezierInput &input, pointOnCubicBezierOutput &output)

  算塞曲上的点
- void generateBezierPathInFrenet (const generateBezierPathInFrenetParam &param, const generateBezierPathInFrenetInput &input, generateBezierPathInFrenetOutput &output)

  在*frenet*中生成塞曲
- void clearPathList (const clearPathListParam &param, const clearPathListInput &input, clearPathListOutput &output)

  清除上次生成的迹
- void getTrajectoryPlanningResult (const getTrajectoryPlanningResultParam &param, const getTrajectoryPlanningResultInput &input, getTrajectoryPlanningResultOutput &output)

  完成了多条迹和速度曲的生成，并化到笛卡坐系中，作出了迹的
- void getDistance (const getDistanceParam &param, const getDistanceInput &input, getDistanceOutput &output)

  算两点之的距离
- void pointToLineDistance (const pointToLineDistanceParam &param, const pointToLineDistanceInput &input, pointToLineDistanceOutput &output)

  算点到段的距离
- void angleRegulate (const angleRegulateParam &param, const angleRegulateInput &input, angleRegulateOutput &output)

  化角度到*0-180*中，便于比
- void getCurrentPosition (const getCurrentPositionParam &param, const getCurrentPositionInput &input, getCurrentPositionOutput &output)

  找距离位置最近的路点
- void generateSpeedList (const generateSpeedListParam &param, const generateSpeedListInput &input, generateSpeedListOutput &output)

  根据道路信息等，采生成速度曲

- void generateTrajectoryList (const generateTrajectoryListParam &param, const generateTrajectoryListInput &input, generateTrajectoryListOutput &output)

  生成采迹，供后

- void processTrafficLight (const processTrafficLightParam &param, const processTrafficLightInput &input, processTrafficLightOutput &output)

  根据灯信息更新决策数据

- void getOptimalTrajectoryIndex (const getOptimalTrajectoryIndexParam &param, const getOptimalTrajectoryIndexInput &input, getOptimalTrajectoryIndexOutput &output)

  估迹，并取最迹索引

- void trajectoryCollisionCheck (const trajectoryCollisionCheckParam &param, const trajectoryCollisionCheckInput &input, trajectoryCollisionCheckOutput &output)

  迹碰撞，*true*表示可行，*false*表示碰撞

- void ableToPassYellowLight (const ableToPassYellowLightParam &param, const ableToPassYellowLightInput &input, ableToPassYellowLightOutput &output)

  判断是否能通黄灯

- void pointCollisionCheck (const pointCollisionCheckParam &param, const pointCollisionCheckInput &input, pointCollisionCheckOutput &output)

  迹点碰撞

- void boundaryCollisionCheck (const boundaryCollisionCheckParam &param, const boundaryCollisionCheckInput &input, boundaryCollisionCheckOutput &output)

  碰撞

- void innerCollisionCheck (const innerCollisionCheckParam &param, const innerCollisionCheckInput &input, innerCollisionCheckOutput &output)

  内部包含

- void changeCurrentPoint (const changeCurrentPointParam &param, const changeCurrentPointInput &input, changeCurrentPointOutput &output)

  判断是否可以道

- void changeLocalToGlobal (const changeLocalToGlobalParam &param, const changeLocalToGlobalInput &input, changeLocalToGlobalOutput &output)

  局部坐迹全局迹

- void initLocalPlanning (const initLocalPlanningParam &param, const initLocalPlanningInput &input, initLocalPlanningOutput &output)

  局部划决策数据初始化

- void fixRoadLaneIndex (const fixRoadLaneIndexParam &param, const fixRoadLaneIndexInput &input, fixRoadLaneIndexOutput &output)

  返回*decisionDate*中*roadID* 和*laneID*的索引

- void calculateCurvature (const calculateCurvatureParam &param, const calculateCurvatureInput &input, calculateCurvatureOutput &output)

  由相三点近似算曲率

- void assessTrajectory (const assessTrajectoryParam &param, const assessTrajectoryInput &input, assessTrajectoryOutput &output)

  返回距离障碍物最近距离，行迹估

- void getMinDistanceOfPoint (const getMinDistanceOfPointParam &param, const getMinDistanceOfPointInput &input, getMinDistanceOfPointOutput &output)

  返回点距离障碍物最近距离

- void findMin (const findMinParam &param, const findMinInput &input, findMinOutput &output)

  返回*double*数最小

- void findMax (const findMaxParam &param, const findMaxInput &input, findMaxOutput &output)

  返回*double*数最大

- void getReferenceLine (const getReferenceLineParam &param, const getReferenceLineInput &input, getReferenceLineOutput &output)

  取局部划参考迹

- void getFrenetLocationPoint (const getFrenetLocationPointParam &param, const getFrenetLocationPointInput &input, getFrenetLocationPointOutput &output)

  取当前位置在*frenet*坐系中坐
- void frenet2Cartesian (const frenet2CartesianParam &param, const frenet2CartesianInput &input, frenet2CartesianOutput &output)

  将*frenet*坐系坐自坐系坐，*xy*右前
- void generateBezierPathListInFrenet (const generateBezierPathListInFrenetParam &param, const generateBezierPathListInFrenetInput &input, generateBezierPathListInFrenetOutput &output)

  在*frenet*坐系中生成多条塞曲
- void hasPrecedingVehicle (const hasPrecedingVehicleParam &param, const hasPrecedingVehicleInput &input, hasPrecedingVehicleOutput &output)

  判断前方是否有需要跟随的
- void cruiseController (const cruiseControllerParam &param, const cruiseControllerInput &input, cruiseControllerOutput &output)

  *acc*景下，目速度算
- void **generateSegmentedBezierPathListInFrenet** (const generateSegmentedBezierPathListInFrenetParam &param, const generateSegmentedBezierPathListInFrenetInput &input, generateSegmentedBezierPathListInFrenetOutput &output)

5.6.1 描述

是一个名LocalPlanning的namespace。

5.6.2 函数明

5.6.2.1 ableToPassYellowLight()

```
void LocalPlanning::ableToPassYellowLight (
            const ableToPassYellowLightParam & param,
            const ableToPassYellowLightInput & input,
            ableToPassYellowLightOutput & output )
```

判断是否能通黄灯

参数

[IN]	param 无
[IN]	input 迹，剩余，到停止前的距离
[OUT]	output 是否通

@cn_name: 判断黄灯能否通

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.2 angleRegulate()

```
void LocalPlanning::angleRegulate (
            const angleRegulateParam & param,
            const angleRegulateInput & input,
            angleRegulateOutput & output )
```

化角度到0-180中，便于比

参数

[IN]	param 无
[IN]	input 原角度
[O↩ UT]	output 化后的角度

@cn_name: 正化角度

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.3 assessTrajectory()

```
void LocalPlanning::assessTrajectory (
            const assessTrajectoryParam & param,
            const assessTrajectoryInput & input,
            assessTrajectoryOutput & output )
```

返回距离障碍物最近距离，行迹估

参数

[IN]	param 格障碍物，imu，路障碍物
[IN]	input 迹
[O↩ UT]	output 距离障碍物最近距离

@cn_name: 迹估

@granularity: composite //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.4 boundaryCollisionCheck()

```
void LocalPlanning::boundaryCollisionCheck (
            const boundaryCollisionCheckParam & param,
```

```
            const boundaryCollisionCheckInput & input,
            boundaryCollisionCheckOutput & output )
```

碰撞

参数

[IN]	param 无
[IN]	input 段点
[OUT]	output 是否碰撞

@cn_name: 碰撞

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.5 calculateCurvature()

```
void LocalPlanning::calculateCurvature (
            const calculateCurvatureParam & param,
            const calculateCurvatureInput & input,
            calculateCurvatureOutput & output )
```

由相三点近似算曲率

参数

[IN]	param 无
[IN]	input 点，前一点，后一点
[OUT]	output 曲率

@cn_name: 算曲率

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.6 changeCurrentPoint()

```
void LocalPlanning::changeCurrentPoint (
            const changeCurrentPointParam & param,
            const changeCurrentPointInput & input,
            changeCurrentPointOutput & output )
```

判断是否可以道

参数

[IN]	param 当前道路，当前道
[IN]	input 原划瞄点道Id，原划瞄点路点Id，原划瞄点，向左是向右道
[OUT]	output 新瞄点道Id，新划瞄点路点Id，新划瞄点，是否道

@cn_name: 判断是否可以道

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.7 changeLocalToGlobal()

```
void LocalPlanning::changeLocalToGlobal (
            const changeLocalToGlobalParam & param,
            const changeLocalToGlobalInput & input,
            changeLocalToGlobalOutput & output )
```

局部坐迹全局迹

参数

[IN]	param 无
[IN]	input 局部坐迹，局部坐迹原地的全局点
[OUT]	output 全局迹

@cn_name: 局部坐迹全局坐迹

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.8 clearPathList()

```
void LocalPlanning::clearPathList (
            const clearPathListParam & param,
            const clearPathListInput & input,
            clearPathListOutput & output )
```

清除上次生成的迹

参数

[IN]	param 无
[IN]	input 存迹的向量
[OUT]	output 判断果

@cn_name: 清除迹

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.9 coordTran2D()

```
void LocalPlanning::coordTran2D (
            const coordTran2DParam & param,
            const coordTran2DInput & input,
            coordTran2DOutput & output )
```

全局坐局部坐

参数

[IN]	param 无
[IN]	input 全局坐度、角度，初始化局部坐xy、角度
[OUT]	output 局部坐、角度

@cn_name: 全局坐自局部坐

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.10 courseAngleRevise()

```
void LocalPlanning::courseAngleRevise (
            const courseAngleReviseParam & param,
            const courseAngleReviseInput & input,
            courseAngleReviseOutput & output )
```

逆旋坐系90度，向0度成向北0度

参数

[IN]	param 无
[IN]	input 原始角度
[OUT]	output 化角度

@cn_name: 坐系逆旋90度

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.11 cruiseController()

```
void LocalPlanning::cruiseController (
            const cruiseControllerParam & param,
            const cruiseControllerInput & input,
            cruiseControllerOutput & output )
```

acc景下，目速度算

参数

[IN]	param imu，路障碍物
[IN]	input 前索引，距前距离
[OUT]	output 期速度

@cn_name: 算acc目速度

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.12 findMax()

```
void LocalPlanning::findMax (
            const findMaxParam & param,
            const findMaxInput & input,
            findMaxOutput & output )
```

返回double数最大

参数

[IN]	param 无
[IN]	input double数
[OUT]	output double数最大

@cn_name: 返回double数最大

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.13 findMin()

```
void LocalPlanning::findMin (
            const findMinParam & param,
            const findMinInput & input,
            findMinOutput & output )
```

返回double数最小

参数

[IN]	param 无
[IN]	input double数
[OUT]	output double数最小

@cn_name: 返回double数最小

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.14 fixRoadLaneIndex()

```
void LocalPlanning::fixRoadLaneIndex (
            const fixRoadLaneIndexParam & param,
            const fixRoadLaneIndexInput & input,
            fixRoadLaneIndexOutput & output )
```

返回decisionDate中roadID 和laneID的索引

参数

[IN]	param 地
[IN]	input 决策数据
[OUT]	output roadID 和laneID的索引

@cn_name: 得道路和道的索引

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.15 frenet2Cartesian()

```
void LocalPlanning::frenet2Cartesian (
            const frenet2CartesianParam & param,
```

```
            const frenet2CartesianInput & input,
            frenet2CartesianOutput & output )
```

将frenet坐系坐自坐系坐，xy右前

参数

[IN]	param 无
[IN]	input 坐点sl，初始化xy，参考迹，上次搜索的索引，本次需返回的坐点的索引
[OUT]	output 坐点xy，坐点的索引

@cn_name: frenet坐系局部坐系

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.16 gaussConvert()

```
void LocalPlanning::gaussConvert (
            const gaussConvertParam & param,
            const gaussConvertInput & input,
            gaussConvertOutput & output )
```

度全局坐

参数

[IN]	param :无
[IN]	input :坐度，初始化全局坐xy
[OUT]	output :全局坐

@cn_name: 度全局坐

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.17 generateBezierPathInFrenet()

```
void LocalPlanning::generateBezierPathInFrenet (
            const generateBezierPathInFrenetParam & param,
            const generateBezierPathInFrenetInput & input,
            generateBezierPathInFrenetOutput & output )
```

在frenet中生成塞曲

参数

[IN]	param 无
[IN]	input 起点、点、初始化迹
[O↩ UT]	output 生成迹

@cn_name: 生成塞曲

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.18 generateBezierPathListInFrenet()

```
void LocalPlanning::generateBezierPathListInFrenet (
            const generateBezierPathListInFrenetParam & param,
            const generateBezierPathListInFrenetInput & input,
            generateBezierPathListInFrenetOutput & output )
```

在frenet坐系中生成多条塞曲

参数

[IN]	param 无
[IN]	input 参考迹，在frenet坐系下坐，初始化的迹向量数
[O↩ UT]	output 迹向量数

@cn_name: 在frenet坐系中生成多条塞曲

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: plannings

5.6.2.19 generateSpeedList()

```
void LocalPlanning::generateSpeedList (
            const generateSpeedListParam & param,
            const generateSpeedListInput & input,
            generateSpeedListOutput & output )
```

根据道路信息等，采生成速度曲

参数

[IN]	param 路障碍物，imu，停止点
[IN]	input 速，决策数据
[O↩ UT]	output 更新后的决策数据

@cn_name: 生成速度曲

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.20 generateTrajectoryList()

```
void LocalPlanning::generateTrajectoryList (
            const generateTrajectoryListParam & param,
            const generateTrajectoryListInput & input,
            generateTrajectoryListOutput & output )
```

生成采迹，供后

参数

[IN]	param 无
[IN]	input 决策数据
[OUT]	output 更新后的决策数据

@cn_name: 生成采曲

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.21 getAngle()

```
void LocalPlanning::getAngle (
            const getAngleParam & param,
            const getAngleInput & input,
            getAngleOutput & output )
```

算坐系中两点角度

参数

[IN]	param 无
[IN]	input 两点XY坐
[OUT]	output 两点角度

@cn_name: 算两点角度

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.22 getAngleDiff()

```
void LocalPlanning::getAngleDiff (
            const getAngleDiffParam & param,
            const getAngleDiffInput & input,
            getAngleDiffOutput & output )
```

算两角度之差，并正化

参数

[IN]	param 无
[IN]	input 两角度
[OUT]	output 两角度之差

@cn_name: 算两角度之差

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.23 getCurrentPosition()

```
void LocalPlanning::getCurrentPosition (
            const getCurrentPositionParam & param,
            const getCurrentPositionInput & input,
            getCurrentPositionOutput & output )
```

找距离位置最近的路点

参数

[IN]	param imu，地
[IN]	input 决策数据
[OUT]	output 更新的决策具

@cn_name: 找距离位置最近的路点

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.24 getDistance()

```
void LocalPlanning::getDistance (
            const getDistanceParam & param,
            const getDistanceInput & input,
            getDistanceOutput & output )
```

算两点之的距离

参数

[IN]	param 无
[IN]	input 两点XY坐
[OUT]	output 两点之的距离

@cn_name: 算两点之的距离

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.25 getFrenetLocationPoint()

```
void LocalPlanning::getFrenetLocationPoint (
            const getFrenetLocationPointParam & param,
            const getFrenetLocationPointInput & input,
            getFrenetLocationPointOutput & output )
```

取当前位置在frenet坐系中坐

参数

[IN]	param 无
[IN]	input 决策数据，参考迹，当前位置
[OUT]	output 更新后的决策数据

@cn_name: 取在frenet坐系中坐

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.26 getMinDistanceOfPoint()

```
void LocalPlanning::getMinDistanceOfPoint (
            const getMinDistanceOfPointParam & param,
```

```
            const getMinDistanceOfPointInput & input,
            getMinDistanceOfPointOutput & output )
```

返回点距离障碍物最近距离

参数

[IN]	param 格障碍物，imu，路障碍物
[IN]	input 点
[OUT]	output 距离障碍物最近距离

@cn_name: 算点距离障碍物最近距离（一般景）

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.27 getNextRoadLaneId()

```
void LocalPlanning::getNextRoadLaneId (
            const getNextRoadLaneIdParam & param,
            const getNextRoadLaneIdInput & input,
            getNextRoadLaneIdOutput & output )
```

从全局划中，取后道和道路Id

参数

[IN]	param 存全局划道和道路的向量
[IN]	input 自当前道和道路Id
[OUT]	output 后道和道路Id

@cn_name: 取后道和道路Id

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.28 getOptimalTrajectoryIndex()

```
void LocalPlanning::getOptimalTrajectoryIndex (
            const getOptimalTrajectoryIndexParam & param,
            const getOptimalTrajectoryIndexInput & input,
            getOptimalTrajectoryIndexOutput & output )
```

估迹，并取最迹索引

参数

[IN]	param 格障碍物，交通灯，imu，路障碍物
[IN]	input 决策数据
[OUT]	output 更新后的决策数据，最迹索引

@cn_name: 取最迹索引

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.29 getPlanningDistance()

```
void LocalPlanning::getPlanningDistance (
            const getPlanningDistanceParam & param,
            const getPlanningDistanceInput & input,
            getPlanningDistanceOutput & output )
```

根据速度、曲率取划距离

参数

[IN]	param 无
[IN]	input 速度、曲率
[OUT]	output 划距离

@cn_name: 算划距离

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.30 getPlanningPoint()

```
void LocalPlanning::getPlanningPoint (
            const getPlanningPointParam & param,
            const getPlanningPointInput & input,
            getPlanningPointOutput & output )
```

根据信息取划瞄点

参数

[IN]	param 当前位置，决策数据，地
[IN]	input 划距离，朝向角，后道Id
[OUT]	output 划瞄点

@cn_name: 算划瞄点

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.31 getReferenceLine()

```
void LocalPlanning::getReferenceLine (
            const getReferenceLineParam & param,
            const getReferenceLineInput & input,
            getReferenceLineOutput & output )
```

取局部划参考迹

参数

[IN]	param 地
[IN]	input 决策数据，当前位置
[OUT]	output 参考迹

@cn_name: 取参考迹

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.32 getTrajectoryPlanningResult()

```
void LocalPlanning::getTrajectoryPlanningResult (
            const getTrajectoryPlanningResultParam & param,
            const getTrajectoryPlanningResultInput & input,
            getTrajectoryPlanningResultOutput & output )
```

完成了多条迹和速度曲的生成，并化到笛卡坐系中，作出了迹的

参数

[IN]	param imu，格障碍物，地，交通灯，路障碍物
[IN]	input 决策数据
[OUT]	output 更新后的决策数据，是否成功生成迹

@cn_name: 取迹划果

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.33 hasPrecedingVehicle()

```
void LocalPlanning::hasPrecedingVehicle (
            const hasPrecedingVehicleParam & param,
            const hasPrecedingVehicleInput & input,
            hasPrecedingVehicleOutput & output )
```

判断前方是否有需要跟随的

参数

[IN]	param imu，路障碍物
[IN]	input 无
[OUT]	output 判断果

@cn_name: 判断是否有需要跟随的

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.34 inArea()

```
void LocalPlanning::inArea (
            const inAreaParam & param,
            const inAreaInput & input,
            inAreaOutput & output )
```

判断点是否在区域内

参数

[IN]	param 无
[IN]	input 目区域中心XY坐，判断点XY坐
[OUT]	output 判断果

@cn_name: 判断点是否在区域内

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.35 initLocalPlanning()

```
void LocalPlanning::initLocalPlanning (
            const initLocalPlanningParam & param,
            const initLocalPlanningInput & input,
            initLocalPlanningOutput & output )
```

局部划决策数据初始化

参数

[IN]	param 无
[IN]	input 决策数据
[OUT]	output 初始化后的决策数据

@cn_name: 初始化局部划

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.36 innerCollisionCheck()

```
void LocalPlanning::innerCollisionCheck (
            const innerCollisionCheckParam & param,
            const innerCollisionCheckInput & input,
            innerCollisionCheckOutput & output )
```

内部包含

参数

[IN]	param 无
[IN]	input 点和障碍物点
[OUT]	output 是否包含

@cn_name: 内部包含

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.37 limitPlanningDistance()

```
void LocalPlanning::limitPlanningDistance (
            const limitPlanningDistanceParam & param,
```

```
            const limitPlanningDistanceInput & input,
            limitPlanningDistanceOutput & output )
```

限制划距离，防止突

参数

[IN]	param 无
[IN]	input 划划距离，上次划距离
[OUT]	output 更新后的划距离

@cn_name: 限制划距离

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.38 loadStopPoints()

```
void LocalPlanning::loadStopPoints (
            const loadStopPointsParam & param,
            const loadStopPointsInput & input,
            loadStopPointsOutput & output )
```

从文件中加停止点

参数

[IN]	param 文件地址
[IN]	input 停止点、停止点路点、X偏移、Y偏移
[OUT]	output 更新后的停止点、更新后的停止点路点

@cn_name: 加停止点

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.39 loadStopPointsFromYaml()

```
void LocalPlanning::loadStopPointsFromYaml (
            const loadStopPointsFromYamlParam & param,
            const loadStopPointsFromYamlInput & input,
            loadStopPointsFromYamlOutput & output )
```

从文件中加停止点

参数

[IN]	param 地、文件地址
[IN]	input 停止点、停止点路点、X偏移、Y偏移
[OUT]	output 更新的停止点、更新的停止点路点

@cn_name: 从Yaml文件中加停止点

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.40 pointCollisionCheck()

```
void LocalPlanning::pointCollisionCheck (
            const pointCollisionCheckParam & param,
            const pointCollisionCheckInput & input,
            pointCollisionCheckOutput & output )
```

迹点碰撞

参数

[IN]	param 无
[IN]	input 点，障碍物中心点，障碍物
[OUT]	output 是否碰撞

@cn_name: 迹点碰撞

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.41 pointDistance()

```
void LocalPlanning::pointDistance (
            const pointDistanceParam & param,
            const pointDistanceInput & input,
            pointDistanceOutput & output )
```

算两点之的距离

参数

[IN]	param 无
[IN]	input 两点的XY坐
[OUT]	output 距离

@cn_name: 算两点之的距离

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.42 pointOnCubicBezier()

```
void LocalPlanning::pointOnCubicBezier (
            const pointOnCubicBezierParam & param,
            const pointOnCubicBezierInput & input,
            pointOnCubicBezierOutput & output )
```

算塞曲上的点

参数

[IN]	param 无
[IN]	input 塞曲控制点，参数
[OUT]	output 判断果

@cn_name: 算塞曲上的点

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.43 pointToLineDistance()

```
void LocalPlanning::pointToLineDistance (
            const pointToLineDistanceParam & param,
            const pointToLineDistanceInput & input,
            pointToLineDistanceOutput & output )
```

算点到段的距离

参数

[IN]	param 无
[IN]	input 段起点，段点，点
[OUT]	output 点到段的距离

@cn_name: 算点到段的距离

@granularity: composite //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

5.6.2.44 processTrafficLight()

```
void LocalPlanning::processTrafficLight (
            const processTrafficLightParam & param,
            const processTrafficLightInput & input,
            processTrafficLightOutput & output )
```

根据灯信息更新决策数据

参数

[IN]	param 格障碍物，交通灯
[IN]	input 决策数据
[OUT]	output 更新后的决策数据，灯信息

@cn_name: 理交通信号灯

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.45 restart()

```
void LocalPlanning::restart (
            const restartParam & param,
            const restartInput & input,
            restartOutput & output )
```

停后，判断能否重新启

参数

[IN]	param 格障碍物
[IN]	input 速
[OUT]	output 判断果

@cn_name: 判断能否重新启

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.46 stop()

```
void LocalPlanning::stop (
            const stopParam & param,
            const stopInput & input,
            stopOutput & output )
```

停修改速度曲

参数

[IN]	param 无
[IN]	input 决策数据，当前速
[OUT]	output 更新的决策数据

@cn_name: 理停速度

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.47 stopPointJudge()

```
void LocalPlanning::stopPointJudge (
            const stopPointJudgeParam & param,
            const stopPointJudgeInput & input,
            stopPointJudgeOutput & output )
```

判断是否在停区域

参数

[IN]	param imu，停止点
[IN]	input 无
[OUT]	output 判断果

@cn_name: 判断停

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

5.6.2.48 trajectoryCollisionCheck()

```
void LocalPlanning::trajectoryCollisionCheck (
            const trajectoryCollisionCheckParam & param,
```

```
const trajectoryCollisionCheckInput & input,
trajectoryCollisionCheckOutput & output )
```

迹碰撞，true表示可行，false表示碰撞

参数

[IN]	param 障碍物
[IN]	input 迹
[OUT]	output 是否碰撞

@cn_name: 迹碰撞

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

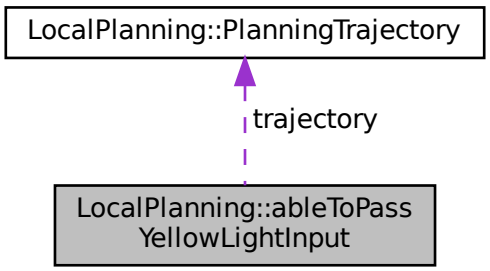
@tag: planning

Chapter 6

明

6.1 LocalPlanning::ableToPassYellowLightInput构体 参考

LocalPlanning::ableToPassYellowLightInput 的作:



Public 属性

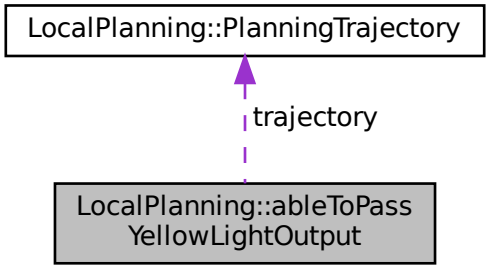
- PlanningTrajectory **trajectory**
- double **remaining_time**
- double **lane_length_before_intersection**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.2 LocalPlanning::ableToPassYellowLightOutput构体 参考

LocalPlanning::ableToPassYellowLightOutput 的作:



Public 属性

- PlanningTrajectory **trajectory**
- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.3 LocalPlanning::ableToPassYellowLightParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.4 LocalPlanning::angleRegulateInput构体 参考

Public 属性

- double **angleInput**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.5 LocalPlanning::angleRegulateOutput构体 参考

Public 属性

- double **angleOutput**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

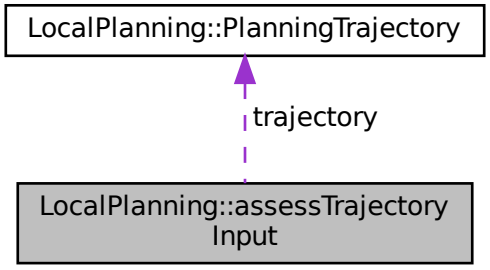
6.6 LocalPlanning::angleRegulateParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.7 LocalPlanning::assessTrajectoryInput构体 参考

LocalPlanning::assessTrajectoryInput 的作:



Public 属性

- PlanningTrajectory **trajectory**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.8 LocalPlanning::assessTrajectoryOutput构体 参考

Public 属性

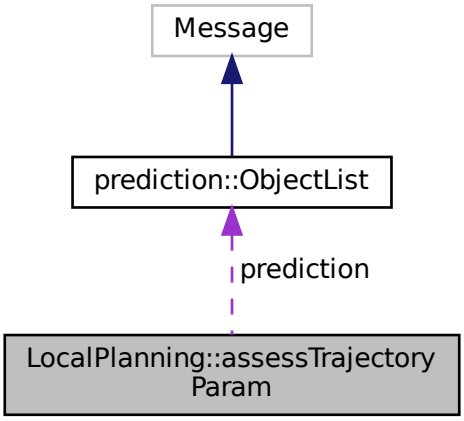
- double **minDistance**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.9 LocalPlanning::assessTrajectoryParam构体 参考

LocalPlanning::assessTrajectoryParam 的作:



Public 属性

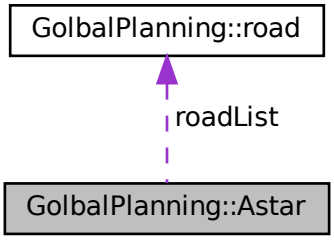
- prediction::ObjectList **prediction**
- pc::Imu **imu**
- std::vector< infopack::ObjectsProto > **objectsCmd**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.10 GolbalPlanning::Astar 参考

GolbalPlanning::Astar 的作:



Public 属性

- road * **roadList**
- list< int > **path**
- vector< pair< int, int > > **pathLanes**
- int **size** = 10000
- std::priority_queue< node, std::vector< node >, cmp > **openList**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/global↩ Planning_m _2.cpp

6.11 GolbalPlanning::AstarCalcFInput构体 参考

Public 属性

- int **tmp**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.12 GolbalPlanning::AstarCalcFOutput构体 参考

Public 属性

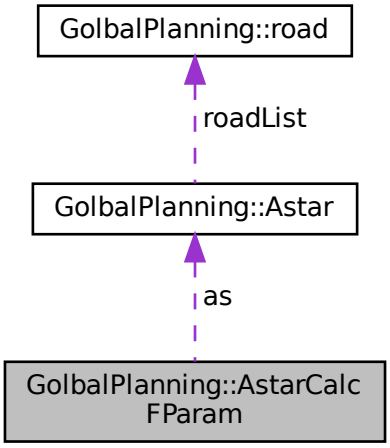
- double **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.13 GolbalPlanning::AstarCalcFParam构体 参考

GolbalPlanning::AstarCalcFParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.14 GolbalPlanning::AstarCalcGInput构体 参考

Public 属性

- int **tmp**
- int **from**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.15 GolbalPlanning::AstarCalcGOutput构体 参考

Public 属性

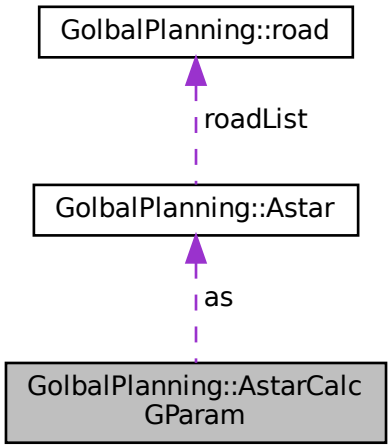
- double **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.16 GolbalPlanning::AstarCalcGParam构体 参考

GolbalPlanning::AstarCalcGParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.17 GolbalPlanning::AstarCalcHInput构体 参考

Public 属性

- int **tmp**
- int **destination**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.18 GolbalPlanning::AstarCalcHOutput构体 参考

Public 属性

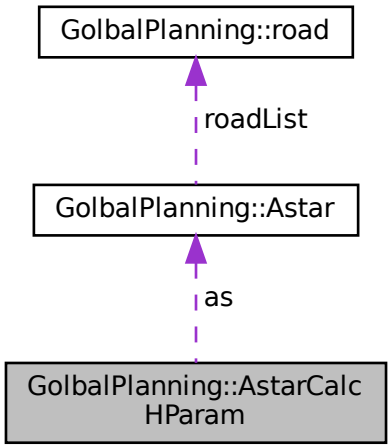
- double **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.19 GolbalPlanning::AstarCalcHParam构体 参考

GolbalPlanning::AstarCalcHParam 的作:



Public 属性

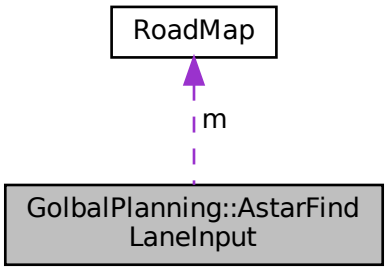
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.20 GolbalPlanning::AstarFindLaneInput构体 参考

GolbalPlanning::AstarFindLaneInput 的作:



Public 属性

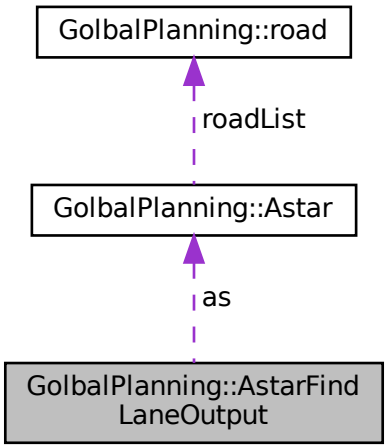
- RoadMap **m**
- list< int > **path**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.21 GolbalPlanning::AstarFindLaneOutput构体 参考

GolbalPlanning::AstarFindLaneOutput 的作:



Public 属性

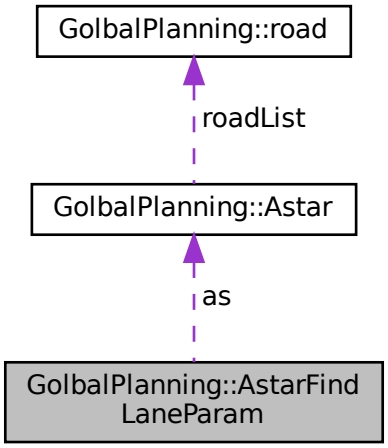
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.22 GolbalPlanning::AstarFindLaneParam构体 参考

GolbalPlanning::AstarFindLaneParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.23 GolbalPlanning::AstarFindLaneRInput构体 参考

Public 属性

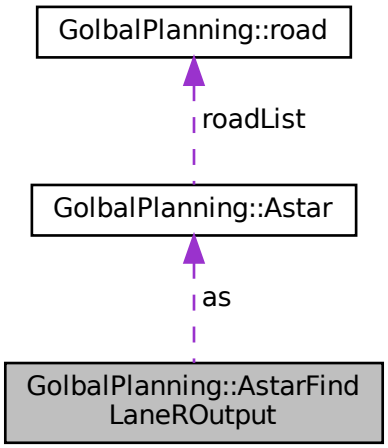
- vector< Road > **roads_**
- list< int > **path_**
- vector< std::pair< int, int > > **pathLanes_**
- int **lane_id_**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.24 GolbalPlanning::AstarFindLaneROutput构体 参考

GolbalPlanning::AstarFindLaneROutput 的作:



Public 属性

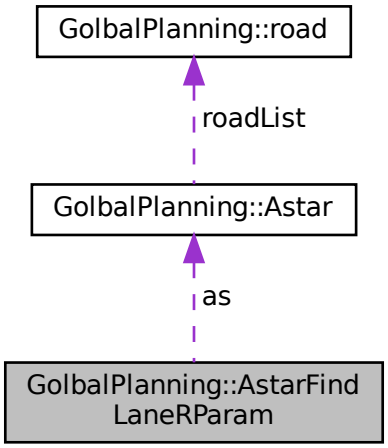
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.25 GolbalPlanning::AstarFindLaneRParam构体 参考

GolbalPlanning::AstarFindLaneRParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.26 GolbalPlanning::AstarFindPathInput构体 参考

Public 属性

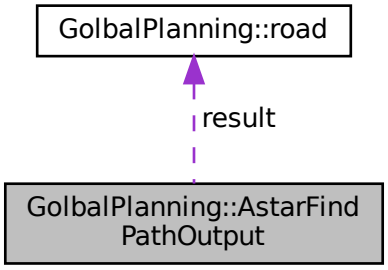
- int **origin**
- int **destination**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.27 GolbalPlanning::AstarFindPathOutput构体 参考

GolbalPlanning::AstarFindPathOutput 的作:



Public 属性

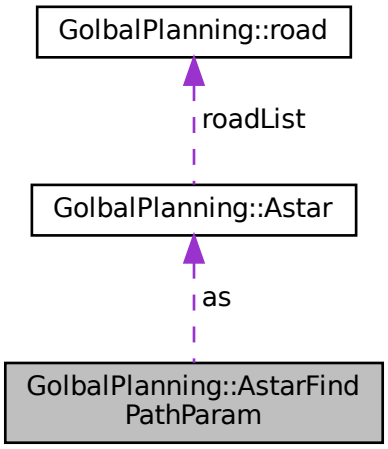
- road * **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.28 GolbalPlanning::AstarFindPathParam构体 参考

GolbalPlanning::AstarFindPathParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.29 GolbalPlanning::AstarGetLeastFInput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.30 GolbalPlanning::AstarGetLeastFOutput构体 参考

Public 属性

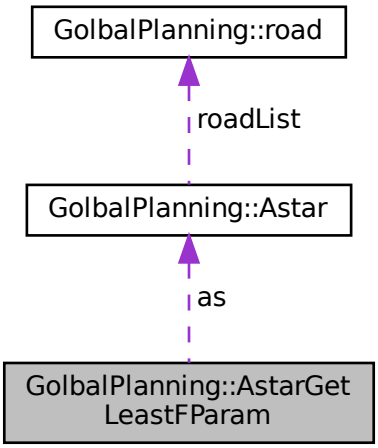
- int **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.31 GolbalPlanning::AstarGetLeastFParam构体 参考

GolbalPlanning::AstarGetLeastFParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.32 GolbalPlanning::AstarGetNextRoadInput构体 参考

Public 属性

- int **temp**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.33 GolbalPlanning::AstarGetNextRoadOutput构体 参考

Public 属性

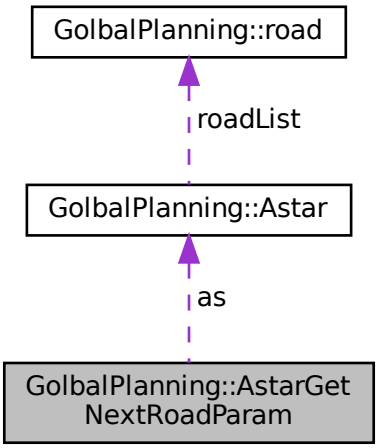
- std::vector< int > **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.34 GolbalPlanning::AstarGetNextRoadParam构体 参考

GolbalPlanning::AstarGetNextRoadParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.35 GolbalPlanning::AstarGetPathInput构体 参考

Public 属性

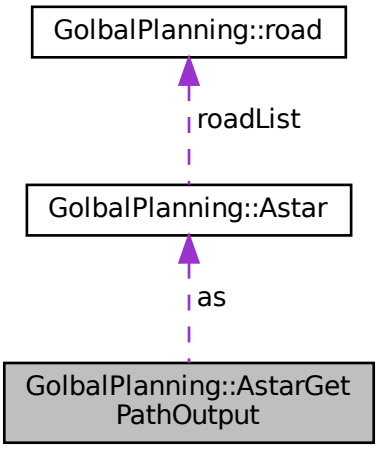
- int **origin**
- int **destination**
- int **originPointId**
- int **destinationPointId**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.36 GolbalPlanning::AstarGetPathOutput构体 参考

GolbalPlanning::AstarGetPathOutput 的作:



Public 属性

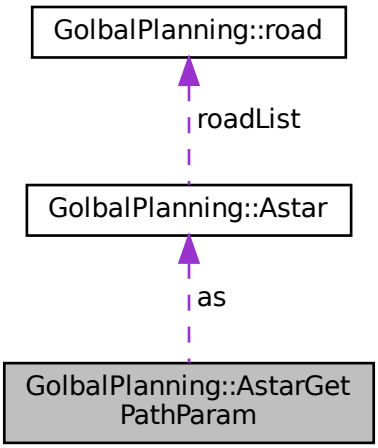
- std::list< int > **path**
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.37 GolbalPlanning::AstarGetPathParam构体 参考

GolbalPlanning::AstarGetPathParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.38 GolbalPlanning::AstarInitLinkInput构体 参考

Public 属性

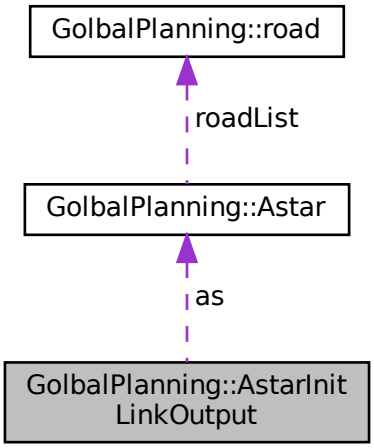
- int **from**
- int **to**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.39 GolbalPlanning::AstarInitLinkOutput构体 参考

GolbalPlanning::AstarInitLinkOutput 的作:



Public 属性

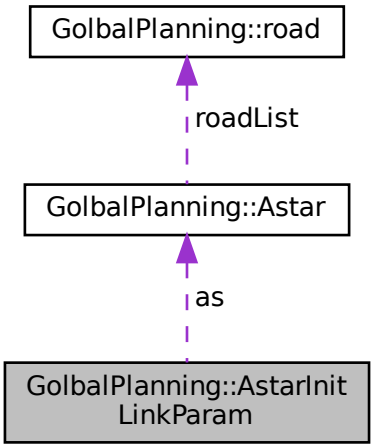
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.40 GolbalPlanning::AstarInitLinkParam构体 参考

GolbalPlanning::AstarInitLinkParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.41 GolbalPlanning::AstarInitRoadInput构体 参考

Public 属性

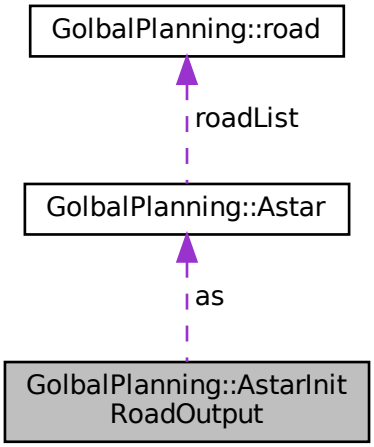
- int **number**
- double **xStart**
- double **yStart**
- double **xEnd**
- double **yEnd**
- double **length**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.42 GolbalPlanning::AstarInitRoadOutput构体 参考

GolbalPlanning::AstarInitRoadOutput 的作:



Public 属性

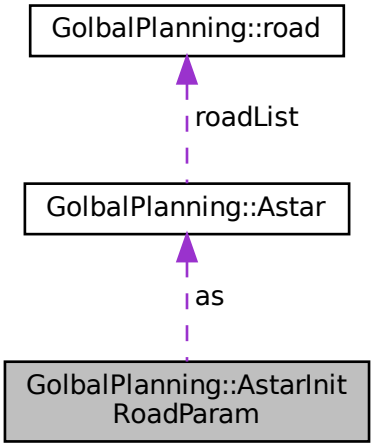
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.43 GolbalPlanning::AstarInitRoadParam构体 参考

GolbalPlanning::AstarInitRoadParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

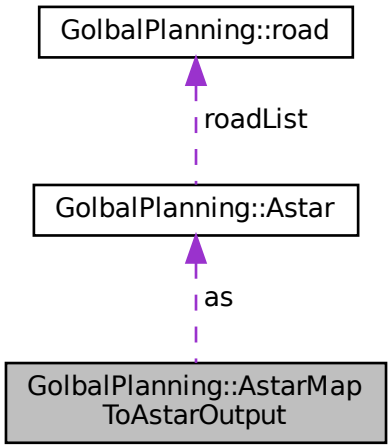
6.44 GolbalPlanning::AstarMapToAstarInput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.45 GolbalPlanning::AstarMapToAstarOutput构体 参考

GolbalPlanning::AstarMapToAstarOutput 的作:



Public 属性

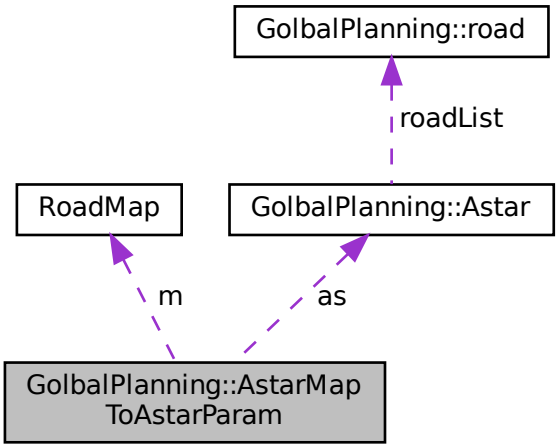
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.46 GolbalPlanning::AstarMapToAstarParam构体 参考

GolbalPlanning::AstarMapToAstarParam 的作:



Public 属性

- RoadMap **m**
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.47 GolbalPlanning::AstarModuleSelfCheckInput构体 参考

Public 属性

- std::list< int > **path**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

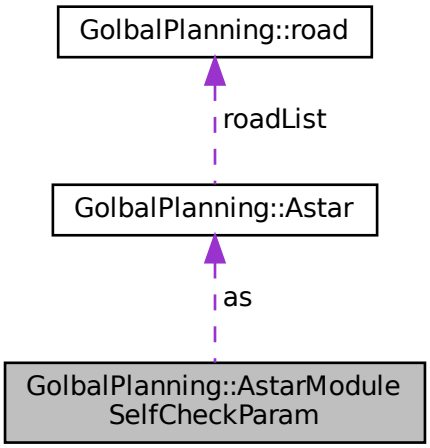
6.48 GolbalPlanning::AstarModuleSelfCheckOutput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.49 GolbalPlanning::AstarModuleSelfCheckParam构体 参考

GolbalPlanning::AstarModuleSelfCheckParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.50 GolbalPlanning::AstarModuleSelfCheckPrintInput构体 参考

Public 属性

- vector< pair< int, int > > **pathLanes**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

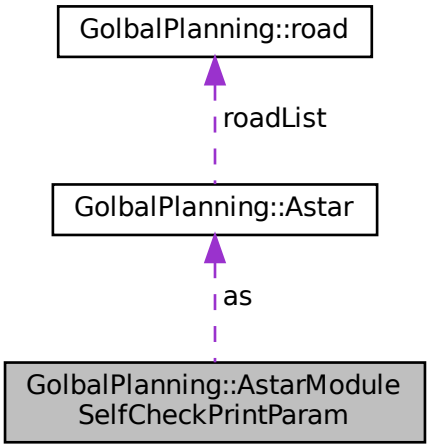
6.51 GolbalPlanning::AstarModuleSelfCheckPrintOutput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.52 GolbalPlanning::AstarModuleSelfCheckPrintParam构体 参考

GolbalPlanning::AstarModuleSelfCheckPrintParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.53 GolbalPlanning::AstarRecursiveFindLaneInput构体 参考

Public 属性

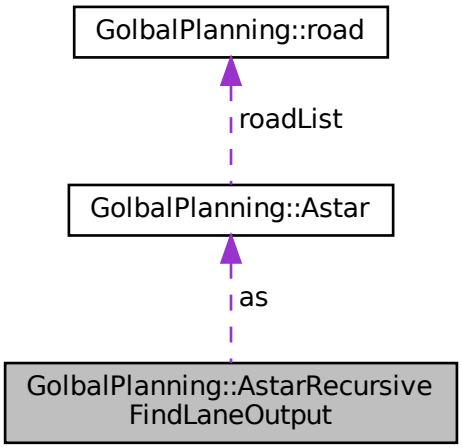
- std::vector< Road > **roads_**
- std::list< int > **path_**
- std::vector< std::tuple< int, int, int > > **optional_lanes_**
- int **road_id_**
- int **lane_id_**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.54 GolbalPlanning::AstarRecursiveFindLaneOutput构体 参考

GolbalPlanning::AstarRecursiveFindLaneOutput 的作:



Public 属性

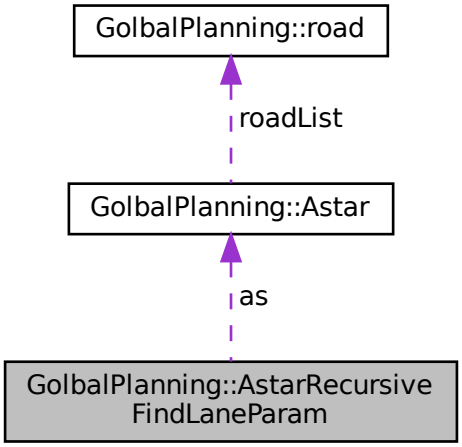
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.55 GolbalPlanning::AstarRecursiveFindLaneParam构体 参考

GolbalPlanning::AstarRecursiveFindLaneParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

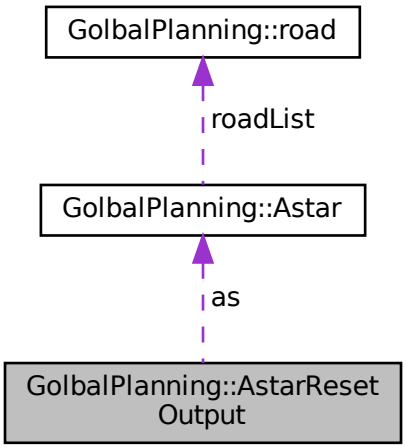
6.56 GolbalPlanning::AstarResetInput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.57 GolbalPlanning::AstarResetOutput构体 参考

GolbalPlanning::AstarResetOutput 的作:



Public 属性

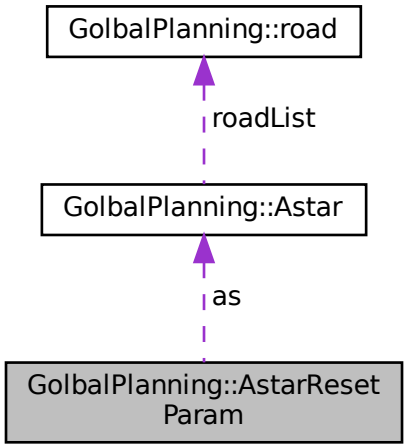
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.58 GolbalPlanning::AstarResetParam构体 参考

GolbalPlanning::AstarResetParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.59 ZtGeographyCoordinateTransform::BL2XYInput构体 参考

Public 属性

- double **lat**
- double **lon**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform↩ _m.h

6.60 ZtGeographyCoordinateTransform::BL2XYOutput构体 参考

Public 属性

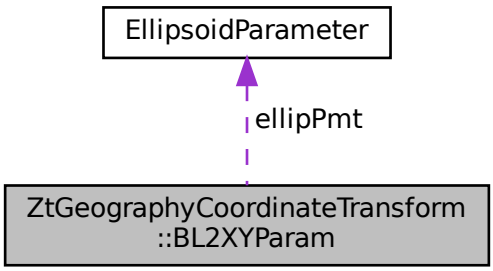
- double **x**
- double **y**
- bool **isValidCoord**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform↩_m.h

6.61 ZtGeographyCoordinateTransform::BL2XYParam构体 参考

ZtGeographyCoordinateTransform::BL2XYParam 的作:



Public 属性

- EllipsoidParameter **ellipPmt**
- double **meridianLine** = -360
- char **projType** = 'g'

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform↩_m.h

6.62 ZtGeographyCoordinateTransform::BLH2XYZInput构体 参考

Public 属性

- double **lat**
- double **lon**
- double **ht**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform↩_m.h

6.63 ZtGeographyCoordinateTransform::BLH2XYZOutput构体 参考

Public 属性

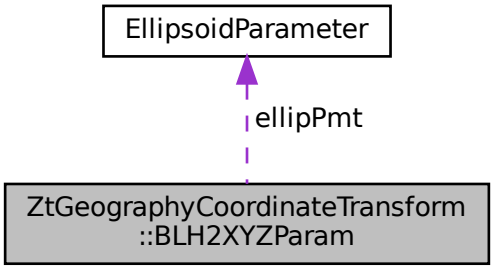
- double **x**
- double **y**
- double **z**
- bool **isValidCoord**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform↩_m.h

6.64 ZtGeographyCoordinateTransform::BLH2XYZParam构体 参考

ZtGeographyCoordinateTransform::BLH2XYZParam 的作:



Public 属性

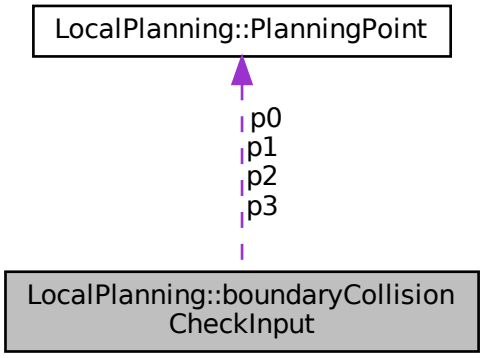
- EllipsoidParameter **ellipPmt**
- double **meridianLine** = -360
- char **projType** = 'g'

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform↩_m.h

6.65 LocalPlanning::boundaryCollisionCheckInput构体 参考

LocalPlanning::boundaryCollisionCheckInput 的作:



Public 属性

- PlanningPoint **p0**
- PlanningPoint **p1**
- PlanningPoint **p2**
- PlanningPoint **p3**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩Planning_m.hpp

6.66 LocalPlanning::boundaryCollisionCheckOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.67 LocalPlanning::boundaryCollisionCheckParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.68 Box构体 参考

Public 属性

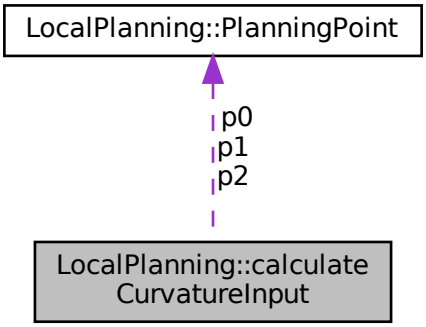
- float **x_min**
- float **y_min**
- float **z_min**
- float **x_max**
- float **y_max**
- float **z_max**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/classicLidarDetector.h

6.69 LocalPlanning::calculateCurvatureInput构体 参考

LocalPlanning::calculateCurvatureInput 的作:



Public 属性

- PlanningPoint **p0**
- PlanningPoint **p1**
- PlanningPoint **p2**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.70 LocalPlanning::calculateCurvatureOutput构体 参考

Public 属性

- double **curvature**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

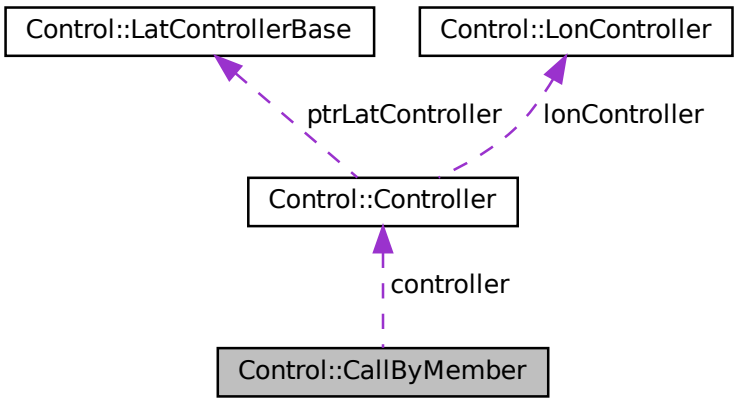
6.71 LocalPlanning::calculateCurvatureParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.72 Control::CallByMember 参考

Control::CallByMember 的作:



Public 成函数

- **CallByMember** (std::string configFile)
- void **run** (State s, const Traj &trajRaw)

Public 属性

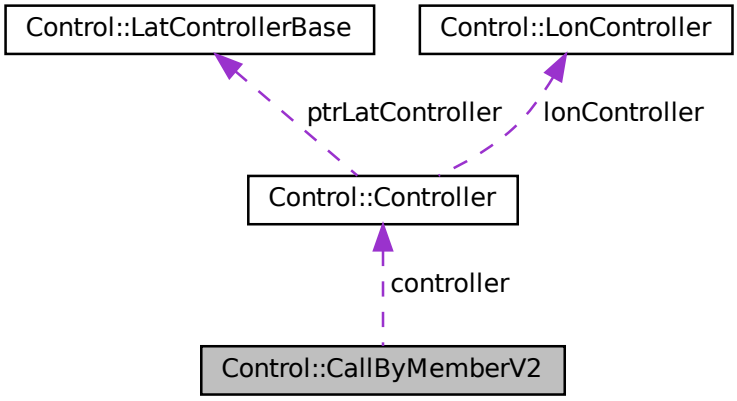
- Controller **controller**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/control↩ Example.cc

6.73 Control::CallByMemberV2 参考

Control::CallByMemberV2 的作:



Public 成函数

- **CallByMemberV2** (std::string configFile)
- void **run** (State s, const Traj &trajRaw)

Public 属性

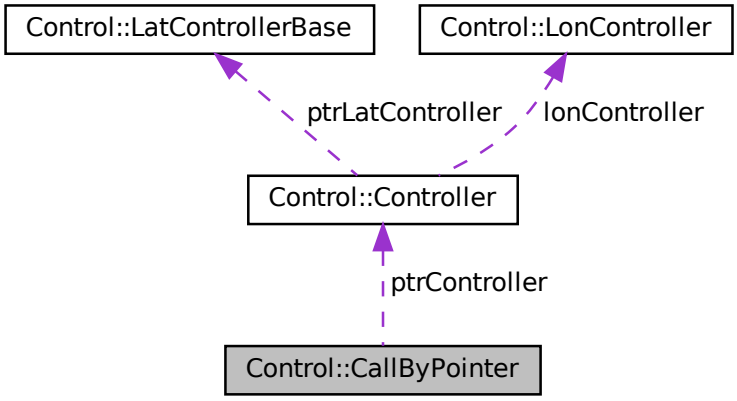
- Controller **controller**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/control↩ Example.cc

6.74 Control::CallByPointer 参考

Control::CallByPointer 的作:



Public 成函数

- **CallByPointer** (std::string configFile)
- void **run** (State s, const Traj &trajRaw)

Public 属性

- Controller * **ptrController**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/control↩ Example.cc

6.75 LaneDetector::CameraInfo构体 参考

Camera Calibration info.

```
#include <InversePerspectiveMapping.hpp>
```

Public 属性

- cv::Point2f focalLength

  focal length in x and y
- cv::Point2f opticalCenter

  optical center coordinates in image frame (origin is (0,0) at top left)
- float cameraHeight

  height of camera above ground
- float pitch

  pitch angle in radians (+ve downwards)
- float yaw

  yaw angle in radians (+ve clockwise)
- float imageWidth

  width of images
- float imageHeight

  height of images

6.75.1 描述

Camera Calibration info.

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_cv/include/Inverse↩ PerspectiveMapping.hpp

6.76 CameraInfoParserInfo构体 参考

Public 属性

- const char ∗ **help_help**
- const char ∗ **version_help**
- double **focalLengthX_arg**
- char ∗ **focalLengthX_orig**
- const char ∗ **focalLengthX_help**
- double **focalLengthY_arg**
- char ∗ **focalLengthY_orig**
- const char ∗ **focalLengthY_help**
- double **opticalCenterX_arg**
- char ∗ **opticalCenterX_orig**
- const char ∗ **opticalCenterX_help**
- double **opticalCenterY_arg**
- char ∗ **opticalCenterY_orig**
- const char ∗ **opticalCenterY_help**
- double **cameraHeight_arg**
- char ∗ **cameraHeight_orig**
- const char ∗ **cameraHeight_help**
- double **pitch_arg**
- char ∗ **pitch_orig**
- const char ∗ **pitch_help**

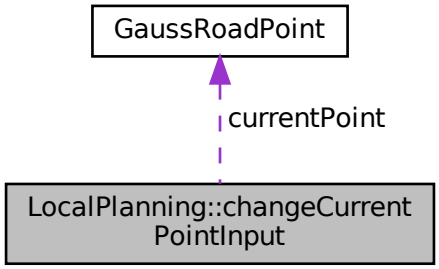
- double **yaw_arg**
- char * **yaw_orig**
- const char * **yaw_help**
- double **imageWidth_arg**
- char * **imageWidth_orig**
- const char * **imageWidth_help**
- double **imageHeight_arg**
- char * **imageHeight_orig**
- const char * **imageHeight_help**
- int **help_given**
- int **version_given**
- int **focalLengthX_given**
- int **focalLengthY_given**
- int **opticalCenterX_given**
- int **opticalCenterY_given**
- int **cameraHeight_given**
- int **pitch_given**
- int **yaw_given**
- int **imageWidth_given**
- int **imageHeight_given**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_cv/include/CameraInfoOpt.h

6.77 LocalPlanning::changeCurrentPointInput构体 参考

LocalPlanning::changeCurrentPointInput 的作:



Public 属性

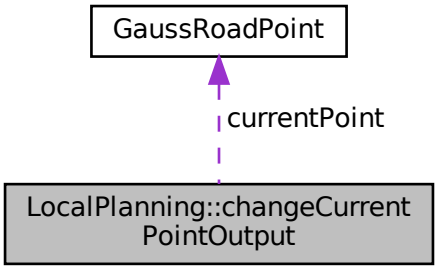
- int **laneId**
- int **pointIndex**
- GaussRoadPoint **currentPoint**
- bool **isLeft**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.78 LocalPlanning::changeCurrentPointOutput构体 参考

LocalPlanning::changeCurrentPointOutput 的作:



Public 属性

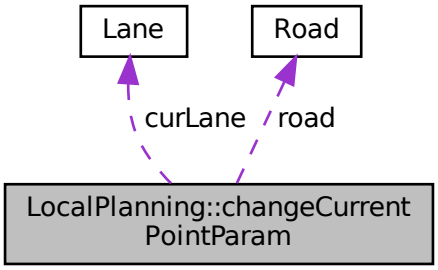
- int **laneId**
- int **pointIndex**
- GaussRoadPoint **currentPoint**
- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.79 LocalPlanning::changeCurrentPointParam构体 参考

LocalPlanning::changeCurrentPointParam 的作:



Public 属性

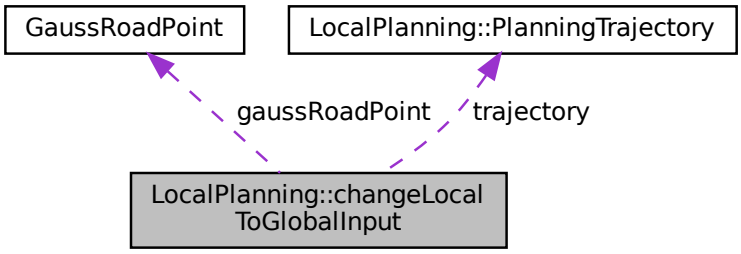
- Road **road**
- Lane **curLane**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.80 LocalPlanning::changeLocalToGlobalInput构体 参考

LocalPlanning::changeLocalToGlobalInput 的作:



Public 属性

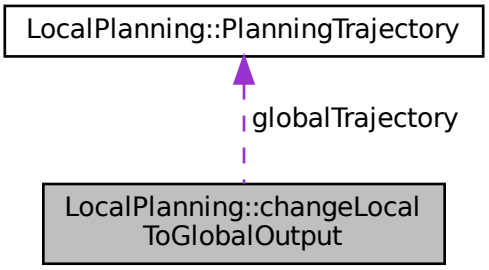
- PlanningTrajectory **trajectory**
- GaussRoadPoint **gaussRoadPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.81 LocalPlanning::changeLocalToGlobalOutput构体 参考

LocalPlanning::changeLocalToGlobalOutput 的作:



Public 属性

- PlanningTrajectory **globalTrajectory**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

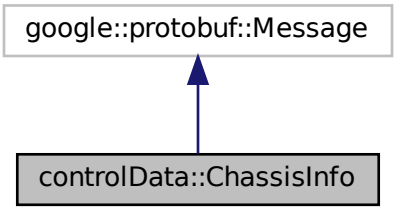
6.82 LocalPlanning::changeLocalToGlobalParam构体 参考

构体的文档由以下文件生成:

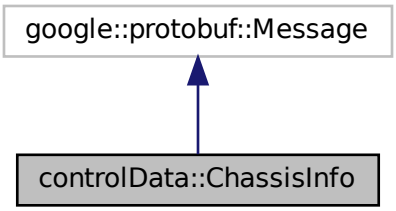
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.83 controlData::ChassisInfo 参考

controlData::ChassisInfo 承系:



controlData::ChassisInfo 的作:



Public 成函数

- **ChassisInfo** (const ChassisInfo &from)
- ChassisInfo & **operator=** (const ChassisInfo &from)
- void **Swap** (ChassisInfo ∗other)
- ChassisInfo ∗ **New** () const final
- ChassisInfo ∗ **New** (::google::protobuf::Arena ∗arena) const final
- void **CopyFrom** (const ::google::protobuf::Message &from) final
- void **MergeFrom** (const ::google::protobuf::Message &from) final
- void **CopyFrom** (const ChassisInfo &from)
- void **MergeFrom** (const ChassisInfo &from)
- void **Clear** () final
- bool **IsInitialized** () const final
- size_t **ByteSizeLong** () const final
- bool **MergePartialFromCodedStream** (::google::protobuf::io::CodedInputStream ∗input) final
- void **SerializeWithCachedSizes** (::google::protobuf::io::CodedOutputStream ∗output) const final
- ::google::protobuf::uint8 ∗ **InternalSerializeWithCachedSizesToArray** (bool deterministic, ::google↩ ::protobuf::uint8 ∗target) const final

- int **GetCachedSize** () const final
- ::google::protobuf::Metadata **GetMetadata** () const final
- void **clear_speed** ()
- float **speed** () const
- void **set_speed** (float value)
- void **clear_steeringangle** ()
- float **steeringangle** () const
- void **set_steeringangle** (float value)
- void **clear_mode** ()
- ::google::protobuf::int32 **mode** () const
- void **set_mode** (::google::protobuf::int32 value)

静 Public 成函数

- static const ::google::protobuf::Descriptor * **descriptor** ()
- static const ChassisInfo & **default_instance** ()
- static void **InitAsDefaultInstance** ()
- static const ChassisInfo * **internal_default_instance** ()

静 Public 属性

- static constexpr int **kIndexInFileMessages**
- static const int **kSpeedFieldNumber** = 1
- static const int **kSteeringAngleFieldNumber** = 2
- static const int **kModeFieldNumber** = 3

友元

- struct **::protobuf_control_2eproto::TableStruct**
- void **swap** (ChassisInfo &a, ChassisInfo &b)

6.83.1 成量明

6.83.1.1 kIndexInFileMessages

```
constexpr int controlData::ChassisInfo::kIndexInFileMessages [static], [constexpr]
```

初始:
=
1

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/control.pb.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/control.pb.cc

6.84 controlData::ChassisInfoDefaultTypeInternal 参考

Public 属性

- ::google::protobuf::internal::ExplicitlyConstructed< ChassisInfo > **_instance**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/control.pb.cc

6.85 LocalPlanning::clearPathListInput构体 参考

Public 属性

- std::vector< PlanningTrajectory > **pathList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.86 LocalPlanning::clearPathListOutput构体 参考

Public 属性

- std::vector< PlanningTrajectory > **pathList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.87 LocalPlanning::clearPathListParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.88 lidarTools::ClusterDistance构体 参考

Public 成函数

- bool **operator**> (const ClusterDistance &other) const

Public 属性

- int **cluster1**
- int **cluster2**
- float **distance**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/lidarTools.h

6.89 ClusterResult构体 参考

Public 属性

- std::vector< int > **labels**
- pcl::PointCloud< pcl::PointXYZI >::Ptr **centroids**
- float **total_error**
- int **num_clusters**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/classicLidarDetector.h

6.90 lidarTools::ClusterResult构体 参考

Public 属性

- std::vector< int > **labels**
- pcl::PointCloud< pcl::PointXYZI >::Ptr **centroids** = pcl::PointCloud<pcl::PointXYZI>::Ptr(new pcl::Point↩ Cloud<pcl::PointXYZI>)
- float **total_error**
- int **num_clusters**
- std::vector< pcl::PointIndices > **cluster_indices**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/lidarTools.h

6.91 GolbalPlanning::cmp 参考

Public 成函数

- bool **operator()** (node a, node b) const

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.92 Color构体 参考

Public 成函数

- **Color** (float setR, float setG, float setB)

Public 属性

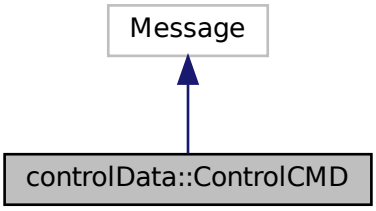
- float **r**
- float **g**
- float **b**

构体的文档由以下文件生成:

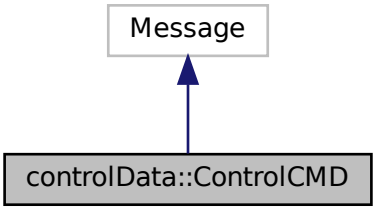
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/classicLidarDetector.h

6.93 controlData::ControlCMD 参考

controlData::ControlCMD 承系:



controlData::ControlCMD 的作:



Public 成函数

- **ControlCMD** (const ControlCMD &from)
- ControlCMD & **operator=** (const ControlCMD &from)
- void **Swap** (ControlCMD ∗other)
- ControlCMD ∗ **New** () const final
- ControlCMD ∗ **New** (::google::protobuf::Arena ∗arena) const final
- void **CopyFrom** (const ::google::protobuf::Message &from) final
- void **MergeFrom** (const ::google::protobuf::Message &from) final
- void **CopyFrom** (const ControlCMD &from)
- void **MergeFrom** (const ControlCMD &from)
- void **Clear** () final
- bool **IsInitialized** () const final
- size_t **ByteSizeLong** () const final
- bool **MergePartialFromCodedStream** (::google::protobuf::io::CodedInputStream ∗input) final
- void **SerializeWithCachedSizes** (::google::protobuf::io::CodedOutputStream ∗output) const final
- ::google::protobuf::uint8 ∗ **InternalSerializeWithCachedSizesToArray** (bool deterministic, ::google↩ ::protobuf::uint8 ∗target) const final
- int **GetCachedSize** () const final
- ::google::protobuf::Metadata **GetMetadata** () const final
- void **clear_targetsteeringangle** ()
- float **targetsteeringangle** () const
- void **set_targetsteeringangle** (float value)
- void **clear_targetspeed** ()
- float **targetspeed** () const
- void **set_targetspeed** (float value)

静 Public 成函数

- static const ::google::protobuf::Descriptor ∗ **descriptor** ()
- static const ControlCMD & **default_instance** ()
- static void **InitAsDefaultInstance** ()
- static const ControlCMD ∗ **internal_default_instance** ()

静 Public 属性

- static constexpr int **kIndexInFileMessages**
- static const int **kTargetSteeringAngleFieldNumber** = 1
- static const int **kTargetSpeedFieldNumber** = 2

友元

- struct **::protobuf_control_2eproto::TableStruct**
- void **swap** (ControlCMD &a, ControlCMD &b)

6.93.1 成量明

6.93.1.1 kIndexInFileMessages

```
constexpr int controlData::ControlCMD::kIndexInFileMessages  [static], [constexpr]
```

初始:
=
0

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/control.pb.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/control.pb.cc

6.94 Control::ControlCMD构体 参考

Public 属性

- double **steer**
- double **speed**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlData.hp

6.95 controlData::ControlCMDDefaultTypeInternal 参考

Public 属性

- ::google::protobuf::internal::ExplicitlyConstructed< ControlCMD > **_instance**

的文档由以下文件生成:

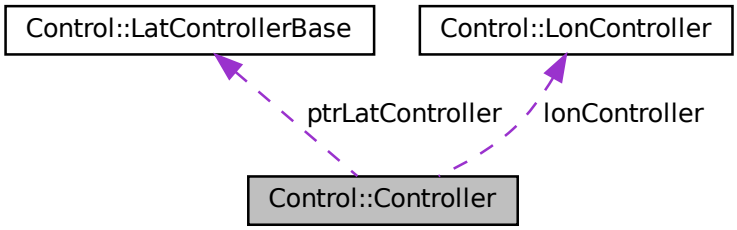
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/control.pb.cc

6.96 Control::Controller 参考

横向控制器基 控制模主要模。内部分横向控制和向控制

```
#include <controlController.hpp>
```

Control::Controller 的作:



Public 成函数

- **Controller** (std::string configFile)
- Controller & **operator=** (const Controller &controller)
- ControlCMD calcControlCMD (State s, const Traj &trajRaw)
  接口：生成控制指令
- bool **moduleSelfCheck** (double steer, double speed)
- void **moduleSelfCheckPrint** (double steer, double speed)

Public 属性

- LatControllerBase * **ptrLatController**
- LonController **lonController**

6.96.1 描述

横向控制器基 控制模主要模。内部分横向控制和向控制

6.96.2 成函数明

6.96.2.1 calcControlCMD()

```
ControlCMD Control::Controller::calcControlCMD (
            State s,
            const Traj & trajRaw )
```

接口：生成控制指令

参数

s	当前状
Traj	决策出【】迹

返回

ControlCMD 控制指令

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlControll
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/controlController.c

6.97 ControllerPID 参考

Public 成函数

- void **update** ()

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/controllerPID.h

6.98 LocalPlanning::coordTran2DInput构体 参考

Public 属性

- double **dX**
- double **dY**
- double **dPhi**
- double **dX0**
- double **dY0**
- double **dPhi0**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.99 LocalPlanning::coordTran2DOutput构体 参考

Public 属性

- double **dX**
- double **dY**
- double **dPhi**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.100 LocalPlanning::coordTran2DParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.101 LocalPlanning::courseAngleReviseInput构体 参考

Public 属性

- double **ang**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.102 LocalPlanning::courseAngleReviseOutput构体 参考

Public 属性

- double **ang**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

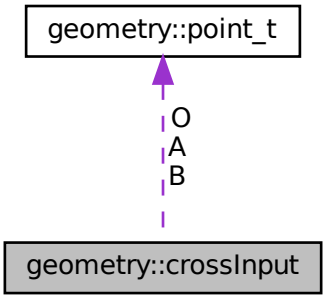
6.103 LocalPlanning::courseAngleReviseParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.104 geometry::crossInput构体 参考

geometry::crossInput 的作:



Public 属性

- point_t **O**
- point_t **A**
- point_t **B**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩ _m.h

6.105 geometry::crossOutput构体 参考

Public 属性

- double **value**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩ _m.h

6.106 geometry::crossParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.107 LocalPlanning::cruiseControllerInput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩Planning_m.hpp

6.108 LocalPlanning::cruiseControllerOutput构体 参考

Public 属性

- double **velocity**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩Planning_m.hpp

6.109 LocalPlanning::cruiseControllerParam构体 参考

Public 属性

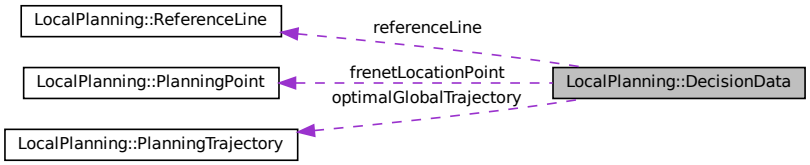
- pc::Imu **imu**
- std::vector< infopack::ObjectsProto > **objectsCmd**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩Planning_m.hpp

6.110 LocalPlanning::DecisionData构体 参考

LocalPlanning::DecisionData 的作:



Public 属性

- std::vector< PlanningTrajectory > **finalPathList**
- ReferenceLine **referenceLine**
- PlanningPoint **frenetLocationPoint**
- std::vector< std::vector< double > > **speedList**
- std::vector< PlanningTrajectory > **controlTrajectoryList**
- std::vector< int32_t > **feasibleTrajectoryIndexList**
- PlanningTrajectory **optimalGlobalTrajectory**
- int32_t **optimalTrajectoryIndex**
- int32_t **currentId** = 0
- int32_t **currentLaneId** = 0
- int32_t **currentIndex** = 0
- std::tuple< int32_t, int32_t > **nextId** = std::tuple<int32_t, int32_t>(0, 0)

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.111 DGPS构体 参考

Public 属性

- double **longitude**
- double **latitude**
- double **gaussX**
- double **gaussY**
- int **psValid**
- double **time**
- double **velocity**
- double **yaw**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/stdThreadDemo.cpp

6.112 EllipsoidParameter构体 参考

Public 成函数

- **EllipsoidParameter** (double ia, double ib)
- **EllipsoidParameter** (double ia, double ib)

Public 属性

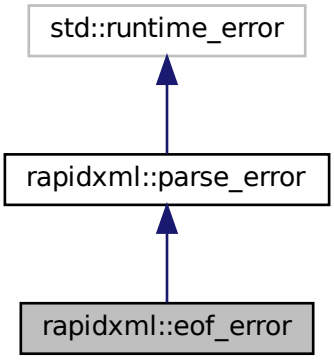
- double **a**
- double **b**
- double **f**
- double **e2**
- double **ep2**
- double **c**
- double **a0**
- double **a2**
- double **a4**
- double **a6**

构体的文档由以下文件生成:

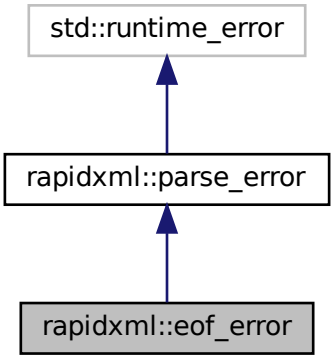
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform.↩ h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform↩ _m.h

6.113 rapidxml::eof_error 参考

rapidxml::eof_error 承系:



rapidxml::eof_error 的作:



Public 成函数

- **eof_error** (const char ∗what, void ∗where)

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml.hpp

6.114 Control::FG_EVAL 参考

Public 成函数

- **FG_EVAL** (Matf traj_ref, double T, double wheelbase, double dt)
- typedef **CPPAD_TESTVECTOR** (AD< double >) ADvector
- void **operator()** (ADvector &fg, const ADvector &vars)

Public 属性

- Matf **traj_ref**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlMPCCo
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/controlMPCContro

6.115 rapidxml::file< Ch > 模板 参考

Represents data loaded from a file.

```
#include <rapidxml_utils.hpp>
```

Public 成函数

- file (const char *filename)
- file (std::basic_istream< Ch > &stream)
- Ch * data ()
- const Ch * data () const
- std::size_t size () const

6.115.1 描述

template<class Ch = char>
class rapidxml::file< Ch >

Represents data loaded from a file.

6.115.2 构造及析构函数明

6.115.2.1 file() [1/2]

```
template<class Ch = char>
rapidxml::file< Ch >::file (
            const char * filename )  [inline]
```

Loads file into the memory. Data will be automatically destroyed by the destructor.

参数

filename	Filename to load.

6.115.2.2 file() [2/2]

```
template<class Ch = char>
rapidxml::file< Ch >::file (
            std::basic_istream< Ch > & stream )  [inline]
```

Loads file into the memory. Data will be automatically destroyed by the destructor

参数

stream	Stream to load from

6.115.3 成函数明

6.115.3.1 data() [1/2]

```
template<class Ch = char>
Ch* rapidxml::file< Ch >::data ( ) [inline]
```

Gets file data.

返回

Pointer to data of file.

6.115.3.2 data() [2/2]

```
template<class Ch = char>
const Ch* rapidxml::file< Ch >::data ( ) const [inline]
```

Gets file data.

返回

Pointer to data of file.

6.115.3.3 size()

```
template<class Ch = char>
std::size_t rapidxml::file< Ch >::size ( ) const [inline]
```

Gets file data size.

返回

Size of file data, in characters.

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml_utils.hpp

6.116 LocalPlanning::findMaxInput构体 参考

Public 属性

- std::vector< double > **array**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.117 LocalPlanning::findMaxOutput构体 参考

Public 属性

- double **value**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.118 LocalPlanning::findMaxParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.119 LocalPlanning::findMinInput构体 参考

Public 属性

- std::vector< double > **array**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.120 LocalPlanning::findMinOutput构体 参考

Public 属性

- double **value**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.121 LocalPlanning::findMinParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.122 LocalPlanning::fixRoadLaneIndexInput构体 参考

LocalPlanning::fixRoadLaneIndexInput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.123 LocalPlanning::fixRoadLaneIndexOutput构体 参考

Public 属性

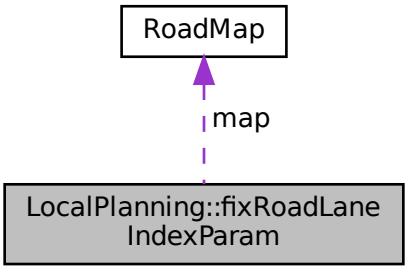
- std::tuple< int32_t, int32_t > **indexTuple**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.124 LocalPlanning::fixRoadLaneIndexParam构体 参考

LocalPlanning::fixRoadLaneIndexParam 的作:



Public 属性

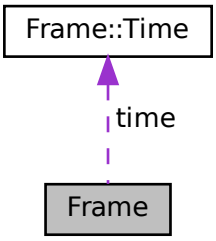
- RoadMap **map**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.125 Frame构体 参考

Frame 的作:



- struct Time

Public 属性

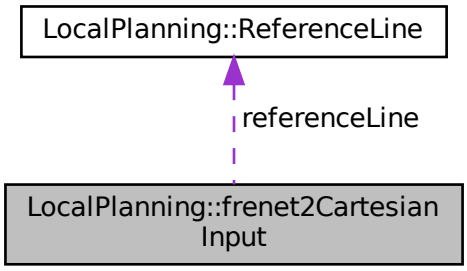
- time_t **time_stamp**
- int **frameId**
- int **Id**
- double **imu_x**
- double **imu_y**
- double **imu_yaw**
- struct Frame::Time **time**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/baseLidarObjectDetector.h

6.126 LocalPlanning::frenet2CartesianInput构体 参考

LocalPlanning::frenet2CartesianInput 的作:



Public 属性

- double **s**
- double **l**
- double **x**
- double **y**
- ReferenceLine **referenceLine**
- int **lastIndex**
- int **prevIndex**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/localPlanning_m.hpp

6.127 LocalPlanning::frenet2CartesianOutput构体 参考

Public 属性

- double **x**
- double **y**
- int **lastIndex**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.128 LocalPlanning::frenet2CartesianParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.129 LocalPlanning::gaussConvertInput构体 参考

Public 属性

- double **longitude1**
- double **latitude1**
- double **dNorth_X**
- double **dEast_Y**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.130 LocalPlanning::gaussConvertOutput构体 参考

Public 属性

- double **dNorth_X**
- double **dEast_Y**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.131 LocalPlanning::gaussConvertParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.132 GaussRoadPoint构体 参考

Public 属性

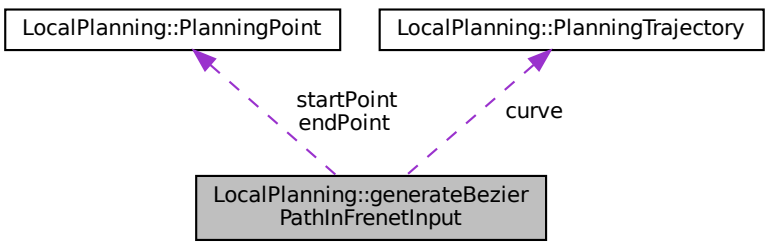
- double **GaussX**
- double **GaussY**
- double **yaw**
- double **curvature**
- double **s**
- double **speedMax**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.133 LocalPlanning::generateBezierPathInFrenetInput构体 参考

LocalPlanning::generateBezierPathInFrenetInput 的作:



Public 属性

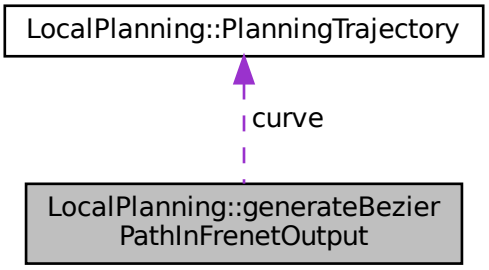
- PlanningPoint **startPoint**
- PlanningPoint **endPoint**
- PlanningTrajectory **curve**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.134 LocalPlanning::generateBezierPathInFrenetOutput构体 参考

LocalPlanning::generateBezierPathInFrenetOutput 的作:



Public 属性

- PlanningTrajectory **curve**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

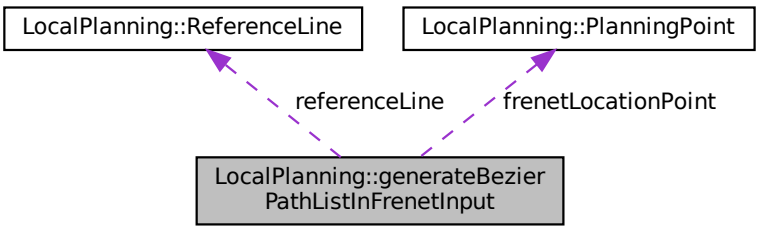
6.135 LocalPlanning::generateBezierPathInFrenetParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.136 LocalPlanning::generateBezierPathListInFrenetInput构体 参考

LocalPlanning::generateBezierPathListInFrenetInput 的作:



Public 属性

- ReferenceLine **referenceLine**
- PlanningPoint **frenetLocationPoint**
- std::vector< PlanningTrajectory > **pathList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.137 LocalPlanning::generateBezierPathListInFrenetOutput构体 参考

Public 属性

- std::vector< PlanningTrajectory > **pathList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

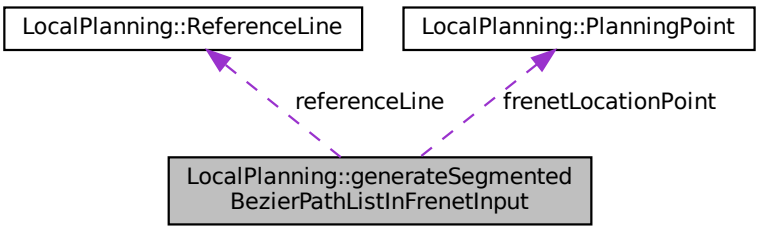
6.138 LocalPlanning::generateBezierPathListInFrenetParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.139 LocalPlanning::generateSegmentedBezierPathListInFrenetInput构体 参考

LocalPlanning::generateSegmentedBezierPathListInFrenetInput 的作:



Public 属性

- ReferenceLine **referenceLine**
- PlanningPoint **frenetLocationPoint**
- std::vector< PlanningTrajectory > **pathList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.140 LocalPlanning::generateSegmentedBezierPathListInFrenetOutput构体 参考

Public 属性

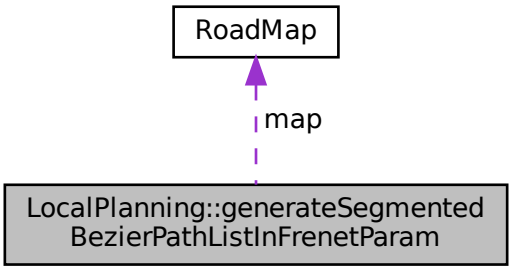
- std::vector< PlanningTrajectory > **pathList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.141 LocalPlanning::generateSegmentedBezierPathListInFrenetParam 构体 参考

LocalPlanning::generateSegmentedBezierPathListInFrenetParam 的作:



Public 属性

- RoadMap **map**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.142 LocalPlanning::generateSpeedListInput构体 参考

LocalPlanning::generateSpeedListInput 的作:

Public 属性

- double **velocity**
- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.143 LocalPlanning::generateSpeedListOutput构体 参考

LocalPlanning::generateSpeedListOutput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.144 LocalPlanning::generateSpeedListParam构体 参考

Public 属性

- std::vector< infopack::ObjectsProto > **objectsCmd**
- pc::Imu **imu**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.145 LocalPlanning::generateTrajectoryListInput构体 参考

LocalPlanning::generateTrajectoryListInput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.146 LocalPlanning::generateTrajectoryListOutput构体 参考

LocalPlanning::generateTrajectoryListOutput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.147 LocalPlanning::generateTrajectoryListParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.148 LocalPlanning::getAngleDiffInput构体 参考

Public 属性

- double **ang0**
- double **ang1**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.149 LocalPlanning::getAngleDiffOutput构体 参考

Public 属性

- double **ret**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.150 LocalPlanning::getAngleDiffParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.151 LocalPlanning::getAngleInput构体 参考

Public 属性

- double **x0**
- double **y0**
- double **x1**
- double **y1**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.152 LocalPlanning::getAngleOutput构体 参考

Public 属性

- double **ang**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.153 LocalPlanning::getAngleParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.154 LocalPlanning::getCurrentPositionInput构体 参考

LocalPlanning::getCurrentPositionInput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.155 LocalPlanning::getCurrentPositionOutput构体 参考

LocalPlanning::getCurrentPositionOutput 的作:

Public 属性

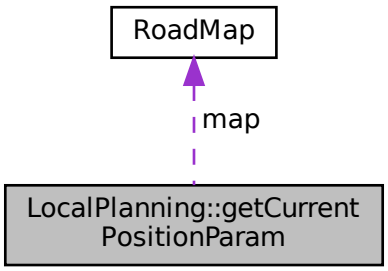
- DecisionData **decisionData**
- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.156 LocalPlanning::getCurrentPositionParam构体 参考

LocalPlanning::getCurrentPositionParam 的作:



Public 属性

- pc::Imu **imu**
- RoadMap **map**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.157 LocalPlanning::getDistanceInput构体 参考

Public 属性

- double **x1**
- double **y1**
- double **x2**
- double **y2**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.158 LocalPlanning::getDistanceOutput构体 参考

Public 属性

- double **dis**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.159 LocalPlanning::getDistanceParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.160 GetEndPointByRoadIDInput构体 参考

Public 属性

- int **roadID**
- std::vector< Road > **roads**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.161 GetEndPointByRoadIDOutput构体 参考

Public 属性

- double **GaussX**
- double **GaussY**
- bool **isValidStartPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

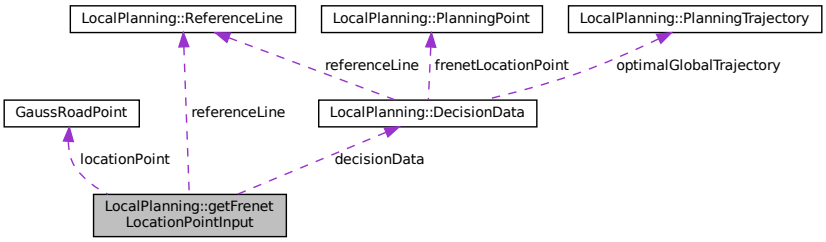
6.162 GetEndPointByRoadIDParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.163 LocalPlanning::getFrenetLocationPointInput构体 参考

LocalPlanning::getFrenetLocationPointInput 的作:



Public 属性

- DecisionData **decisionData**
- ReferenceLine **referenceLine**
- GaussRoadPoint **locationPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.164 LocalPlanning::getFrenetLocationPointOutput构体 参考

LocalPlanning::getFrenetLocationPointOutput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

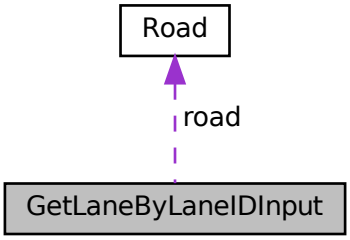
6.165 LocalPlanning::getFrenetLocationPointParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.166 GetLaneByLaneIDInput构体 参考

GetLaneByLaneIDInput 的作:



Public 属性

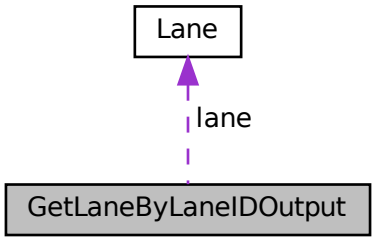
- int **laneID**
- Road **road**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.167 GetLaneByLaneIDOutput构体 参考

GetLaneByLaneIDOutput 的作:



Public 属性

- Lane & **lane**
- bool **isValidLine**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

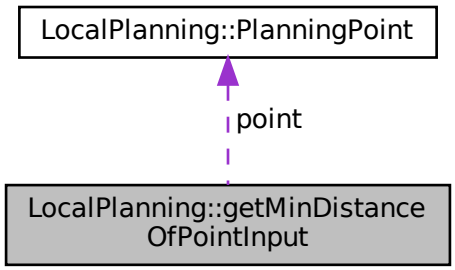
6.168 GetLaneByLaneIDParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.169 LocalPlanning::getMinDistanceOfPointInput构体 参考

LocalPlanning::getMinDistanceOfPointInput 的作:



Public 属性

- PlanningPoint **point**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.170 LocalPlanning::getMinDistanceOfPointOutput构体 参考

Public 属性

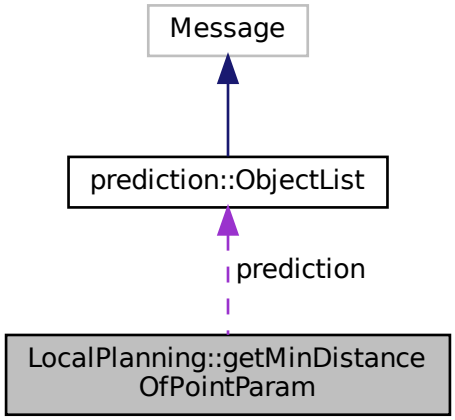
- double **minDistance**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.171 LocalPlanning::getMinDistanceOfPointParam构体 参考

LocalPlanning::getMinDistanceOfPointParam 的作:



Public 属性

- prediction::ObjectList **prediction**
- pc::Imu **imu**
- std::vector< infopack::ObjectsProto > **objectsCmd**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.172 LocalPlanning::getNextRoadLaneIdInput构体 参考

Public 属性

- std::tuple< int32_t, int32_t > **currentId**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.173 LocalPlanning::getNextRoadLaneIdOutput构体 参考

Public 属性

- std::tuple< int32_t, int32_t > **nextRoadLaneId**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.174 LocalPlanning::getNextRoadLaneIdParam构体 参考

Public 属性

- std::vector< std::tuple< int32_t, int32_t > > **routingList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.175 LocalPlanning::getOptimalTrajectoryIndexInput构体 参考

LocalPlanning::getOptimalTrajectoryIndexInput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.176 LocalPlanning::getOptimalTrajectoryIndexOutput构体 参考

LocalPlanning::getOptimalTrajectoryIndexOutput 的作:

Public 属性

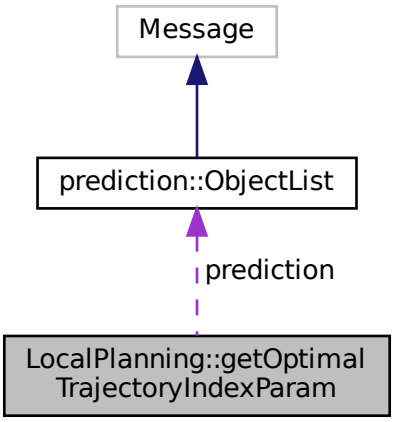
- DecisionData **decisionData**
- int32_t **index**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.177 LocalPlanning::getOptimalTrajectoryIndexParam构体 参考

LocalPlanning::getOptimalTrajectoryIndexParam 的作:



Public 属性

- prediction::ObjectList **prediction**
- infopack::TrafficLight **trafficLight**
- pc::Imu **imu**
- std::vector< infopack::ObjectsProto > **objectsCmd**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.178 LocalPlanning::getPlanningDistanceInput构体 参考

Public 属性

- double **velocity**
- double **curvature**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.179 LocalPlanning::getPlanningDistanceOutput构体 参考

Public 属性

- double **distance**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

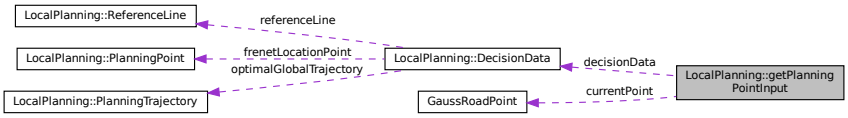
6.180 LocalPlanning::getPlanningDistanceParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.181 LocalPlanning::getPlanningPointInput构体 参考

LocalPlanning::getPlanningPointInput 的作:



Public 属性

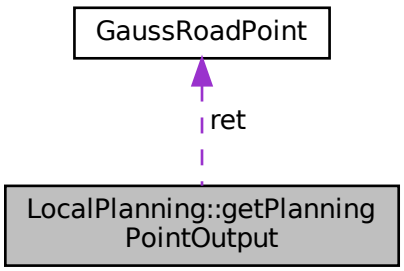
- GaussRoadPoint **currentPoint**
- DecisionData **decisionData**
- double **distance**
- double **yaw**
- std::tuple< int32_t, int32_t > **nextId**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.182 LocalPlanning::getPlanningPointOutput构体 参考

LocalPlanning::getPlanningPointOutput 的作:



Public 属性

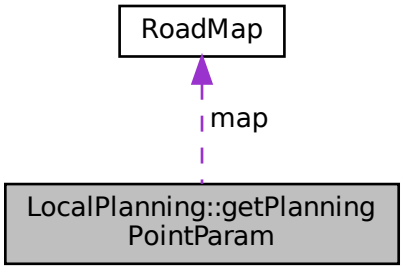
- GaussRoadPoint **ret**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.183 LocalPlanning::getPlanningPointParam构体 参考

LocalPlanning::getPlanningPointParam 的作:



Public 属性

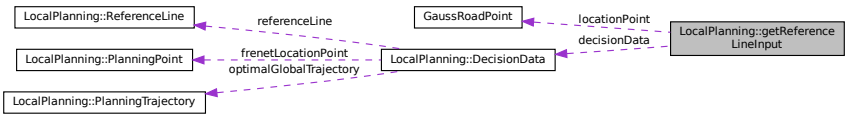
- RoadMap **map**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.184 LocalPlanning::getReferenceLineInput构体 参考

LocalPlanning::getReferenceLineInput 的作:



Public 属性

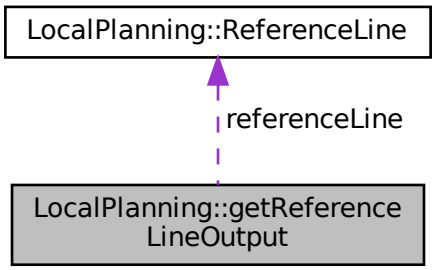
- DecisionData **decisionData**
- GaussRoadPoint **locationPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.185 LocalPlanning::getReferenceLineOutput构体 参考

LocalPlanning::getReferenceLineOutput 的作:



Public 属性

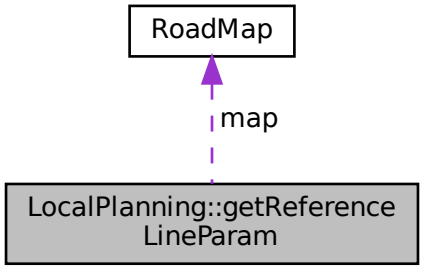
- ReferenceLine **referenceLine**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.186 LocalPlanning::getReferenceLineParam构体 参考

LocalPlanning::getReferenceLineParam 的作:



Public 属性

- RoadMap **map**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.187 GetRoadByRoadIDInput构体 参考

Public 属性

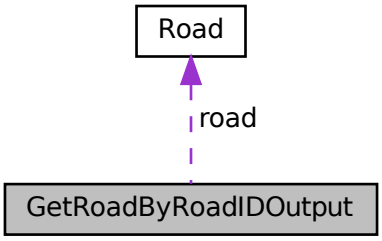
- std::vector< Road > **roads**
- int **roadID**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.188 GetRoadByRoadIDOutput构体 参考

GetRoadByRoadIDOutput 的作:



Public 属性

- Road **road**
- bool **isValidRoad**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

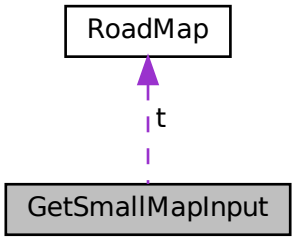
6.189 GetRoadByRoadIDParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.190 GetSmallMapInput构体 参考

GetSmallMapInput 的作:



Public 属性

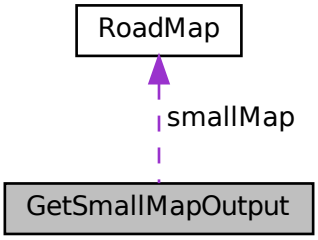
- RoadMap **t**
- double **x**
- double **y**
- double **r**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.cpp

6.191 GetSmallMapOutput构体 参考

GetSmallMapOutput 的作:



Public 属性

- RoadMap **smallMap**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.cpp

6.192 GetSmallMapParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.cpp

6.193 GetStartPointByRoadIDInput构体 参考

Public 属性

- int **roadID**
- std::vector< Road > **roads**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.194 GetStartPointByRoadIDOutput构体 参考

Public 属性

- double **GaussX**
- double **GaussY**
- bool **isValidStartPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

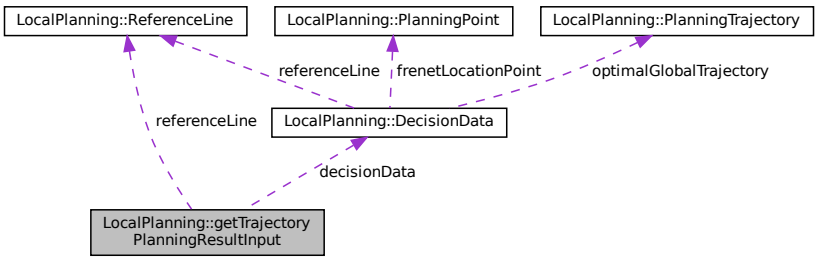
6.195 GetStartPointByRoadIDParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.196 LocalPlanning::getTrajectoryPlanningResultInput构体 参考

LocalPlanning::getTrajectoryPlanningResultInput 的作:



Public 属性

- double **velocity**
- DecisionData **decisionData**
- ReferenceLine **referenceLine**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.197 LocalPlanning::getTrajectoryPlanningResultOutput构体 参考

LocalPlanning::getTrajectoryPlanningResultOutput 的作:

Public 属性

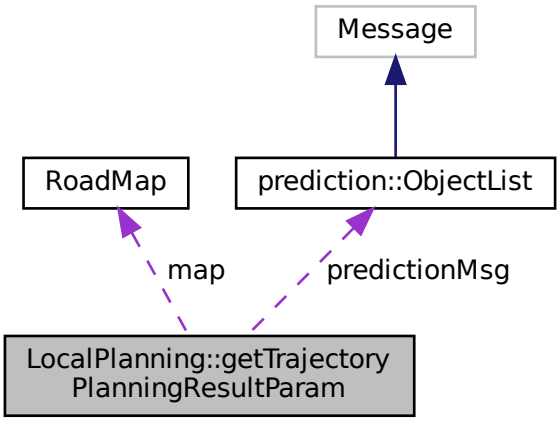
- DecisionData **decisionData**
- int32_t **index**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.198 LocalPlanning::getTrajectoryPlanningResultParam构体 参考

LocalPlanning::getTrajectoryPlanningResultParam 的作:



Public 属性

- pc::Imu **imu**
- prediction::ObjectList **predictionMsg**
- RoadMap **map**
- infopack::TrafficLight **trafficLight**
- std::vector< infopack::ObjectsProto > **objectsCmd**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.199 PointCloudsDetector::GridInfos构体 参考

Public 属性

- int **point_num** = 0
- int **isvisited** = 0
- int **property** = 0
- float **mean_height** = -1000.0
- float **max_height** = -1000.0
- float **min_height** = 1000.0
- float **plane_height** = -1000.0
- float **normal_vector_x** = 0.0
- float **normal_vector_y** = 0.0
- float **normal_vector_z** = 0.0
- std::vector< int > **point_indices**
- std::vector< int > **ground_indices**
- std::vector< int > **nonground_indices**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/classicLidarDetector.h

6.200 LocalPlanning::hasPrecedingVehicleInput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.201 LocalPlanning::hasPrecedingVehicleOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.202 LocalPlanning::hasPrecedingVehicleParam构体 参考

Public 属性

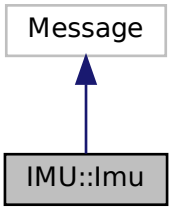
- pc::Imu **imu**
- std::vector< infopack::ObjectsProto > **objectsCmd**

构体的文档由以下文件生成:

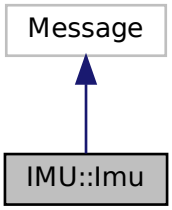
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.203 IMU::Imu 参考

IMU::Imu 承系:



IMU::Imu 的作:



Public 成函数

- **Imu** (const Imu &from)
- Imu & **operator=** (const Imu &from)
- void **Swap** (Imu ∗other)
- Imu ∗ **New** () const final
- Imu ∗ **New** (::google::protobuf::Arena ∗arena) const final
- void **CopyFrom** (const ::google::protobuf::Message &from) final
- void **MergeFrom** (const ::google::protobuf::Message &from) final
- void **CopyFrom** (const Imu &from)
- void **MergeFrom** (const Imu &from)
- void **Clear** () final
- bool **IsInitialized** () const final
- size_t **ByteSizeLong** () const final
- bool **MergePartialFromCodedStream** (::google::protobuf::io::CodedInputStream ∗input) final
- void **SerializeWithCachedSizes** (::google::protobuf::io::CodedOutputStream ∗output) const final
- ::google::protobuf::uint8 ∗ **InternalSerializeWithCachedSizesToArray** (bool deterministic, ::google↩ ::protobuf::uint8 ∗target) const final
- int **GetCachedSize** () const final
- ::google::protobuf::Metadata **GetMetadata** () const final
- void **clear_longitude** ()
- double **longitude** () const
- void **set_longitude** (double value)
- void **clear_latitude** ()
- double **latitude** () const
- void **set_latitude** (double value)
- void **clear_gaussx** ()
- double **gaussx** () const
- void **set_gaussx** (double value)
- void **clear_gaussy** ()
- double **gaussy** () const
- void **set_gaussy** (double value)
- void **clear_time** ()
- double **time** () const
- void **set_time** (double value)
- void **clear_velocity** ()
- double **velocity** () const
- void **set_velocity** (double value)

- void **clear_yaw** ()
- double **yaw** () const
- void **set_yaw** (double value)
- void **clear_gpsvalid** ()
- ::google::protobuf::int32 **gpsvalid** () const
- void **set_gpsvalid** (::google::protobuf::int32 value)
- **Imu** (const Imu &from)
- Imu & **operator=** (const Imu &from)
- void **Swap** (Imu *other)
- Imu * **New** () const final
- Imu * **New** (::google::protobuf::Arena *arena) const final
- void **CopyFrom** (const ::google::protobuf::Message &from) final
- void **MergeFrom** (const ::google::protobuf::Message &from) final
- void **CopyFrom** (const Imu &from)
- void **MergeFrom** (const Imu &from)
- void **Clear** () final
- bool **IsInitialized** () const final
- size_t **ByteSizeLong** () const final
- bool **MergePartialFromCodedStream** (::google::protobuf::io::CodedInputStream *input) final
- void **SerializeWithCachedSizes** (::google::protobuf::io::CodedOutputStream *output) const final
- ::google::protobuf::uint8 * **InternalSerializeWithCachedSizesToArray** (bool deterministic, ::google↩ ::protobuf::uint8 *target) const final
- int **GetCachedSize** () const final
- ::google::protobuf::Metadata **GetMetadata** () const final
- void **clear_longitude** ()
- double **longitude** () const
- void **set_longitude** (double value)
- void **clear_latitude** ()
- double **latitude** () const
- void **set_latitude** (double value)
- void **clear_gaussx** ()
- double **gaussx** () const
- void **set_gaussx** (double value)
- void **clear_gaussy** ()
- double **gaussy** () const
- void **set_gaussy** (double value)
- void **clear_time** ()
- double **time** () const
- void **set_time** (double value)
- void **clear_velocity** ()
- double **velocity** () const
- void **set_velocity** (double value)
- void **clear_yaw** ()
- double **yaw** () const
- void **set_yaw** (double value)
- void **clear_gpsvalid** ()
- ::google::protobuf::int32 **gpsvalid** () const
- void **set_gpsvalid** (::google::protobuf::int32 value)

静 Public 成函数

- static const ::google::protobuf::Descriptor * **descriptor** ()
- static const Imu & **default_instance** ()
- static void **InitAsDefaultInstance** ()
- static const Imu * **internal_default_instance** ()
- static const ::google::protobuf::Descriptor * **descriptor** ()
- static const Imu & **default_instance** ()
- static void **InitAsDefaultInstance** ()
- static const Imu * **internal_default_instance** ()

静 Public 属性

- static constexpr int **kIndexInFileMessages**
- static const int **kLongitudeFieldNumber** = 1
- static const int **kLatitudeFieldNumber** = 2
- static const int **kGaussXFieldNumber** = 3
- static const int **kGaussYFieldNumber** = 4
- static const int **kTimeFieldNumber** = 6
- static const int **kVelocityFieldNumber** = 7
- static const int **kYawFieldNumber** = 8
- static const int **kGpsValidFieldNumber** = 5

友元

- struct **::protobuf_imu_2eproto::TableStruct**
- void **swap** (Imu &a, Imu &b)
- void **swap** (Imu &a, Imu &b)

6.203.1 成量明

6.203.1.1 kIndexInFileMessages

```
static constexpr int IMU::Imu::kIndexInFileMessages  [static], [constexpr]
```

初始:
=
0

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩coms_livox/include/imu.pb.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/imu.pb.cc

6.204 IMU::ImuDefaultTypeInternal 参考

Public 属性

- ::google::protobuf::internal::ExplicitlyConstructed< Imu > **_instance**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/imu.pb.cc

6.205 LocalPlanning::inAreaInput构体 参考

Public 属性

- double **longitudeX**
- double **latitudeY**
- double **targetX**
- double **targetY**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.206 LocalPlanning::inAreaOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.207 LocalPlanning::inAreaParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.208 LocalPlanning::initLocalPlanningInput构体 参考

LocalPlanning::initLocalPlanningInput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.209 LocalPlanning::initLocalPlanningOutput构体 参考

LocalPlanning::initLocalPlanningOutput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

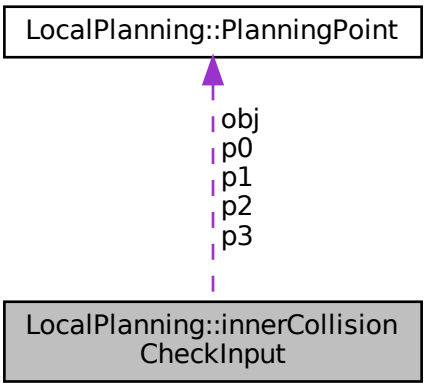
6.210 LocalPlanning::initLocalPlanningParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.211 LocalPlanning::innerCollisionCheckInput构体 参考

LocalPlanning::innerCollisionCheckInput 的作:



Public 属性

- PlanningPoint **p0**
- PlanningPoint **p1**
- PlanningPoint **p2**
- PlanningPoint **p3**
- PlanningPoint **obj**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.212 LocalPlanning::innerCollisionCheckOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.213 LocalPlanning::innerCollisionCheckParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.214 geometry::inOrOutPolygonInput构体 参考

Public 属性

- int **nvert**
- double ∗ **vertx**
- double ∗ **verty**
- double **testx**
- double **testy**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩ _m.h

6.215 geometry::inOrOutPolygonOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩ _m.h

6.216 geometry::inOrOutPolygonParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩ _m.h

6.217 InputStruct01构体 参考

classicLidarDetectorFind_Z_value 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **cloud**

6.217.1 描述

classicLidarDetectorFind_Z_value 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.218 InputStruct02构体 参考

classicLidarDetectorPassThroughFilter 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **cloud**
- double **z_max**
- double **z_min**

6.218.1 描述

classicLidarDetectorPassThroughFilter 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.219 InputStruct03构体 参考

classicLidarDetectorPointcloud_to_grid 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **cloud**
- double **map_resolution**

6.219.1 描述

classicLidarDetectorPointcloud_to_grid 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.220 InputStruct04构体 参考

classicLidarDetectorInputcloud 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- int **i**

6.220.1 描述

classicLidarDetectorInputcloud 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.221 InputStruct05构体 参考

classicLidarDetectorInitCamera 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- CameraAngle **setAngle**

6.221.1 描述

classicLidarDetectorInitCamera 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.222 InputStruct06构体 参考

classicLidarDetectorInitView 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.222.1 描述

classicLidarDetectorInitView 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.223 InputStruct07构体 参考

classicLidarDetectorCheckAlive 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.223.1 描述

classicLidarDetectorCheckAlive 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.224 InputStruct08构体 参考

classicLidarDetectorReceiveCloud 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **input**

6.224.1 描述

classicLidarDetectorReceiveCloud 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.225 InputStruct09构体 参考

classicLidarDetectorRPointCloud 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- const pcl::PointCloud< pcl::PointXYZI >::Ptr & **cloud**

6.225.1 描述

classicLidarDetectorRPointCloud 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

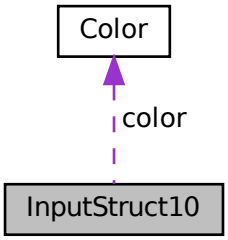
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.226 InputStruct10构体 参考

classicLidarDetectorRenderPointCloud 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

InputStruct10 的作:



Public 属性

- const pcl::PointCloud< pcl::PointXYZI >::Ptr & **cloud**
- std::string **name**
- Color **color**

6.226.1 描述

classicLidarDetectorRenderPointCloud 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.227 InputStruct11构体 参考

classicLidarDetectorSwapXAndY 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **cloud**

6.227.1 描述

classicLidarDetectorSwapXAndY 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.228 InputStruct12构体 参考

classicLidarDetectorSwapYAndX 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **cloud**

6.228.1 描述

classicLidarDetectorSwapYAndX 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.229 InputStruct13构体 参考

classicLidarDetectorNumPoints 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **cloud**

6.229.1 描述

classicLidarDetectorNumPoints 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.230 InputStruct14构体 参考

classicLidarDetectorGetViewer 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.230.1 描述

classicLidarDetectorGetViewer 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.231 InputStruct15构体 参考

classicLidarDetectorClipPlane 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **src_cloud**
- const Eigen::Vector4f **plane**
- bool **negative**

6.231.1 描述

classicLidarDetectorClipPlane 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.232 InputStruct16构体 参考

classicLidarDetectorFilterCloud 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **cloud**
- float **filterRes_x**
- float **filterRes_y**
- float **filterRes_z**
- Eigen::Vector4f **minPoint**
- Eigen::Vector4f **maxPoint**

6.232.1 描述

classicLidarDetectorFilterCloud 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.233 InputStruct17构体 参考

classicLidarDetectorRmvNearPoints 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **cloud**
- Eigen::Vector4f **minPoint**
- Eigen::Vector4f **maxPoint**

6.233.1 描述

classicLidarDetectorRmvNearPoints 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.234 InputStruct18构体 参考

classicLidarDetectorRANSAC3d 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **cloud**
- int **maxIterations**
- float **distanceThreshold**

6.234.1 描述

classicLidarDetectorRANSAC3d 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.235 InputStruct19构体 参考

classicLidarDetectorClustering 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **cloud**
- float **clusterTolerance**
- int **minSize**
- int **maxSize**

6.235.1 描述

classicLidarDetectorClustering 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.236 InputStruct20构体 参考

classicLidarDetectorGridGroundFilter 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- const pcl::PointCloud< pcl::PointXYZI >::Ptr **cloud**
- int **normal_estimation_Ksearch**
- float **slope_rate**
- float **normal_thereshold**
- int **grid_height_mode**
- float **car_height**
- float **grid_size**
- float **right_left_distance**
- float **back_distance**
- float **front_distance**

6.236.1 描述

classicLidarDetectorGridGroundFilter 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

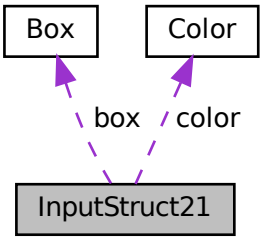
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.237 InputStruct21构体 参考

classicLidarDetectorRenderBox 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

InputStruct21 的作:



Public 属性

- Box **box**
- int **id**
- Color **color** = Color(0, 1, 0)
- float **opacity** = 1

6.237.1 描述

classicLidarDetectorRenderBox 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

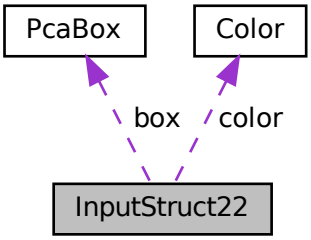
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.238 InputStruct22构体 参考

classicLidarDetectorRenderPcaBox 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

InputStruct22 的作:



Public 属性

- PcaBox **box**
- int **id**
- Color **color** = Color(0, 1, 0)
- float **opacity** = 1

6.238.1 描述

classicLidarDetectorRenderPcaBox 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩coms_livox/include/functionLidarDetector.h

6.239 InputStruct23构体 参考

classicLidarDetectoRun 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.239.1 描述

classicLidarDetectoRun 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩coms_livox/include/functionLidarDetector.h

6.240 InputStruct24构体 参考

classicLidarDetectoTest 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.240.1 描述

classicLidarDetectoTest 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.241 InputStruct25构体 参考

classicLidarDetectoGridRANSAC 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- const pcl::PointCloud< pcl::PointXYZI >::Ptr **cloud**
- float **ransac_threshold**
- int **normal_estimation_Ksearch**
- float **slope_rate**
- float **normal_thereshold**
- int **grid_height_mode**
- float **car_height**
- float **grid_size**
- float **right_left_distance**
- float **back_distance**
- float **front_distance**

6.241.1 描述

classicLidarDetectoGridRANSAC 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.242 InputStruct26构体 参考

classicLidarDetectoGridRANSAC 函数件入的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **point_cloudi**
- int **normal_estimation_Ksearch**
- float **slope_rate**
- float **normal_thereshold**
- int **grid_height_mode**
- float **car_height**
- float **grid_size**
- float **right_left_distance**
- float **back_distance**
- float **front_distance**

6.242.1 描述

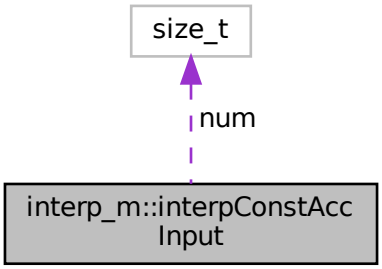
classicLidarDetectoGridRANSAC 函数件入的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.243 interp_m::interpConstAccInput构体 参考

interp_m::interpConstAccInput 的作:



Public 属性

- double **start_speed**
- double **end_speed**
- size_t **num**
- double **acceleration**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/interpolate↩ _m.h

6.244 interp_m::interpConstAccOutput构体 参考

Public 属性

- std::vector< double > **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/interpolate↩ _m.h

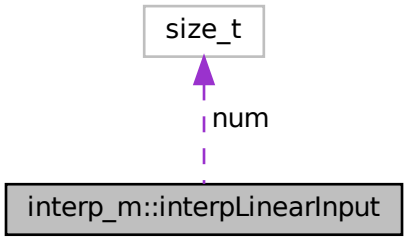
6.245 interp_m::interpConstAccParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/interpolate↩ _m.h

6.246 interp_m::interpLinearInput构体 参考

interp_m::interpLinearInput 的作:



Public 属性

- double **start_speed**
- double **end_speed**
- size_t **num**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/interpolate↩_m.h

6.247 interp_m::interpLinearOutput构体 参考

Public 属性

- std::vector< double > **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/interpolate↩_m.h

6.248 interp_m::interpLinearParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/interpolate↩_m.h

6.249 LaneDetector::IPMInfo构体 参考

```
#include <InversePerspectiveMapping.hpp>
```

Public 属性

- float xLimits [2]

  min and max x-value on ground in world coordinates
- float yLimits [2]

  min and max y-value on ground in world coordinates
- float xScale
- float yScale
- int width

  width
- int height

  height
- float vpPortion
- float ipmLeft

  Left point in original image of region to make IPM for.
- float ipmRight

  Right point in original image of region to make IPM for.
- float ipmTop

  Top point in original image of region to make IPM for.
- float ipmBottom

  Bottom point in original image of region to make IPM for.
- int ipmInterpolation

  interpolation to use for IPM (0: bilinear, 1:nearest neighbor)

6.249.1 描述

Structure to hold the info about IPM transformation

6.249.2 成量明

6.249.2.1 vpPortion

```
float LaneDetector::IPMInfo::vpPortion
```

portion of image height to add to y-coordinate of vanishing point

6.249.2.2 xScale

```
float LaneDetector::IPMInfo::xScale
```

conversion between mm in world coordinate on the ground in x-direction and pixel in image

6.249.2.3 yScale

```
float LaneDetector::IPMInfo::yScale
```

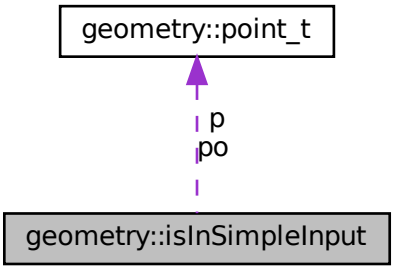
conversion between mm in world coordinate on the ground in y-direction and pixel in image

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_cv/include/InversePerspectiveMapping.hpp

6.250 geometry::isInSimpleInput构体 参考

geometry::isInSimpleInput 的作:



Public 属性

- point_t * **p**
- int **n**
- point_t **po**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry_m.h

6.251 geometry::isInSimpleOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry_m.h

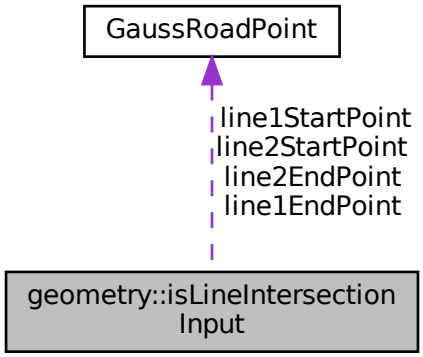
6.252 geometry::isInSimpleParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.253 geometry::isLineIntersectionInput构体 参考

geometry::isLineIntersectionInput 的作:



Public 属性

- GaussRoadPoint **line1StartPoint**
- GaussRoadPoint **line1EndPoint**
- GaussRoadPoint **line2StartPoint**
- GaussRoadPoint **line2EndPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.254 geometry::isLineIntersectionOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

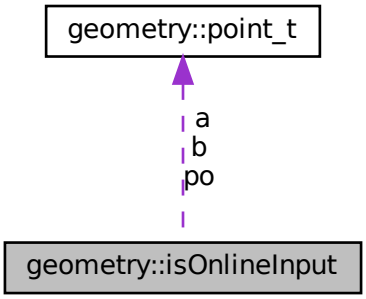
6.255 geometry::isLineIntersectionParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.256 geometry::isOnlineInput构体 参考

geometry::isOnlineInput 的作:



Public 属性

- point_t **a**
- point_t **b**
- point_t **po**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.257 geometry::isOnlineOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

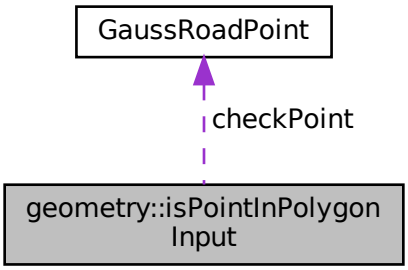
6.258 geometry::isOnlineParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.259 geometry::isPointInPolygonInput构体 参考

geometry::isPointInPolygonInput 的作:



Public 属性

- std::vector< GaussRoadPoint > **pointPolygon**
- GaussRoadPoint **checkPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.260 geometry::isPointInPolygonOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.261 geometry::isPointInPolygonParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.262 geometry::isPolygonsIntersectionInput构体 参考

Public 属性

- std::vector< GaussRoadPoint > **polygon1Point**
- std::vector< GaussRoadPoint > **polygon2Point**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.263 geometry::isPolygonsIntersectionOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.264 geometry::isPolygonsIntersectionParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.265 Jobs 参考

Public 成函数

- **Jobs** (std::vector< void * > &receivingSocketList, std::vector< void * > &sendingSocketList, RoadMap &m, std::vector< GaussRoadPoint > &stopPoints, std::tuple< int32_t, int32_t, int32_t > &stopPointRoadLaneId, std::vector< std::tuple< int32_t, int32_t >> &rList)
- virtual void **subscriber** ()
- virtual void **publisher** ()
- virtual void **processorLocalPlanning** ()
- virtual void **processorGlobalPlanning** ()
- virtual void **processorGlobalPlanningBaseLocal** ()
- virtual void **processorGlobalPlanningBaseServer** ()
- void **FindPointIDatRoad** (GaussRoadPoint roadPoint, int32_t roadID, int32_t &laneID, int32_t &pointID)
- virtual void **request** ()

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/planning.↩hpp
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/planning.cpp
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/planningFigure_m.cpp

6.266 Lane 参考

Public 属性

- int **id**
- std::vector< LaneSuccessorId > **successorId**
- std::vector< int > **leftLaneId**
- std::vector< int > **rightLaneId**
- std::vector< GaussRoadPoint > **gaussRoadPoints**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩Analysis.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩Analysis_m.h

6.267 LaneSuccessorId构体 参考

Public 属性

- int **sucRoadID**
- int **sucLaneID**

构体的文档由以下文件生成:

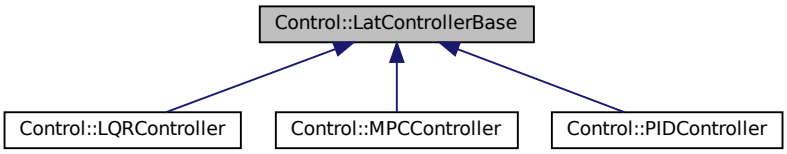
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩Analysis.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩Analysis_m.h

6.268 Control::LatControllerBase 参考

横向控制器基。

```
#include <controlController.hpp>
```

Control::LatControllerBase 承系:



Public 成函数

- **LatControllerBase** (double maxSteer, double ratioWheelToSteer=20, double dt=0.1)
- virtual double **calcSteer** (State s, const Traj &traj, size_t nearestID)=0

Protected 成函数

- double **regulateOutput** (double steer)

Protected 属性

- double **dt**
- double **maxSteer**
- double **ratioWheelToSteer**

6.268.1 描述

横向控制器基。

注解

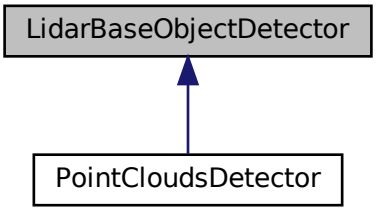
抽象，子控制器必 `calcSteer`算前角方法

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlControll

6.269 LidarBaseObjectDetector 参考

LidarBaseObjectDetector 承系:



Public 成函数

- virtual std::pair< std::vector< int >, std::vector< Frame > > **run** ()=0
- virtual void **receiveCloud** (pcl::PointCloud< pcl::PointXYZI >::Ptr input)=0
- virtual pcl::visualization::PCLVisualizer::Ptr **getViewer** ()=0
- bool **sendResultInit** ()
- void **sendResult** ()

Public 属性

- perception::ObjectList **objResultList**
- pcl::PointCloud< pcl::PointXYZI >::Ptr **input_cloud**
- double **imu_x**
- double **imu_y**
- double **imu_yaw**
- double **imu_v**
- double **imu_rtkMode**
- void * **publisher**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/baseLidarObjectDetector.h

6.270 LocalPlanning::limitPlanningDistanceInput构体 参考

Public 属性

- double **targetDistance**
- double **lastTargetDistance**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.271 LocalPlanning::limitPlanningDistanceOutput构体 参考

Public 属性

- double **targetDistance**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.272 LocalPlanning::limitPlanningDistanceParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.273 line_list构体 参考

line_list 的作:

Public 属性

- char * **string_arg**
- struct line_list * **next**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_cv/src/Camera↩ InfoOpt.c

6.274 LocalPlanning::loadStopPointsFromYamlInput构体 参考

Public 属性

- std::vector< GaussRoadPoint > **rawStopPoints**
- std::tuple< int32_t, int32_t, int32_t > **stopPointRoadLanePointId**
- double **offsetX**
- double **offsetY**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.275 LocalPlanning::loadStopPointsFromYamlOutput构体 参考

Public 属性

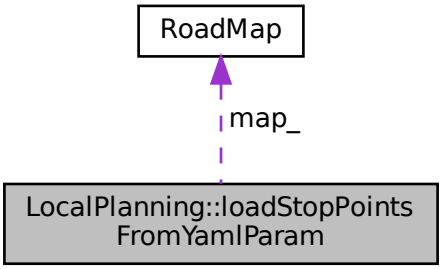
- std::vector< GaussRoadPoint > **rawStopPoints**
- std::tuple< int32_t, int32_t, int32_t > **stopPointRoadLanePointId**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.276 LocalPlanning::loadStopPointsFromYamlParam构体 参考

LocalPlanning::loadStopPointsFromYamlParam 的作:



Public 属性

- RoadMap **map_**
- std::string **fileName**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.277 LocalPlanning::loadStopPointsInput构体 参考

Public 属性

- std::vector< GaussRoadPoint > **rawStopPoints**
- std::tuple< int32_t, int32_t, int32_t > **stopPointRoadLanePointId**
- double **offsetX**
- double **offsetY**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.278 LocalPlanning::loadStopPointsOutput构体 参考

Public 属性

- std::vector< GaussRoadPoint > **rawStopPoints**
- std::tuple< int32_t, int32_t, int32_t > **stopPointRoadLanePointId**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.279 LocalPlanning::loadStopPointsParam构体 参考

Public 属性

- std::string **fileName**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.280 Control::LonController 参考

longitude 向控制器

```
#include <controlController.hpp>
```

Public 成函数

- **LonController** (double maxSpeed)
- double calcSpeed (State s, const Traj &traj, size_t nearestID)

  向控制器 根据迹最近点分配速度

Protected 成函数

- double **regulateOutput** (double speed)

Protected 属性

- double **maxSpeed**

6.280.1 描述

longitude 向控制器

6.280.2 成函数明

6.280.2.1 calcSpeed()

```
double Control::LonController::calcSpeed (
            State s,
            const Traj & traj,
            size_t nearestID )
```

向控制器 根据迹最近点分配速度

参数

s	当前状
traj	待跟踪迹
nearestID	traj中最近点ID

返回

double 期望速度

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlControll
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/controlController.c

6.281 Control::LQR 参考

Public 成函数

- Mat solveDARE (Mat A, Mat B, Mat Q, Mat R)

  solve a discrete time_Algebraic Riccati equation (DARE) 用迭代法求解离散黎卡提方程。
- Mat dLQR (Mat A, Mat B, Mat Q, Mat R)

  Solve the discrete time lqr controller. 求解离散*LQR*控制率 *K. ref Bertsekas, p.151.*

的文档由以下文件生成:

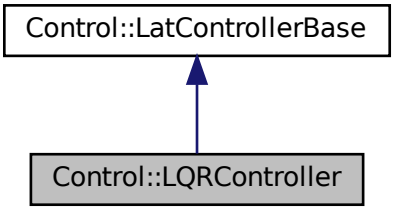
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlLQRCo
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/controlLQRControl

6.282 Control::LQRController 参考

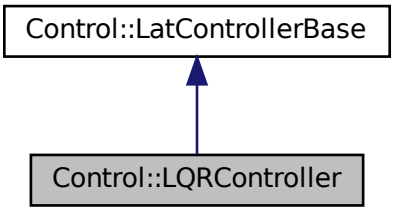
LQR横向控制

```
#include <controlLQRController.hpp>
```

Control::LQRController 承系:



Control::LQRController 的作:



Public 成函数

- **LQRController** (double wheelbase, double maxSteer, double ratioWheelToSteer, double dt=0.1)
- double calcSteer (State s, const Traj &traj, size_t nearestID)

  求解横向控制量

外承的成函数

6.282.1 描述

LQR横向控制

6.282.2 成函数明

6.282.2.1 calcSteer()

```
double Control::LQRController::calcSteer (
            State s,
            const Traj & traj,
            size_t nearestID )  [virtual]
```

求解横向控制量

参数

s	
traj	
nearestID	

返回

double

了 Control::LatControllerBase.

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlLQRCo
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/controlLQRControl

6.283 mapAnalysisInput构体 参考

Public 属性

- std::string **path**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.284 RoadMap::mapAnalysisInput构体 参考

Public 属性

- std::string **path**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.285 RoadMap::mapAnalysisOutput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.286 mapAnalysisOutput构体 参考

Public 属性

- std::vector< Road > **roads**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.287 mapAnalysisParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.288 RoadMap::mapAnalysisParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.289 rapidxml::memory_pool< Ch > 模板 参考

```
#include <rapidxml.hpp>
```

Public 成函数

- memory_pool ()

  Constructs empty pool with default allocator functions.
- ~memory_pool ()
- xml_node< Ch > * allocate_node (node_type type, const Ch *name=0, const Ch *value=0, std::size_t name↩ _size=0, std::size_t value_size=0)
- xml_attribute< Ch > * allocate_attribute (const Ch *name=0, const Ch *value=0, std::size_t name_size=0, std::size_t value_size=0)
- template<typename Sch >
  Ch * allocate_string (const Sch *source=0, std::size_t size=0)
- Ch * **nullstr** ()
- Ch * **xmlns_xml** (std::size_t &xmlns_size)
- Ch * **xmlns_xmlns** (std::size_t &xmlns_size)
- xml_node< Ch > * clone_node (const xml_node< Ch > *source, xml_node< Ch > *result=0)
- void clear ()
- void set_allocator (alloc_func *af, free_func *ff)

6.289.1 描述

template<class Ch = char>
class rapidxml::memory_pool< Ch >

This class is used by the parser to create new nodes and attributes, without overheads of dynamic memory allocation. In most cases, you will not need to use this class directly. However, if you need to create nodes manually or modify names/values of nodes, you are encouraged to use memory_pool of relevant xml_document to allocate the memory. Not only is this faster than allocating them by using `new` operator, but also their lifetime will be tied to the lifetime of document, possibly simplyfing memory management.

Call allocate_node() or allocate_attribute() functions to obtain new nodes or attributes from the pool. You can also call allocate_string() function to allocate strings. Such strings can then be used as names or values of nodes without worrying about their lifetime. Note that there is no `free()` function – all allocations are freed at once when clear() function is called, or when the pool is destroyed.

It is also possible to create a standalone memory_pool, and use it to allocate nodes, whose lifetime will not be tied to any document.

Pool maintains `RAPIDXML_STATIC_POOL_SIZE` bytes of statically allocated memory. Until static memory is exhausted, no dynamic memory allocations are done. When static memory is exhausted, pool allocates additional blocks of memory of size `RAPIDXML_DYNAMIC_POOL_SIZE` each, by using global `new[]` and `delete[]` operators. This behaviour can be changed by setting custom allocation routines. Use set_allocator() function to set them.

Allocations for nodes, attributes and strings are aligned at `RAPIDXML_ALIGNMENT` bytes. This value defaults to the size of pointer on target architecture.

To obtain absolutely top performance from the parser, it is important that all nodes are allocated from a single, contiguous block of memory. Otherwise, cache misses when jumping between two (or more) disjoint blocks of memory can slow down parsing quite considerably. If required, you can tweak `RAPIDXML_STATIC_POOL_SIZE`, `RAPIDXML_DYNAMIC_POOL_SIZE` and `RAPIDXML_ALIGNMENT` to obtain best wasted memory to performance compromise. To do it, define their values before rapidxml.hpp file is included.

参数

Ch	Character type of created nodes.

6.289.2 构造及析构函数明

6.289.2.1 ~memory_pool()

```
template<class Ch = char>
rapidxml::memory_pool< Ch >::~memory_pool ( ) [inline]
```

Destroys pool and frees all the memory. This causes memory occupied by nodes allocated by the pool to be freed. Nodes allocated from the pool are no longer valid.

6.289.3 成函数明

6.289.3.1 allocate_attribute()

```
template<class Ch = char>
xml_attribute<Ch>* rapidxml::memory_pool< Ch >::allocate_attribute (
            const Ch * name = 0,
            const Ch * value = 0,
            std::size_t name_size = 0,
            std::size_t value_size = 0 )  [inline]
```

Allocates a new attribute from the pool, and optionally assigns name and value to it. If the allocation request cannot be accomodated, this function will throw `std::bad_alloc`. If exceptions are disabled by defining RAPIDXML↩_NO_EXCEPTIONS, this function will call rapidxml::parse_error_handler() function.

参数

name	Name to assign to the attribute, or 0 to assign no name.
value	Value to assign to the attribute, or 0 to assign no value.
name_size	Size of name to assign, or 0 to automatically calculate size from name string.
value_size	Size of value to assign, or 0 to automatically calculate size from value string.

返回

Pointer to allocated attribute. This pointer will never be NULL.

6.289.3.2 allocate_node()

```
template<class Ch = char>
xml_node<Ch>* rapidxml::memory_pool< Ch >::allocate_node (
            node_type type,
            const Ch * name = 0,
            const Ch * value = 0,
            std::size_t name_size = 0,
            std::size_t value_size = 0 )  [inline]
```

Allocates a new node from the pool, and optionally assigns name and value to it. If the allocation request cannot be accomodated, this function will throw `std::bad_alloc`. If exceptions are disabled by defining RAPIDXML_N↩O_EXCEPTIONS, this function will call rapidxml::parse_error_handler() function.

参数

type	Type of node to create.
name	Name to assign to the node, or 0 to assign no name.
value	Value to assign to the node, or 0 to assign no value.
name_size	Size of name to assign, or 0 to automatically calculate size from name string.
value_size	Size of value to assign, or 0 to automatically calculate size from value string.

返回

Pointer to allocated node. This pointer will never be NULL.

6.289.3.3 allocate_string()

```
template<class Ch = char>
template<typename Sch >
Ch* rapidxml::memory_pool< Ch >::allocate_string (
            const Sch * source = 0,
            std::size_t size = 0 )  [inline]
```

Allocates a char array of given size from the pool, and optionally copies a given string to it. If the allocation request cannot be accomodated, this function will throw `std::bad_alloc`. If exceptions are disabled by defining RAP↩IDXML_NO_EXCEPTIONS, this function will call rapidxml::parse_error_handler() function.

参数

source	String to initialize the allocated memory with, or 0 to not initialize it.
size	Number of characters to allocate, or zero to calculate it automatically from source string length; if size is 0, source string must be specified and null terminated.

返回

Pointer to allocated char array. This pointer will never be NULL.

6.289.3.4 clear()

```
template<class Ch = char>
void rapidxml::memory_pool< Ch >::clear ( )  [inline]
```

Clears the pool. This causes memory occupied by nodes allocated by the pool to be freed. Any nodes or strings allocated from the pool will no longer be valid.

6.289.3.5 clone_node()

```
template<class Ch = char>
xml_node<Ch>* rapidxml::memory_pool< Ch >::clone_node (
            const xml_node< Ch > * source,
            xml_node< Ch > * result = 0 )  [inline]
```

Clones an xml_node and its hierarchy of child nodes and attributes. Nodes and attributes are allocated from this memory pool. Names and values are not cloned, they are shared between the clone and the source. Result node can be optionally specified as a second parameter, in which case its contents will be replaced with cloned source node. This is useful when you want to clone entire document.

参数

source	Node to clone.
result	Node to put results in, or 0 to automatically allocate result node

返回

Pointer to cloned node. This pointer will never be NULL.

6.289.3.6 set_allocator()

```
template<class Ch = char>
void rapidxml::memory_pool< Ch >::set_allocator (
            alloc_func * af,
            free_func * ff )  [inline]
```

Sets or resets the user-defined memory allocation functions for the pool. This can only be called when no memory is allocated from the pool yet, otherwise results are undefined. Allocation function must not return invalid pointer on failure. It should either throw, stop the program, or use `longjmp()` function to pass control to other place of program. If it returns invalid pointer, results are undefined.

User defined allocation functions must have the following forms:

```
void *allocate(std::size_t size);
void free(void *pointer);
```

参数

af	Allocation function, or 0 to restore default function
ff	Free function, or 0 to restore default function

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml.hpp

6.290 moduleSelfCheckInput构体 参考

Public 属性

- std::vector< Road > **roads**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.291 moduleSelfCheckOutput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.292 moduleSelfCheckParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.293 moduleSelfCheckPrintInput构体 参考

Public 属性

- std::vector< Road > **roads**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.294 moduleSelfCheckPrintOutput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.295 moduleSelfCheckPrintParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

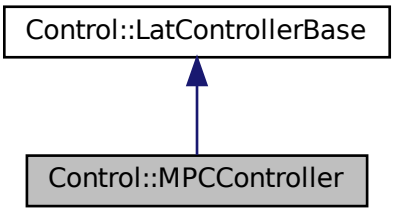
6.296 Control::MPC 参考

的文档由以下文件生成:

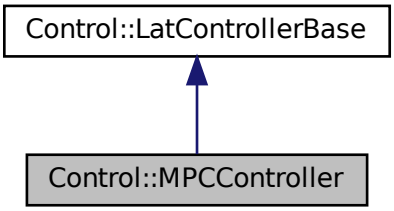
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlMPCCo

6.297 Control::MPCController 参考

Control::MPCController 承系:



Control::MPCController 的作:



Public 成函数

- **MPCController** (std::string configFile, double maxSteer_, double ratioWheelToSteer_, double dt_=0.1)
- double **calcSteer** (State s, const Traj &traj, size_t nearestID)

外承的成函数

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlMPCCo
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/controlMPCContro

6.298 neighborLaneSortInput构体 参考

Public 属性

- std::vector< Road > **roads**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.299 neighborLaneSortOutput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.300 neighborLaneSortParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.301 GolbalPlanning::node 参考

Public 成函数

- **node** (int _number, double _F, int _father)

Public 属性

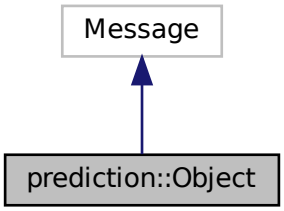
- int **number**
- double **F**
- int **father**

的文档由以下文件生成:

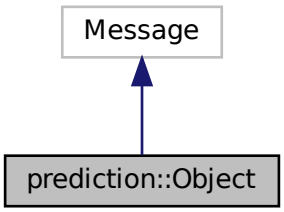
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.302 prediction::Object 参考

prediction::Object 承系:



prediction::Object 的作:



Public 成函数

- **Object** (const Object &from)
- Object & **operator=** (const Object &from)
- void **Swap** (Object ∗other)
- Object ∗ **New** () const final
- Object ∗ **New** (::google::protobuf::Arena ∗arena) const final
- void **CopyFrom** (const ::google::protobuf::Message &from) final
- void **MergeFrom** (const ::google::protobuf::Message &from) final
- void **CopyFrom** (const Object &from)
- void **MergeFrom** (const Object &from)
- void **Clear** () final
- bool **IsInitialized** () const final
- size_t **ByteSizeLong** () const final
- bool **MergePartialFromCodedStream** (::google::protobuf::io::CodedInputStream ∗input) final
- void **SerializeWithCachedSizes** (::google::protobuf::io::CodedOutputStream ∗output) const final
- ::google::protobuf::uint8 ∗ **InternalSerializeWithCachedSizesToArray** (bool deterministic, ::google↩ ::protobuf::uint8 ∗target) const final

- int **GetCachedSize** () const final
- ::google::protobuf::Metadata **GetMetadata** () const final
- int **predictpoint_size** () const
- void **clear_predictpoint** ()
- ::prediction::PredictPoint * **mutable_predictpoint** (int index)
- ::google::protobuf::RepeatedPtrField< ::prediction::PredictPoint > * **mutable_predictpoint** ()
- const ::prediction::PredictPoint & **predictpoint** (int index) const
- ::prediction::PredictPoint * **add_predictpoint** ()
- const ::google::protobuf::RepeatedPtrField< ::prediction::PredictPoint > & **predictpoint** () const
- void **clear_z** ()
- double **z** () const
- void **set_z** (double value)
- void **clear_w** ()
- double **w** () const
- void **set_w** (double value)
- void **clear_h** ()
- double **h** () const
- void **set_h** (double value)
- void **clear_l** ()
- double **l** () const
- void **set_l** (double value)
- void **clear_type** ()
- ::google::protobuf::int32 **type** () const
- void **set_type** (::google::protobuf::int32 value)
- void **clear_trackid** ()
- ::google::protobuf::int32 **trackid** () const
- void **set_trackid** (::google::protobuf::int32 value)

静 Public 成函数

- static const ::google::protobuf::Descriptor * **descriptor** ()
- static const Object & **default_instance** ()
- static void **InitAsDefaultInstance** ()
- static const Object * **internal_default_instance** ()

静 Public 属性

- static constexpr int **kIndexInFileMessages**
- static const int **kPredictPointFieldNumber** = 1
- static const int **kZFieldNumber** = 2
- static const int **kWFieldNumber** = 3
- static const int **kHFieldNumber** = 4
- static const int **kLFieldNumber** = 5
- static const int **kTypeFieldNumber** = 6
- static const int **kTrackIDFieldNumber** = 7

友元

- struct **::protobuf_prediction_2eproto::TableStruct**
- void **swap** (Object &a, Object &b)

6.302.1 成量明

6.302.1.1 kIndexInFileMessages

```
constexpr int prediction::Object::kIndexInFileMessages  [static], [constexpr]
```

初始:

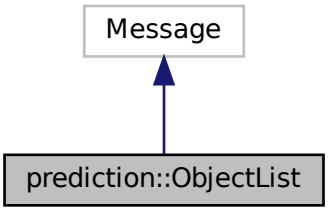
```
=
    1
```

的文档由以下文件生成:

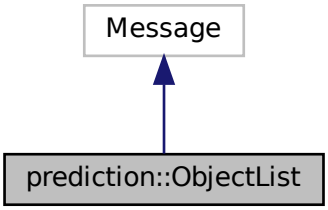
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/prediction.pb.h

6.303 prediction::ObjectList 参考

prediction::ObjectList 承系:



prediction::ObjectList 的作:



Public 成函数

- **ObjectList** (const ObjectList &from)
- ObjectList & **operator=** (const ObjectList &from)
- void **Swap** (ObjectList ∗other)
- ObjectList ∗ **New** () const final
- ObjectList ∗ **New** (::google::protobuf::Arena ∗arena) const final
- void **CopyFrom** (const ::google::protobuf::Message &from) final
- void **MergeFrom** (const ::google::protobuf::Message &from) final
- void **CopyFrom** (const ObjectList &from)
- void **MergeFrom** (const ObjectList &from)
- void **Clear** () final
- bool **IsInitialized** () const final
- size_t **ByteSizeLong** () const final
- bool **MergePartialFromCodedStream** (::google::protobuf::io::CodedInputStream ∗input) final
- void **SerializeWithCachedSizes** (::google::protobuf::io::CodedOutputStream ∗output) const final
- ::google::protobuf::uint8 ∗ **InternalSerializeWithCachedSizesToArray** (bool deterministic, ::google↩ ::protobuf::uint8 ∗target) const final
- int **GetCachedSize** () const final
- ::google::protobuf::Metadata **GetMetadata** () const final
- int **object_size** () const
- void **clear_object** ()
- ::prediction::Object ∗ **mutable_object** (int index)
- ::google::protobuf::RepeatedPtrField< ::prediction::Object > ∗ **mutable_object** ()
- const ::prediction::Object & **object** (int index) const
- ::prediction::Object ∗ **add_object** ()
- const ::google::protobuf::RepeatedPtrField< ::prediction::Object > & **object** () const

静 Public 成函数

- static const ::google::protobuf::Descriptor ∗ **descriptor** ()
- static const ObjectList & **default_instance** ()
- static void **InitAsDefaultInstance** ()
- static const ObjectList ∗ **internal_default_instance** ()

静 Public 属性

- static constexpr int **kIndexInFileMessages**
- static const int **kObjectFieldNumber** = 1

友元

- struct **::protobuf_prediction_2eproto::TableStruct**
- void **swap** (ObjectList &a, ObjectList &b)

6.303.1 成量明

6.303.1.1 kIndexInFileMessages

```
constexpr int prediction::ObjectList::kIndexInFileMessages [static], [constexpr]
```

初始:
=
2

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/prediction.pb.h

6.304 OSMFormat 参考

Public 成函数

- **OSMFormat** (std::string path)
- void **formatConversion** (std::string path)
- void **saveMapToOSM** (std::string path)
- **OSMFormat** (std::string path)
- void **formatConversion** (std::string path)
- void **saveMapToOSM** (std::string path)

Public 属性

- std::vector< OSMNode > **allNodes**
- std::vector< OSMWay > **allWays**
- std::vector< OSMRelation > **allRelations**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis.cpp
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.cpp

6.305 OSMNode 参考

Public 属性

- int **id**
- double **localX**
- double **localY**
- double **lon**
- double **lat**
- double **yaw**
- double **curvature**
- std::string **idString**
- std::string **localXString**
- std::string **localYString**
- std::string **lonString**
- std::string **latString**
- std::string **yawString**
- std::string **curvatureString**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.306 OSMRelation 参考

Public 属性

- int **id**
- std::string **idString**
- std::vector< std::string > **waySet**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.307 OSMWay 参考

Public 属性

- int **id**
- std::string **idString**
- std::vector< std::string > **nodeSet**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.308 OutputStruct01构体 参考

classicLidarDetectorFind_Z_value 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.308.1 描述

classicLidarDetectorFind_Z_value 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.309 OutputStruct02构体 参考

classicLidarDetectorPassThroughFilter 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **output**

6.309.1 描述

classicLidarDetectorPassThroughFilter 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.310 OutputStruct03构体 参考

classicLidarDetectorPointcloud_to_grid 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- vector< int > **Pos_array**

6.310.1 描述

classicLidarDetectorPointcloud_to_grid 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.311 OutputStruct04构体 参考

classicLidarDetectorInputcloud 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **output**

6.311.1 描述

classicLidarDetectorInputcloud 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.312 OutputStruct05构体 参考

classicLidarDetectorInitCamera 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.312.1 描述

classicLidarDetectorInitCamera 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.313 OutputStruct06构体 参考

classicLidarDetectorInitView 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.313.1 描述

classicLidarDetectorInitView 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.314 OutputStruct07构体 参考

classicLidarDetectorCheckAlive 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- bool **isAlive**

6.314.1 描述

classicLidarDetectorCheckAlive 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.315 OutputStruct08构体 参考

classicLidarDetectorReceiveCloud 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.315.1 描述

classicLidarDetectorReceiveCloud 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.316 OutputStruct09构体 参考

classicLidarDetectorRPointCloud 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.316.1 描述

classicLidarDetectorRPointCloud 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.317 OutputStruct10构体 参考

classicLidarDetectorRenderPointCloud 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.317.1 描述

classicLidarDetectorRenderPointCloud 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.318 OutputStruct11构体 参考

classicLidarDetectorSwapXAndY 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.318.1 描述

classicLidarDetectorSwapXAndY 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.319 OutputStruct12构体 参考

classicLidarDetectorSwapYAndX 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.319.1 描述

classicLidarDetectorSwapYAndX 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.320 OutputStruct13构体 参考

classicLidarDetectorNumPoints 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.320.1 描述

classicLidarDetectorNumPoints 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.321 OutputStruct14构体 参考

classicLidarDetectorGetViewer 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::visualization::PCLVisualizer::Ptr **output**

6.321.1 描述

classicLidarDetectorGetViewer 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.322 OutputStruct15构体 参考

classicLidarDetectorClipPlane 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **output**

6.322.1 描述

classicLidarDetectorClipPlane 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.323 OutputStruct16构体 参考

classicLidarDetectorFilterCloud 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **output**

6.323.1 描述

classicLidarDetectorFilterCloud 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.324 OutputStruct17构体 参考

classicLidarDetectorRmvNearPoints 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **output**

6.324.1 描述

classicLidarDetectorRmvNearPoints 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.325 OutputStruct18构体 参考

classicLidarDetectorRANSAC3d 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- pcl::PointCloud< pcl::PointXYZI >::Ptr **output**

6.325.1 描述

classicLidarDetectorRANSAC3d 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.326 OutputStruct19构体 参考

classicLidarDetectorClustering 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- std::vector< pcl::PointCloud< pcl::PointXYZI >::Ptr > **output**

6.326.1 描述

classicLidarDetectorClustering 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.327 OutputStruct20构体 参考

classicLidarDetectorGridGroundFilterSimple 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- std::vector< int > **output**

6.327.1 描述

classicLidarDetectorGridGroundFilterSimple 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.328 OutputStruct21构体 参考

classicLidarDetectorRenderBox 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.328.1 描述

classicLidarDetectorRenderBox 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.329 OutputStruct22构体 参考

classicLidarDetectorRenderPcaBox 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.329.1 描述

classicLidarDetectorRenderPcaBox 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.330 OutputStruct23构体 参考

classicLidarDetectoRun 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- std::pair< std::vector< int >, std::vector< Frame > > **output**

6.330.1 描述

classicLidarDetectoRun 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.331 OutputStruct24构体 参考

classicLidarDetectoTest 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

6.331.1 描述

classicLidarDetectoTest 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.332 OutputStruct25构体 参考

classicLidarDetectoGridRANSAC 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- std::vector< int > **output**

6.332.1 描述

classicLidarDetectoGridRANSAC 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.333 OutputStruct26构体 参考

classicLidarDetectoGridRANSAC 函数件出的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

Public 属性

- std::vector< int > **output**

6.333.1 描述

classicLidarDetectoGridRANSAC 函数件出的参数， 用struct 封装

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.334 patchwork::Params构体 参考

Public 属性

- bool **verbose**
- bool **enable_RNR**
- bool **enable_RVPF**
- bool **enable_TGR**
- int **num_iter**
- int **num_lpr**
- int **num_min_pts**
- int **num_zones**
- int **num_rings_of_interest**
- double **RNR_ver_angle_thr**
- double **RNR_intensity_thr**
- double **sensor_height**
- double **th_seeds**
- double **th_dist**
- double **th_seeds_v**
- double **th_dist_v**
- double **max_range**
- double **min_range**
- double **uprightness_thr**
- double **adaptive_seed_selection_margin**
- double **intensity_thr**
- vector< int > **num_sectors_each_zone**
- vector< int > **num_rings_each_zone**
- int **max_flatness_storage**
- int **max_elevation_storage**
- vector< double > **elevation_thr**
- vector< double > **flatness_thr**

构体的文档由以下文件生成:

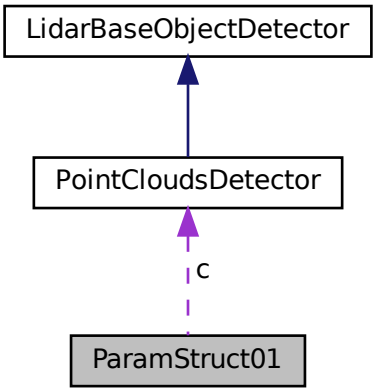
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/patchworkpp.h

6.335 ParamStruct01构体 参考

classicLidarDetectorFind_Z_value 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct01 的作:



Public 属性

- PointCloudsDetector **c**

6.335.1 描述

classicLidarDetectorFind_Z_value 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

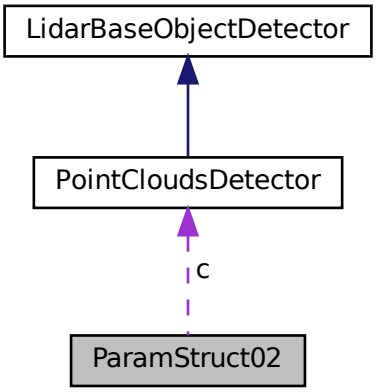
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.336 ParamStruct02构体 参考

classicLidarDetectorPassThroughFilter 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct02 的作:



Public 属性

- PointCloudsDetector **c**

6.336.1 描述

classicLidarDetectorPassThroughFilter 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

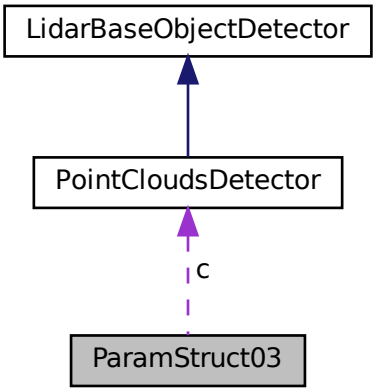
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.337 ParamStruct03构体 参考

classicLidarDetectorPointcloud_to_grid 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct03 的作:



Public 属性

- PointCloudsDetector **c**

6.337.1 描述

classicLidarDetectorPointcloud_to_grid 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

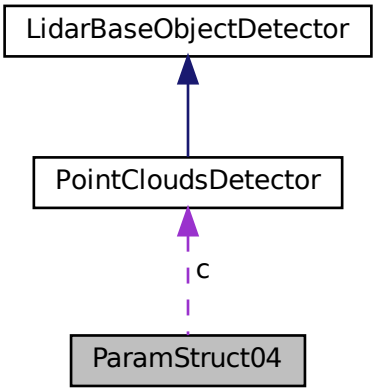
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.338 ParamStruct04构体 参考

classicLidarDetectorInputcloud 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct04 的作:



Public 属性

- PointCloudsDetector **c**

6.338.1 描述

classicLidarDetectorInputcloud 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

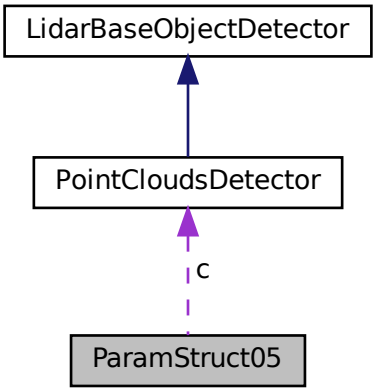
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.339 ParamStruct05构体 参考

classicLidarDetectorInitCamera 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct05 的作:



Public 属性

- PointCloudsDetector **c**

6.339.1 描述

classicLidarDetectorInitCamera 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

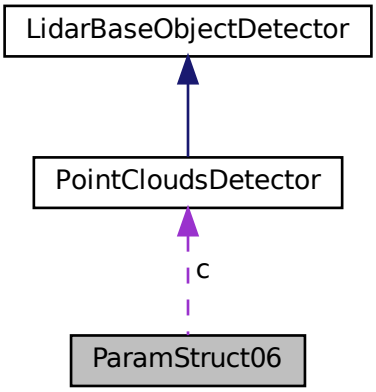
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.340 ParamStruct06构体 参考

classicLidarDetectorFind_Z_value 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct06 的作:



Public 属性

- PointCloudsDetector **c**

6.340.1 描述

classicLidarDetectorFind_Z_value 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

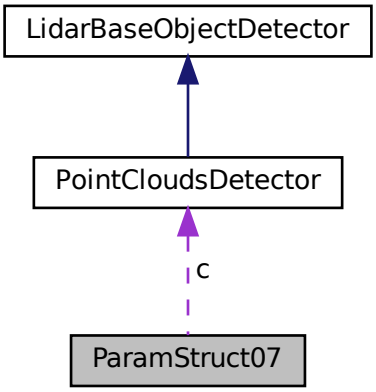
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.341 ParamStruct07构体 参考

classicLidarDetectorCheckAlive 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct07 的作:



Public 属性

- PointCloudsDetector **c**

6.341.1 描述

classicLidarDetectorCheckAlive 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

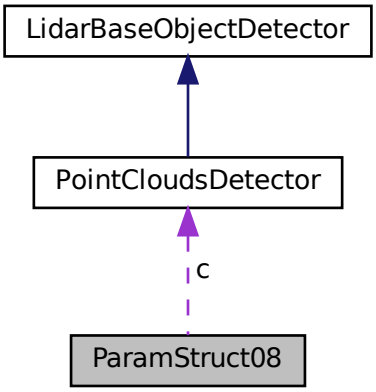
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.342 ParamStruct08构体 参考

classicLidarDetectorReceiveCloud 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct08 的作:



Public 属性

- PointCloudsDetector * **c**

6.342.1 描述

classicLidarDetectorReceiveCloud 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

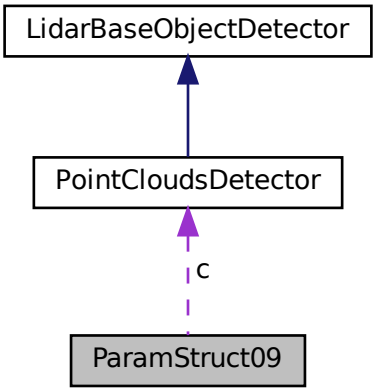
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.343 ParamStruct09构体 参考

classicLidarDetectorRPointCloud 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct09 的作:



Public 属性

- PointCloudsDetector **c**

6.343.1 描述

classicLidarDetectorRPointCloud 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

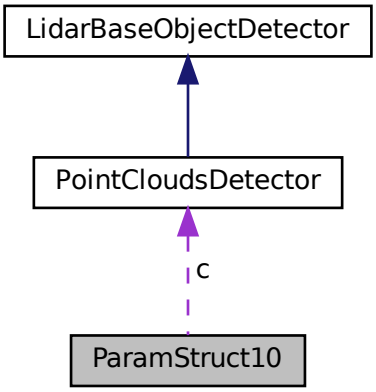
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.344 ParamStruct10构体 参考

classicLidarDetectorRenderPointCloud 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct10 的作:



Public 属性

- PointCloudsDetector * **c**

6.344.1 描述

classicLidarDetectorRenderPointCloud 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

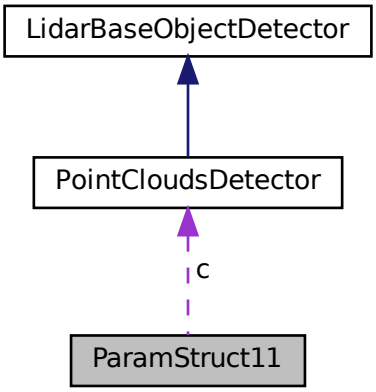
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.345 ParamStruct11构体 参考

classicLidarDetectorSwapXAndY 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct11 的作:



Public 属性

- PointCloudsDetector **c**

6.345.1 描述

classicLidarDetectorSwapXAndY 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

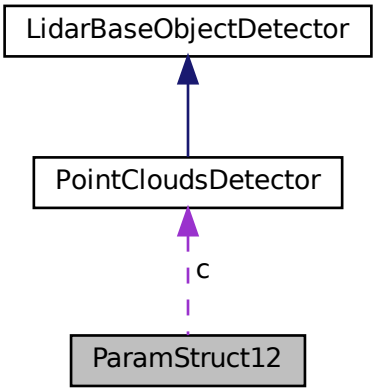
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.346 ParamStruct12构体 参考

classicLidarDetectorSwapYAndX 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct12 的作:



Public 属性

- PointCloudsDetector **c**

6.346.1 描述

classicLidarDetectorSwapYAndX 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

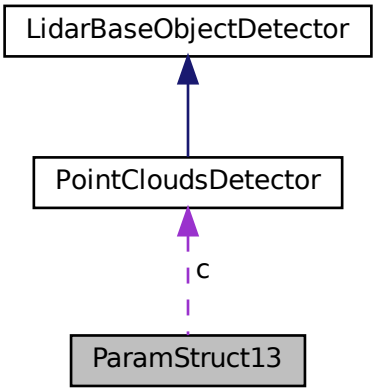
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.347 ParamStruct13构体 参考

classicLidarDetectorNumPoints 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct13 的作:



Public 属性

- PointCloudsDetector **c**

6.347.1 描述

classicLidarDetectorNumPoints 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

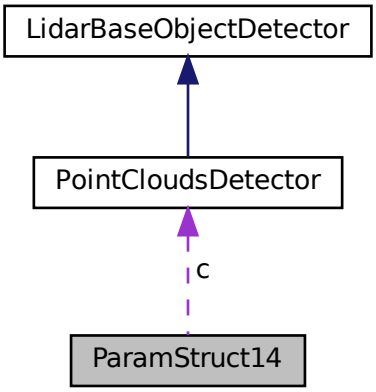
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.348 ParamStruct14构体 参考

classicLidarDetectorGetViewer 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct14 的作:



Public 属性

- PointCloudsDetector **c**

6.348.1 描述

classicLidarDetectorGetViewer 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

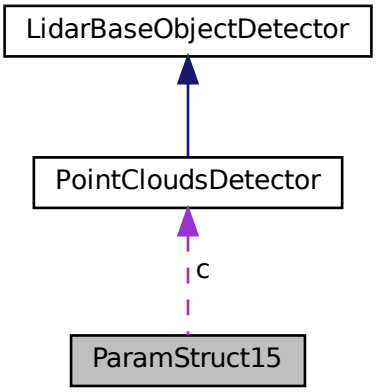
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.349 ParamStruct15构体 参考

classicLidarDetectorClipPlane 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct15 的作:



Public 属性

- PointCloudsDetector **c**

6.349.1 描述

classicLidarDetectorClipPlane 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

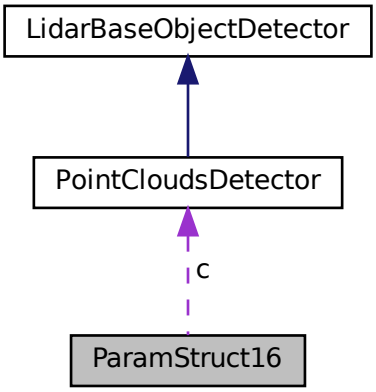
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.350 ParamStruct16构体 参考

classicLidarDetectorFilterCloud 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct16 的作:



Public 属性

- PointCloudsDetector * **c**

6.350.1 描述

classicLidarDetectorFilterCloud 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

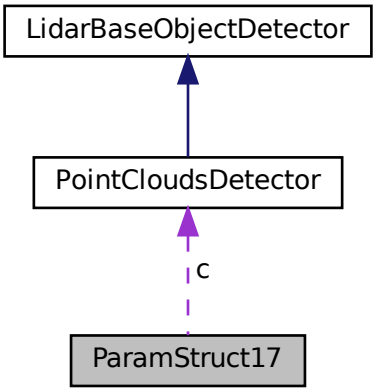
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.351 ParamStruct17构体 参考

classicLidarDetectorRmvNearPoints 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct17 的作:



Public 属性

- PointCloudsDetector **c**

6.351.1 描述

classicLidarDetectorRmvNearPoints 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

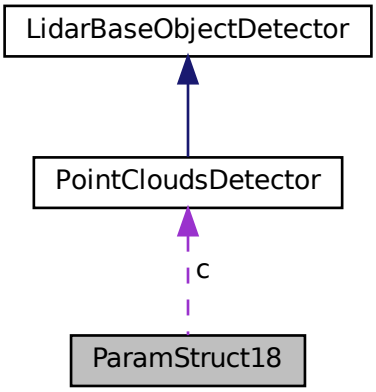
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.352 ParamStruct18构体 参考

classicLidarDetectorRANSAC3d 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct18 的作:



Public 属性

- PointCloudsDetector **c**

6.352.1 描述

classicLidarDetectorRANSAC3d 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

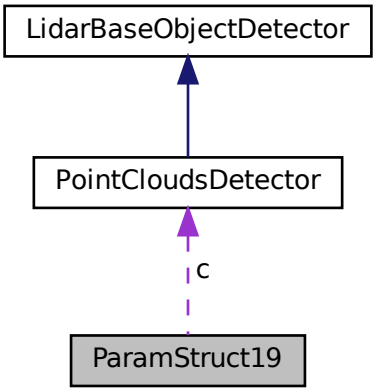
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.353 ParamStruct19构体 参考

classicLidarDetectorClustering 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct19 的作:



Public 属性

- PointCloudsDetector **c**

6.353.1 描述

classicLidarDetectorClustering 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

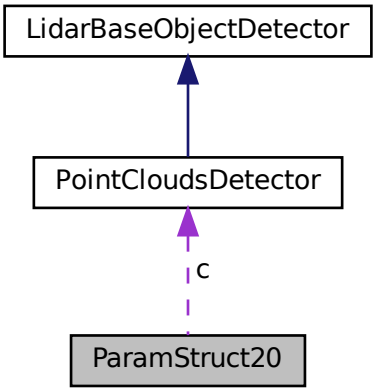
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.354 ParamStruct20构体 参考

classicLidarDetectorGridGroundFilter 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct20 的作:



Public 属性

- PointCloudsDetector * **c**

6.354.1 描述

classicLidarDetectorGridGroundFilter 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

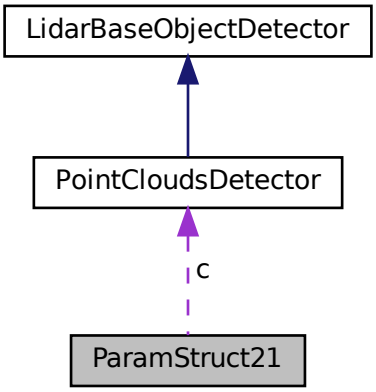
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.355 ParamStruct21构体 参考

classicLidarDetectorRenderBox 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct21 的作:



Public 属性

- PointCloudsDetector **c**

6.355.1 描述

classicLidarDetectorRenderBox 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

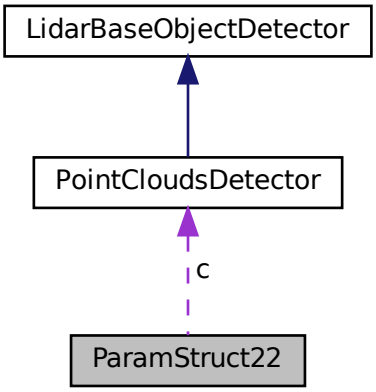
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.356 ParamStruct22构体 参考

classicLidarDetectorRenderPcaBox 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct22 的作:



Public 属性

- PointCloudsDetector **c**

6.356.1 描述

classicLidarDetectorRenderPcaBox 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

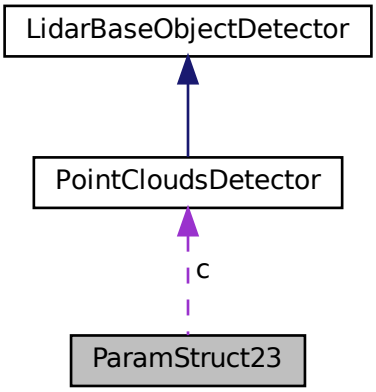
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/functionLidarDetector.h

6.357 ParamStruct23构体 参考

classicLidarDetectoRun 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct23 的作:



Public 属性

- PointCloudsDetector * **c**

6.357.1 描述

classicLidarDetectoRun 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

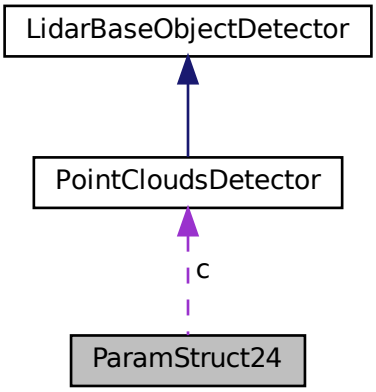
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.358 ParamStruct24构体 参考

classicLidarDetectoTest 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct24 的作:



Public 属性

- PointCloudsDetector **c**

6.358.1 描述

classicLidarDetectoTest 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

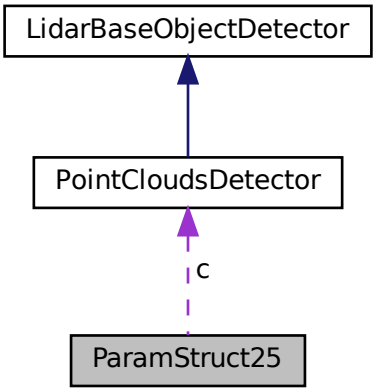
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩coms_livox/include/functionLidarDetector.h

6.359 ParamStruct25构体 参考

classicLidarDetectoGridRANSAC 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct25 的作:



Public 属性

- PointCloudsDetector * **c**

6.359.1 描述

classicLidarDetectoGridRANSAC 函数件可以配置的参数， 用struct 封装

构体的文档由以下文件生成:

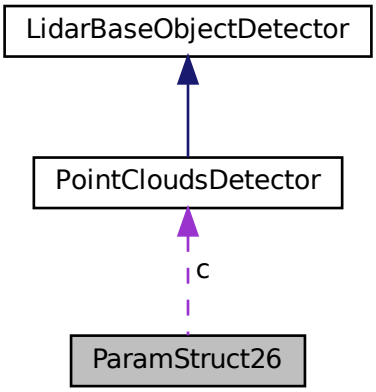
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

6.360 ParamStruct26构体 参考

classicLidarDetectoGridRANSAC 函数件可以配置的参数， 用struct 封装

```
#include <functionLidarDetector.h>
```

ParamStruct26 的作:



Public 属性

- PointCloudsDetector * **c**

6.360.1 描述

classicLidarDetectoGridRANSAC 函数件可以配置的参数， 用struct 封装

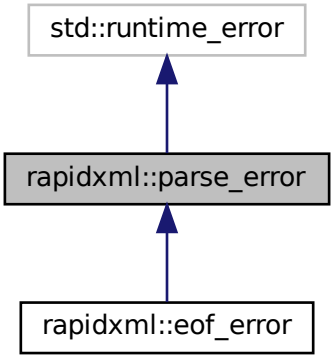
构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/functionLidarDetector.h

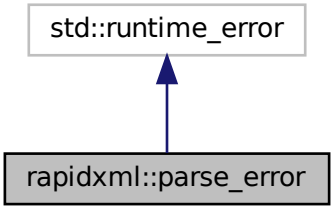
6.361 rapidxml::parse_error 参考

```
#include <rapidxml.hpp>
```

rapidxml::parse_error 承系:



rapidxml::parse_error 的作:



Public 成函数

- parse_error (const char *what, void *where)

  Constructs parse error.
- template<class Ch >
  Ch * where () const

6.361.1 描述

Parse error exception. This exception is thrown by the parser when an error occurs. Use what() function to get human-readable error message. Use where() function to get a pointer to position within source text where error was detected.

If throwing exceptions by the parser is undesirable, it can be disabled by defining RAPIDXML_NO_EXCEPT↩ IONS macro before rapidxml.hpp is included. This will cause the parser to call rapidxml::parse_error_handler() function instead of throwing an exception. This function must be defined by the user.

This class derives from `std::exception` class.

6.361.2 成函数明

6.361.2.1 where()

```
template<class Ch >
Ch* rapidxml::parse_error::where ( ) const  [inline]
```

Gets pointer to character data where error happened. Ch should be the same as char type of xml_document that produced the error.

返回

Pointer to location within the parsed string where error occured.

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml.hpp

6.362 patchwork::PatchWorkpp 参考

Public 型

- typedef std::vector< vector< PointXYZ > > **Ring**
- typedef std::vector< Ring > **Zone**

Public 成函数

- **PatchWorkpp** (patchwork::Params _params)
- void **estimateGround** (Eigen::MatrixXf cloud_in)
- double **getHeight** ()
- double **getTimeTaken** ()
- Eigen::MatrixX3f **getGround** ()
- Eigen::MatrixX3f **getNonground** ()
- Eigen::VectorXi **getGroundIndices** ()
- Eigen::VectorXi **getNongroundIndices** ()
- Eigen::MatrixX3f **getCenters** ()
- Eigen::MatrixX3f **getNormals** ()

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/include/patchworkpp.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_coms_livox/src/patchworkpp.cpp

6.363 PcaBox构体 参考

Public 属性

- Eigen::Vector3f **center**
- Eigen::Quaternionf **rotation**
- float **boxLength**
- float **boxWidth**
- float **boxHeight**
- pcl::PointXYZI **minPoint**
- pcl::PointXYZI **maxPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/classicLidarDetector.h

6.364 Control::PID 参考

PID 控制器

```
#include <controlPIDController.hpp>
```

Public 成函数

- PID (double kp, double ki, double kd, double dt, bool isThreshOn, double errorThresh, double dThresh, bool isdThreshOn, double w)

  Construct a new PID::PID object.
- double **update** (double error)
- double **output** ()
- void **modifyGains** (double kp, double ki, double kd)
- void **resetI** ()

6.364.1 描述

PID 控制器

6.364.2 构造及析构函数明

6.364.2.1 PID()

```
Control::PID::PID (
            double kp,
            double ki,
            double kd,
            double dt,
            bool isThreshOn,
            double errorThresh,
            double dThresh,
            bool isdThreshOn,
            double w = 1 )
```

Construct a new PID::PID object.

参数

kp	
ki	
kd	
dt	控制器周期
isThreshOn	是否打分器
errorThresh	分器上限

的文档由以下文件生成:

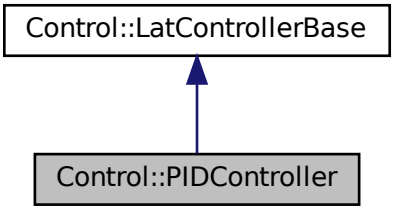
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlPIDCon
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/controlPIDControll

6.365 Control::PIDController 参考

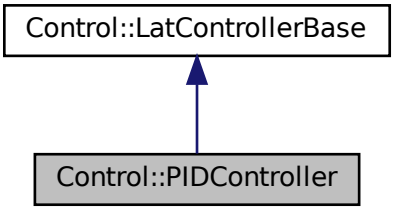
PID横向运控制器

```
#include <controlPIDController.hpp>
```

Control::PIDController 承系:



Control::PIDController 的作:



Public 成函数

- PIDController (std::string configFile, double maxSteer, double ratioWheelToSteer=20, double dt=0.1)

  Construct a new PIDController::PIDController object.
- double calcSteer (State s, const Traj &traj, size_t nearestID)

  算前角
- double calcFeedForwardAngle (State state, const Traj &traj, int nID, int rID)

  根据追踪，算前角，前。推*doc*

外承的成函数

6.365.1 描述

PID横向运控制器

承横向控制虚基LatControllerBase。通前+反生成控制量，算法原理 文档。

参

controlPIDController.cc 横向控制PID的具体方法

6.365.2 构造及析构函数明

6.365.2.1 PIDController()

```
Control::PIDController::PIDController (
            std::string configFile,
            double maxSteer,
            double ratioWheelToSteer = 20,
            double dt = 0.1 )
```

Construct a new PIDController::PIDController object.

参数

configFile	config file name. 控制配置文件地址

6.365.3 成函数明

6.365.3.1 calcFeedForwardAngle()

```
double Control::PIDController::calcFeedForwardAngle (
            State state,
```

```
            const Traj & traj,
            int nID,
            int pID )
```

根据追踪，算前角，前。推doc

参数

state	状
traj	待跟踪迹
nID	当前点ID
pID	瞄点ID

返回

double delta: 前角

6.365.3.2 calcSteer()

```
double Control::PIDController::calcSteer (
            State s,
            const Traj & traj,
            size_t nearestID )  [virtual]
```

算前角

参数

s	状
traj	待跟踪迹

返回

double steer: 前+反前角

了 Control::LatControllerBase.

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlPIDCon
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/controlPIDControll

6.366 LocalPlanning::PlanningPoint构体 参考

Public 属性

- double **x**
- double **y**
- double **angle**
- double **gaussX**
- double **gaussY**
- double **gaussAngle**
- double **s**
- double **l**
- double **frenetAngle**
- double **v**
- double **curvature**
- double **accumS**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.367 LocalPlanning::PlanningTrajectory构体 参考

Public 属性

- std::vector< PlanningPoint > **planningPoints**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.368 geometry::point_t构体 参考

Public 属性

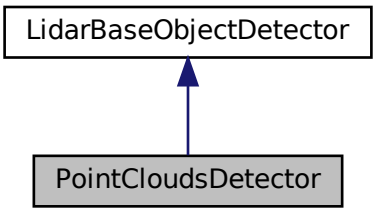
- double **x**
- double **y**

构体的文档由以下文件生成:

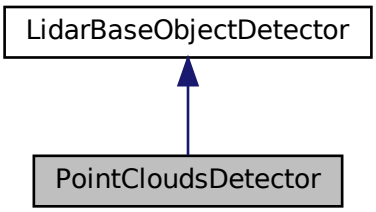
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩ _m.h

6.369 PointCloudsDetector 参考

PointCloudsDetector 承系:



PointCloudsDetector 的作:



- struct GridInfos

Public 成函数

- void **find_Z_value** (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud)
- pcl::PointCloud< pcl::PointXYZI >::Ptr **PassThroughFilter** (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud, double z_max, double z_min)
- vector< int > **Pointcloud_to_grid** (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud, double map_resolution)
- pcl::PointCloud< pcl::PointXYZI >::Ptr **inputcloud** (int i)
- void **initCamera** (CameraAngle setAngle)
- void **initView** ()
- bool **checkAlive** ()
- void **receiveCloud** (pcl::PointCloud< pcl::PointXYZI >::Ptr input) override
- void **rPointCloud** (const pcl::PointCloud< pcl::PointXYZI >::Ptr &cloud)
- void **renderPointCloud** (const pcl::PointCloud< pcl::PointXYZI >::Ptr &cloud, std::string name, Color color)

- Box **setBoundingBox** (pcl::PointCloud< pcl::PointXYZI >::Ptr cluster)
- PcaBox **setPcaBox** (Frame frame)
- Frame **object** (pcl::PointCloud< pcl::PointXYZI >::Ptr cluster, double trackingDis)
- std::vector< Frame > **ObstacleBounding** (std::vector< pcl::PointCloud< pcl::PointXYZI >::Ptr > cluster↩ Cloud, double trackingDis)
- void **renderBox** (Box box, int id, Color color=Color(0, 1, 0), float opacity=1)
- void **renderPcaBox** (PcaBox box, int id, Color color=Color(0, 1, 0), float opacity=1)
- void **swapXAndY** (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud)
- void **swapYAndX** (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud)
- void **numPoints** (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud)
- pcl::visualization::PCLVisualizer::Ptr **getViewer** () override
- pcl::PointCloud< pcl::PointXYZI >::Ptr **clipPlane** (pcl::PointCloud< pcl::PointXYZI >::Ptr src_cloud, const Eigen::Vector4f plane, bool negative)
- pcl::PointCloud< pcl::PointXYZI >::Ptr **filterCloud** (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud, float filterRes_x, float filterRes_y, float filterRes_z, Eigen::Vector4f minPoint, Eigen::Vector4f maxPoint)
- pcl::PointCloud< pcl::PointXYZI >::Ptr **rmvNearPoints** (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud, Eigen::Vector4f minPoint, Eigen::Vector4f maxPoint)
- pcl::PointCloud< pcl::PointXYZI >::Ptr **RANSAC3d** (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud, int max↩ Iterations, float distanceThreshold)
- std::vector< pcl::PointCloud< pcl::PointXYZI >::Ptr > **Clustering** (pcl::PointCloud< pcl::PointXYZI >::Ptr cloud, float clusterTolerance, int minSize, int maxSize)
- std::vector< int > **GridGroundFilter** (const pcl::PointCloud< pcl::PointXYZI >::Ptr filtered_cloud, int normal_estimation_Ksearch, float slope_rate, float normal_thereshold, int grid_height_mode, float car_height, float grid_size, float right_left_distance, float back_distance, float front_distance)
- std::vector< int > **GridGroundFilterSimple** (const pcl::PointCloud< pcl::PointXYZI >::Ptr filtered_cloud, int normal_estimation_Ksearch, float slope_rate, float normal_thereshold, int grid_height_mode, float car_height, float grid_size, float right_left_distance, float back_distance, float front_distance)
- std::vector< int > **GridRANSAC** (const pcl::PointCloud< pcl::PointXYZI >::Ptr filtered_cloud, float ransac_↩ threshold, int normal_estimation_Ksearch, float slope_rate, float normal_thereshold, int grid_height_mode, float car_height, float grid_size, float right_left_distance, float back_distance, float front_distance)
- void **read_pcd_to_eigen** (pcl::PointCloud< pcl::PointXYZI >::Ptr point_cloudi, Eigen::MatrixXf &cloud, float right_left_distance, float back_distance, float front_distance)
- void **processPointClouds** (pcl::PointCloud< pcl::PointXYZI >::Ptr point_cloudi, float right_left_distance, float back_distance, float front_distance)
- std::vector< int > **GridPatchworkpp** (pcl::PointCloud< pcl::PointXYZI >::Ptr point_cloudi, int normal_↩ estimation_Ksearch, float slope_rate, float normal_thereshold, int grid_height_mode, float car_height, float grid_size, float right_left_distance, float back_distance, float front_distance)
- std::pair< std::vector< int >, std::vector< Frame > > **run** () override
- void **test** ()

Public 属性

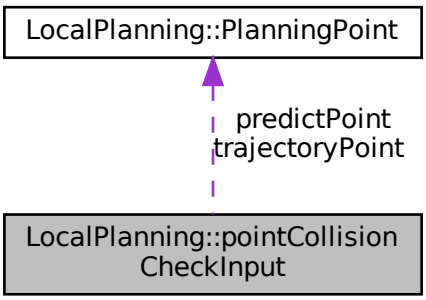
- boost::shared_ptr< pcl::visualization::PCLVisualizer > **viewer**
- double **max_z**
- double **min_z**
- int **wi** = 0
- int **frame_id** = 0
- int **time_id** = 0
- int **id** = 0
- pcl::PointCloud< pcl::PointXYZI >::Ptr **ground_point**
- pcl::PointCloud< pcl::PointXYZI >::Ptr **nonground_point**
- std::vector< Frame > **frames**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/classicLidarDetector.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/src/classicLidarDetector.cpp

6.370 LocalPlanning::pointCollisionCheckInput构体 参考

LocalPlanning::pointCollisionCheckInput 的作:



Public 属性

- PlanningPoint **trajectoryPoint**
- PlanningPoint **predictPoint**
- double **w**
- double **l**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.371 LocalPlanning::pointCollisionCheckOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.372 LocalPlanning::pointCollisionCheckParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.373 LocalPlanning::pointDistanceInput构体 参考

Public 属性

- double **x1**
- double **x2**
- double **y1**
- double **y2**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.374 LocalPlanning::pointDistanceOutput构体 参考

Public 属性

- double **distance**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.375 LocalPlanning::pointDistanceParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.376 LocalPlanning::pointOnCubicBezierInput构体 参考

Public 属性

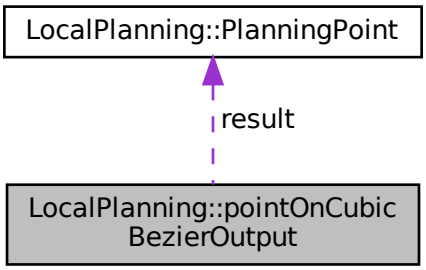
- std::vector< PlanningPoint > **cp**
- double **t**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.377 LocalPlanning::pointOnCubicBezierOutput构体 参考

LocalPlanning::pointOnCubicBezierOutput 的作:



Public 属性

- PlanningPoint **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

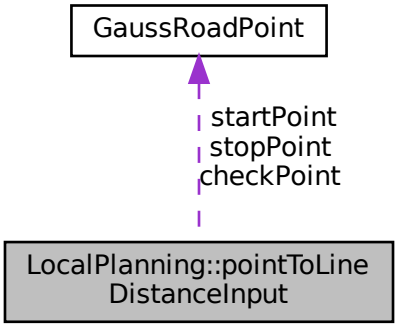
6.378 LocalPlanning::pointOnCubicBezierParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.379 LocalPlanning::pointToLineDistanceInput构体 参考

LocalPlanning::pointToLineDistanceInput 的作:



Public 属性

- GaussRoadPoint **startPoint**
- GaussRoadPoint **stopPoint**
- GaussRoadPoint **checkPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.380 LocalPlanning::pointToLineDistanceOutput构体 参考

Public 属性

- double **dis**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

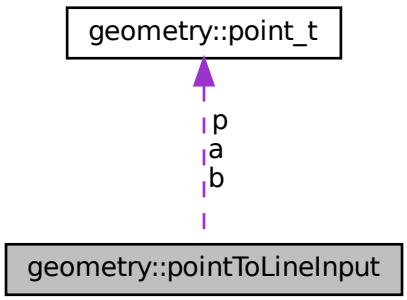
6.381 LocalPlanning::pointToLineDistanceParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.382 geometry::pointToLineInput构体 参考

geometry::pointToLineInput 的作:



Public 属性

- point_t **a**
- point_t **b**
- point_t **p**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.383 geometry::pointToLineOutput构体 参考

Public 属性

- double **value**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

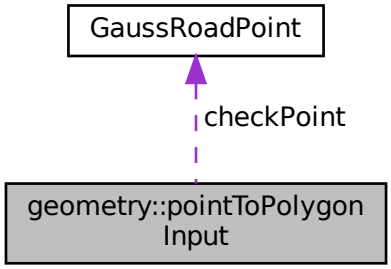
6.384 geometry::pointToLineParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.385 geometry::pointToPolygonInput构体 参考

geometry::pointToPolygonInput 的作:



Public 属性

- std::vector< GaussRoadPoint > **pointPolygon**
- GaussRoadPoint **checkPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.386 geometry::pointToPolygonOutput构体 参考

Public 属性

- double **dis**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.387 geometry::pointToPolygonParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

6.388 patchwork::PointXYZ构体 参考

Public 成函数

- **PointXYZ** (float _x, float _y, float _z, int _idx=-1)

Public 属性

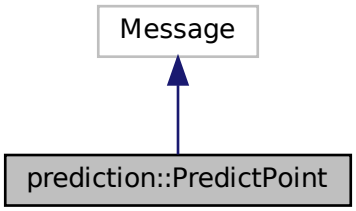
- float **x**
- float **y**
- float **z**
- int **idx**

构体的文档由以下文件生成:

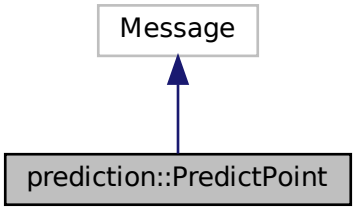
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/patchworkpp.h

6.389 prediction::PredictPoint 参考

prediction::PredictPoint 承系:



prediction::PredictPoint 的作:



Public 成函数

- **PredictPoint** (const PredictPoint &from)
- PredictPoint & **operator=** (const PredictPoint &from)
- void **Swap** (PredictPoint *other)
- PredictPoint * **New** () const final
- PredictPoint * **New** (::google::protobuf::Arena *arena) const final
- void **CopyFrom** (const ::google::protobuf::Message &from) final
- void **MergeFrom** (const ::google::protobuf::Message &from) final
- void **CopyFrom** (const PredictPoint &from)
- void **MergeFrom** (const PredictPoint &from)
- void **Clear** () final
- bool **IsInitialized** () const final
- size_t **ByteSizeLong** () const final
- bool **MergePartialFromCodedStream** (::google::protobuf::io::CodedInputStream *input) final
- void **SerializeWithCachedSizes** (::google::protobuf::io::CodedOutputStream *output) const final
- ::google::protobuf::uint8 * **InternalSerializeWithCachedSizesToArray** (bool deterministic, ::google↩ ::protobuf::uint8 *target) const final
- int **GetCachedSize** () const final
- ::google::protobuf::Metadata **GetMetadata** () const final
- void **clear_x** ()
- double **x** () const
- void **set_x** (double value)
- void **clear_y** ()
- double **y** () const
- void **set_y** (double value)
- void **clear_vx** ()
- double **vx** () const
- void **set_vx** (double value)
- void **clear_vy** ()
- double **vy** () const
- void **set_vy** (double value)

静 Public 成函数

- static const ::google::protobuf::Descriptor * **descriptor** ()
- static const PredictPoint & **default_instance** ()
- static void **InitAsDefaultInstance** ()
- static const PredictPoint * **internal_default_instance** ()

静 Public 属性

- static constexpr int **kIndexInFileMessages**
- static const int **kXFieldNumber** = 1
- static const int **kYFieldNumber** = 2
- static const int **kVxFieldNumber** = 3
- static const int **kVyFieldNumber** = 4

友元

- struct **::protobuf_prediction_2eproto::TableStruct**
- void **swap** (PredictPoint &a, PredictPoint &b)

6.389.1 成量明

6.389.1.1 kIndexInFileMessages

```
constexpr int prediction::PredictPoint::kIndexInFileMessages  [static], [constexpr]
```

初始:
=
0

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩coms_livox/include/prediction.pb.h

6.390 LocalPlanning::processTrafficLightInput构体 参考

LocalPlanning::processTrafficLightInput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩Planning_m.hpp

6.391 LocalPlanning::processTrafficLightOutput构体 参考

LocalPlanning::processTrafficLightOutput 的作:

Public 属性

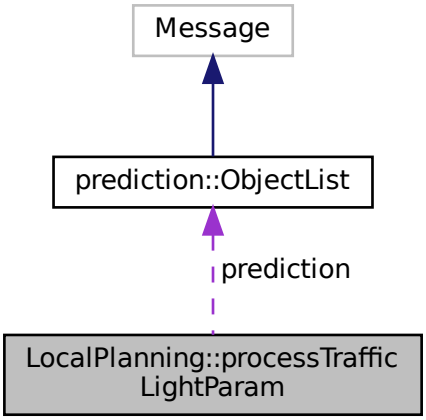
- DecisionData **decisionData**
- int **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.392 LocalPlanning::processTrafficLightParam构体 参考

LocalPlanning::processTrafficLightParam 的作:



Public 属性

- prediction::ObjectList **prediction**
- infopack::TrafficLight **trafficLight**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.393 LocalPlanning::ReferenceLine构体 参考

Public 属性

- std::vector< PlanningPoint > **referenceLinePoints**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.394 LocalPlanning::restartInput构体 参考

Public 属性

- double **velocity**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.395 LocalPlanning::restartOutput构体 参考

Public 属性

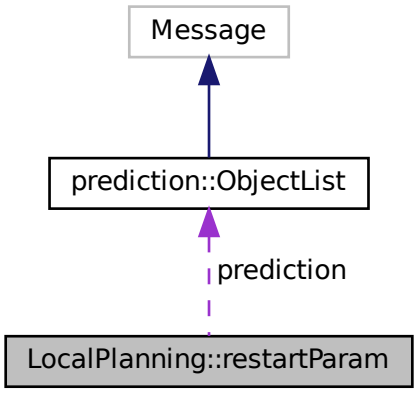
- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.396 LocalPlanning::restartParam构体 参考

LocalPlanning::restartParam 的作:



Public 属性

- prediction::ObjectList **prediction**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.397 patchwork::RevertCandidate构体 参考

Public 成函数

- **RevertCandidate** (int _c_idx, int _s_idx, double _flatness, double _line_var, Eigen::VectorXf _pc_mean, vector< PointXYZ > _ground)

Public 属性

- int **concentric_idx**
- int **sector_idx**
- double **ground_flatness**
- double **line_variable**
- Eigen::VectorXf **pc_mean**
- vector< PointXYZ > **regionwise_ground**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/patchworkpp.h

6.398 Road 参考

Public 属性

- int **id**
- bool **isLaneChange**
- std::vector< Lane > **lanes**
- std::vector< int > **successorId**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h

6.399 GolbalPlanning::road构体 参考

Public 属性

- int **number**
- double **xBegin**
- double **yBegin**
- double **xEnd**
- double **yEnd**
- double **length**
- std::vector< int > **to**
- int **father**
- int **isInList** = -2
- double **F**
- double **G**
- double **H**
- int **numLanes**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

6.400 RoadMap 参考

- struct mapAnalysisInput
- struct mapAnalysisOutput
- struct mapAnalysisParam

Public 成函数

- **RoadMap** (std::string path)
- void **mapAnalysis** (std::string path)
- void **neighborLaneSort** ()
- void **moduleSelfCheck** ()
- void **moduleSelfCheckPrint** ()
- bool **GetRoadByRoadID** (int roadID, Road &road)
- bool **GetLaneByLaneID** (int laneID, Road road, Lane &lane)
- bool **GetStartPointByRoadID** (int roadID, double &GaussX, double &GaussY)
- bool **GetEndPointByRoadID** (int roadID, double &GaussX, double &GaussY)
- **RoadMap** (std::string path)
- void **mapAnalysis** (const mapAnalysisParam &param, const mapAnalysisInput &input, const mapAnalysisOutput &output)
- void **neighborLaneSort** ()
- void **moduleSelfCheck** ()
- void **moduleSelfCheckPrint** ()
- bool **GetRoadByRoadID** (int roadID, Road &road)
- bool **GetLaneByLaneID** (int laneID, Road road, Lane &lane)
- bool **GetStartPointByRoadID** (int roadID, double &GaussX, double &GaussY)
- bool **GetEndPointByRoadID** (int roadID, double &GaussX, double &GaussY)

Public 属性

- std::vector< Road > **roads**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis.cpp
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/localization_Map↩ Analysis_m.cpp

6.401 Control::Spline 参考

Public 成函数

- **Spline** (Vec_d x_, Vec_d y_)
- double **calc** (double t)
- double **calcD** (double t)
- double **calcDD** (double t)

Public 属性

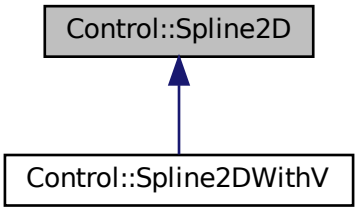
- Vec_d **x**
- Vec_d **y**
- int **nx**
- Vec_d **h**
- Vec_d **a**
- Vec_d **b**
- Vec_d **c**
- Vec_d **d**

的文档由以下文件生成:

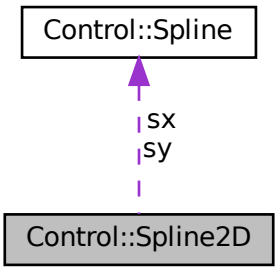
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlCubicSp

6.402 Control::Spline2D 参考

Control::Spline2D 承系:



Control::Spline2D 的作:



Public 成函数

- **Spline2D** (Vec_d x, Vec_d y)
- Poi_d **calcPosition** (double s_t)
- double **calcCurvature** (double s_t)
- double **calcYaw** (double s_t)

Public 属性

- Spline **sx**
- Spline **sy**
- Vec_d **s**

Protected 成函数

- Vec_d **calcS** (Vec_d x, Vec_d y)

的文档由以下文件生成:

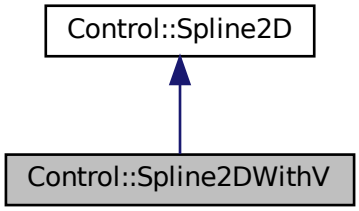
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlCubicSp

6.403 Control::Spline2DWithV 参考

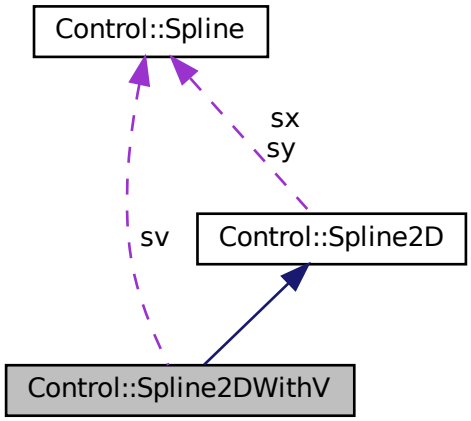
TrajRaw -> Traj。插所需

```
#include <controlCubicSpline.h>
```

Control::Spline2DWithV 承系:



Control::Spline2DWithV 的作:



Public 成函数

- **Spline2DWithV** (Vec_d x, Vec_d y, Vec_d v)
- double **calcV** (double s_t)

Public 属性

- Spline **sv**

外承的成函数

6.403.1 描述

TrajRaw -> Traj。插所需

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlCubicSp

6.404 Control::State构体 参考

入位置信息

```
#include <controlData.hpp>
```

Public 属性

- double **x**
- double **y**
- double **yaw**
- double **v**
- double **rtkMode**

6.404.1 描述

入位置信息

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlData.hp

6.405 State构体 参考

Public 属性

- double **x**
- double **y**
- double **yaw**
- double **v**
- double **rtkMode**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/control↩ Transfer.cpp

6.406 protobuf_control_2eproto::StaticDescriptorInitializer构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/control.pb.cc

6.407 protobuf_imu_2eproto::StaticDescriptorInitializer构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/imu.pb.cc

6.408 SteeringController 参考

Public 成函数

- void **setPath** ()
- void **calculateSteering** ()
- double **getSteering** ()

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/steeringController.h

6.409 LocalPlanning::stopInput构体 参考

LocalPlanning::stopInput 的作:

Public 属性

- DecisionData **decisionData**
- double **velocity**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.410 LocalPlanning::stopOutput构体 参考

LocalPlanning::stopOutput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.411 LocalPlanning::stopParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.412 LocalPlanning::stopPointJudgeInput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.413 LocalPlanning::stopPointJudgeOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.414 LocalPlanning::stopPointJudgeParam构体 参考

Public 属性

- pc::Imu **imu**
- std::vector< GaussRoadPoint > **stopPoints**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.415 protobuf_imu_2eproto::TableStruct构体 参考

静 Public 属性

- static const ::google::protobuf::internal::ParseTableField **entries** [ ]
- static const ::google::protobuf::internal::AuxillaryParseTableField **aux** [ ]
- static const ::google::protobuf::internal::ParseTable **schema** [1]
- static const ::google::protobuf::internal::FieldMetadata **field_metadata** [ ]
- static const ::google::protobuf::internal::SerializationTable **serialization_table** [ ]
- static const ::google::protobuf::uint32 **offsets** [ ]

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/imu.pb.h

6.416 protobuf_control_2eproto::TableStruct构体 参考

静 Public 属性

- static const ::google::protobuf::internal::ParseTableField **entries** [ ]
- static const ::google::protobuf::internal::AuxillaryParseTableField **aux** [ ]
- static const ::google::protobuf::internal::ParseTable **schema** [2]
- static const ::google::protobuf::internal::FieldMetadata **field_metadata** [ ]
- static const ::google::protobuf::internal::SerializationTable **serialization_table** [ ]
- static const ::google::protobuf::uint32 **offsets** [ ]

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/control.pb.h

6.417 protobuf_prediction_2eproto::TableStruct构体 参考

静 Public 属性

- static const ::google::protobuf::internal::ParseTableField **entries** [ ]
- static const ::google::protobuf::internal::AuxillaryParseTableField **aux** [ ]
- static const ::google::protobuf::internal::ParseTable **schema** [3]
- static const ::google::protobuf::internal::FieldMetadata **field_metadata** [ ]
- static const ::google::protobuf::internal::SerializationTable **serialization_table** [ ]
- static const ::google::protobuf::uint32 **offsets** [ ]

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/prediction.pb.h

6.418 ThreadJobs 参考

Public 成函数

- pcl::visualization::PCLVisualizer::Ptr **getViewer** ()
- void **recvPointCloud16_0** ()
- void **recvPointCloud16_1** ()
- void **recvPointCloud128** ()
- void **recvImu** ()
- void **processCloudPoints** ()
- long int **cvtThreadId2Long** (thread::id id)
- void **checkThreadStatus** (int tStatus, thread::id id)
- void **recvCallback** ()
- void **sendCallback** ()
- **ThreadJobs** (std::string configFile)
- void subTraj ()

  接收并更新迹
- void recvStateAndPub ()

6.418.1 描述

整个的作用是接收路径和状信息，并算控制命令，然后将命令送出去。 由于路径信息必在状信息之前接收和理，因此使用isTrajInit量来确保路径信息已被接收和理。中使用了一个互斥 （trajMtx），以确保在算控制命令不会未准好的迹信息。

6.418.2 成函数明

6.418.2.1 recvStateAndPub()

```
void ThreadJobs::recvStateAndPub ( ) [inline]
```

函数在一个无限循中运行，它从socketSubState接收状信息，并使用parseStateData()函数将接收到的数据解析状信息， 然后使用calcControlCMD()函数算控制命令，并将命令通socketPub送出去。

6.418.2.2 subTraj()

```
void ThreadJobs::subTraj ( ) [inline]
```

接收并更新迹

函数在一个无限循中运行，它从socketSubTraj接收路径信息，并使用parseTrajData()函数将接收到的数据解析迹信息，然后将isTrajInit置true。 函数在路径信息接收之前等待，因此它需要先收到路径信息，然后才能接收和理状信息。

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/src/perceptionLidarObjectDetection.cpp
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/stdThreadDemo.cpp
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/control↩ MainZMQ.cc

6.419 Frame::Time构体 参考

Public 属性

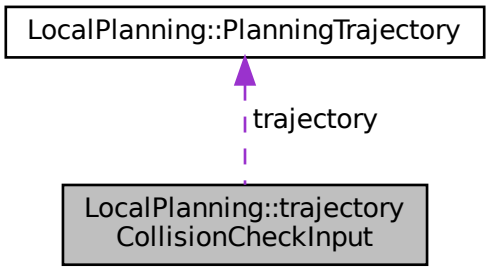
- int **timeId**
- int **objectId**
- int **type**
- double **x**
- double **y**
- double **z**
- double **yaw**
- double **vel**
- double **lidar_x**
- double **lidar_y**
- double **lidar_z**
- double **lidar_yaw**
- double **w**
- double **l**
- double **h**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/baseLidarObjectDetector.h

6.420 LocalPlanning::trajectoryCollisionCheckInput构体 参考

LocalPlanning::trajectoryCollisionCheckInput 的作:



Public 属性

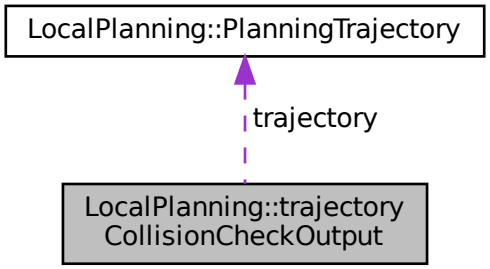
- PlanningTrajectory **trajectory**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.421 LocalPlanning::trajectoryCollisionCheckOutput构体 参考

LocalPlanning::trajectoryCollisionCheckOutput 的作:



Public 属性

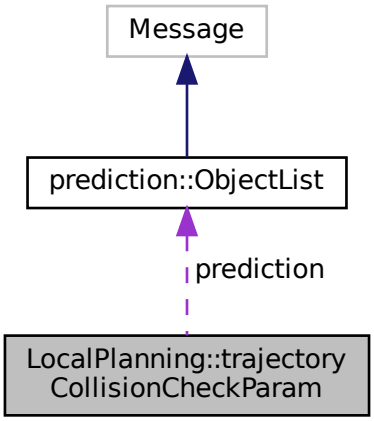
- PlanningTrajectory **trajectory**
- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.422 LocalPlanning::trajectoryCollisionCheckParam构体 参考

LocalPlanning::trajectoryCollisionCheckParam 的作:



Public 属性

- prediction::ObjectList **prediction**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

6.423 TrajPoint构体 参考

Public 属性

- double **x**
- double **y**
- double **yaw**
- double **v**
- double **k**
- double **s**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/control↩ Transfer.cpp

6.424 Control::TrajPoint构体 参考

Public 属性

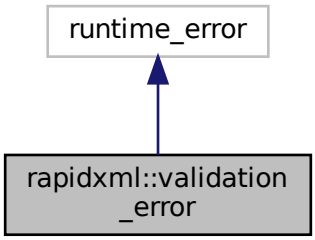
- double **x**
- double **y**
- double **yaw**
- double **v**
- double **k**
- double **s**

构体的文档由以下文件生成:

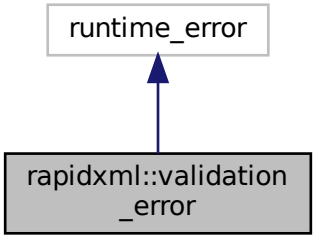
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlData.hp

6.425 rapidxml::validation_error 参考

rapidxml::validation_error 承系:



rapidxml::validation_error 的作:



Public 成函数

- **validation_error** (const char *what)

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml.hpp

6.426 Vect3构体 参考

Public 成函数

- **Vect3** (double setX, double setY, double setZ)
- Vect3 **operator+** (const Vect3 &vec)

Public 属性

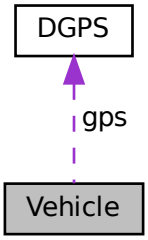
- double **x**
- double **y**
- double **z**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_lidar_↩ coms_livox/include/classicLidarDetector.h

6.427 Vehicle 参考

Vehicle 的作:



Public 成函数

- void **VehicleDynamic** (double &x, double &y, double &theta, double D, double delta)
- double **mod2pi** (double x)

Public 属性

- DGPS **gps**
- double **dBaseLength**
- double **dTargetSteeringAngle**
- double **dTargetSpeed**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/stdThreadDemo.cpp

6.428 rapidxml::xml_attribute< Ch > 模板 参考

```
#include <rapidxml.hpp>
```

Public 成函数

- xml_attribute ()
- xml_document< Ch > * document () const
- Ch * **xmlns** () const
- std::size_t **xmlns_size** () const
- xml_attribute< Ch > * previous_attribute (const Ch *name=0, std::size_t name_size=0, bool case_sensitive=true) const
- xml_attribute< Ch > * next_attribute (const Ch *name=0, std::size_t name_size=0, bool case_sensitive=true) const
- Ch * **local_name** () const
- std::size_t **local_name_size** () const

友元

- class **xml_node< Ch >**

6.428.1 描述

template<class Ch = char>
class rapidxml::xml_attribute< Ch >

Class representing attribute node of XML document. Each attribute has name and value strings, which are available through name() and value() functions (inherited from xml_base). Note that after parse, both name and value of attribute will point to interior of source text used for parsing. Thus, this text must persist in memory for the lifetime of attribute.

参数

Ch	Character type to use.

6.428.2 构造及析构函数明

6.428.2.1 xml_attribute()

```
template<class Ch = char>
rapidxml::xml_attribute< Ch >::xml_attribute ( ) [inline]
```

Constructs an empty attribute with the specified type. Consider using memory_pool of appropriate xml_document if allocating attributes manually.

6.428.3 成函数明

6.428.3.1 document()

```
template<class Ch = char>
xml_document<Ch>* rapidxml::xml_attribute< Ch >::document ( ) const [inline]
```

Gets document of which attribute is a child.

返回

Pointer to document that contains this attribute, or 0 if there is no parent document.

6.428.3.2 next_attribute()

```
template<class Ch = char>
xml_attribute<Ch>* rapidxml::xml_attribute< Ch >::next_attribute (
            const Ch * name = 0,
            std::size_t name_size = 0,
            bool case_sensitive = true ) const  [inline]
```

Gets next attribute, optionally matching attribute name.

参数

name	Name of attribute to find, or 0 to return next attribute regardless of its name; this string doesn't have to be zero-terminated if name_size is non-zero
name_size	Size of name, in characters, or 0 to have size calculated automatically from string
case_sensitive	Should name comparison be case-sensitive; non case-sensitive comparison works properly only for ASCII characters

返回

Pointer to found attribute, or 0 if not found.

6.428.3.3 previous_attribute()

```
template<class Ch = char>
xml_attribute<Ch>* rapidxml::xml_attribute< Ch >::previous_attribute (
            const Ch * name = 0,
            std::size_t name_size = 0,
            bool case_sensitive = true ) const  [inline]
```

Gets previous attribute, optionally matching attribute name.

参数

name	Name of attribute to find, or 0 to return previous attribute regardless of its name; this string doesn't have to be zero-terminated if name_size is non-zero
name_size	Size of name, in characters, or 0 to have size calculated automatically from string
case_sensitive	Should name comparison be case-sensitive; non case-sensitive comparison works properly only for ASCII characters

返回

Pointer to found attribute, or 0 if not found.

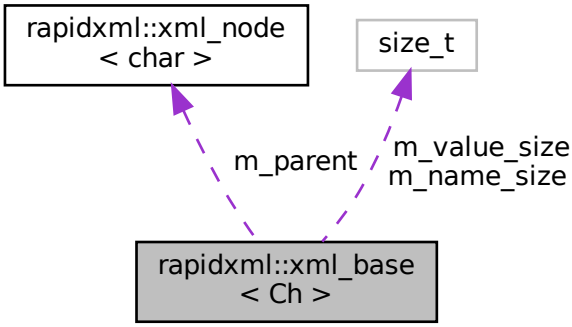
的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml.hpp

6.429 rapidxml::xml_base< Ch > 模板 参考

```
#include <rapidxml.hpp>
```

rapidxml::xml_base< Ch > 的作:



Public 成函数

- Ch ∗ name () const
- std::size_t name_size () const
- Ch ∗ value () const
- std::size_t value_size () const
- void name (const Ch ∗name, std::size_t size)
- void name (const Ch ∗name)
- void value (const Ch ∗value, std::size_t size)
- void value (const Ch ∗value)
- xml_node< Ch > ∗ parent () const

Protected 属性

- Ch ∗ **m_name**
- Ch ∗ **m_value**
- std::size_t **m_name_size**
- std::size_t **m_value_size**
- xml_node< Ch > ∗ **m_parent**

6.429.1 描述

template<class Ch = char>
class rapidxml::xml_base< Ch >

Base class for xml_node and xml_attribute implementing common functions: name(), name_size(), value(), value_size() and parent().

参数

Ch	Character type to use

6.429.2 成函数明

6.429.2.1 name() [1/3]

```
template<class Ch = char>
Ch* rapidxml::xml_base< Ch >::name ( ) const  [inline]
```

Gets name of the node. Interpretation of name depends on type of node. Note that name will not be zero-terminated if rapidxml::parse_no_string_terminators option was selected during parse.

Use name_size() function to determine length of the name.

返回

Name of node, or empty string if node has no name.

6.429.2.2 name() [2/3]

```
template<class Ch = char>
void rapidxml::xml_base< Ch >::name (
            const Ch * name )  [inline]
```

Sets name of node to a zero-terminated string. See also ownership_of_strings and xml_node::name(const Ch *, std::size_t).

参数

name	Name of node to set. Must be zero terminated.

6.429.2.3 name() [3/3]

```
template<class Ch = char>
void rapidxml::xml_base< Ch >::name (
            const Ch * name,
            std::size_t size )  [inline]
```

Sets name of node to a non zero-terminated string. See ownership_of_strings.

Note that node does not own its name or value, it only stores a pointer to it. It will not delete or otherwise free the pointer on destruction. It is reponsibility of the user to properly manage lifetime of the string. The easiest way to achieve it is to use memory_pool of the document to allocate the string - on destruction of the document the string will be automatically freed.

Size of name must be specified separately, because name does not have to be zero terminated. Use name(const Ch *) function to have the length automatically calculated (string must be zero terminated).

参数

name	Name of node to set. Does not have to be zero terminated.
size	Size of name, in characters. This does not include zero terminator, if one is present.

6.429.2.4 name_size()

```
template<class Ch = char>
std::size_t rapidxml::xml_base< Ch >::name_size ( ) const [inline]
```

Gets size of node name, not including terminator character. This function works correctly irrespective of whether name is or is not zero terminated.

返回

Size of node name, in characters.

6.429.2.5 parent()

```
template<class Ch = char>
xml_node<Ch>* rapidxml::xml_base< Ch >::parent ( ) const [inline]
```

Gets node parent.

返回

Pointer to parent node, or 0 if there is no parent.

6.429.2.6 value() [1/3]

```
template<class Ch = char>
Ch* rapidxml::xml_base< Ch >::value ( ) const  [inline]
```

Gets value of node. Interpretation of value depends on type of node. Note that value will not be zero-terminated if rapidxml::parse_no_string_terminators option was selected during parse.

Use value_size() function to determine length of the value.

返回

Value of node, or empty string if node has no value.

6.429.2.7 value() [2/3]

```
template<class Ch = char>
void rapidxml::xml_base< Ch >::value (
            const Ch * value )  [inline]
```

Sets value of node to a zero-terminated string. See also ownership_of_strings and xml_node::value(const Ch *, std::size_t).

参数

value	Vame of node to set. Must be zero terminated.

6.429.2.8 value() [3/3]

```
template<class Ch = char>
void rapidxml::xml_base< Ch >::value (
            const Ch * value,
            std::size_t size )  [inline]
```

Sets value of node to a non zero-terminated string. See ownership_of_strings.

Note that node does not own its name or value, it only stores a pointer to it. It will not delete or otherwise free the pointer on destruction. It is reponsibility of the user to properly manage lifetime of the string. The easiest way to achieve it is to use memory_pool of the document to allocate the string - on destruction of the document the string will be automatically freed.

Size of value must be specified separately, because it does not have to be zero terminated. Use value(const Ch *) function to have the length automatically calculated (string must be zero terminated).

If an element has a child node of type node_data, it will take precedence over element value when printing. If you want to manipulate data of elements using values, use parser flag rapidxml::parse_no_data_nodes to prevent creation of data nodes by the parser.

参数

value	value of node to set. Does not have to be zero terminated.
size	Size of value, in characters. This does not include zero terminator, if one is present.

6.429.2.9 value_size()

```
template<class Ch = char>
std::size_t rapidxml::xml_base< Ch >::value_size ( ) const [inline]
```

Gets size of node value, not including terminator character. This function works correctly irrespective of whether value is or is not zero terminated.

返回

Size of node value, in characters.

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml.hpp

6.430 rapidxml::xml_document< Ch > 模板 参考

```
#include <rapidxml.hpp>
```

Public 成函数

- xml_document ()

  Constructs empty XML document.
- template<int Flags>
  Ch * parse (Ch *text, xml_document< Ch > *parent=0)
- template<int Flags>
  Ch * **parse** (Ch *text, xml_document< Ch > &parent)
- void clear ()
- template<int Flags>
  void fixup (xml_node< Ch > *element, bool recurse)
- void **validate** () const

6.430.1 描述

template<class Ch = char>
class rapidxml::xml_document< Ch >

This class represents root of the DOM hierarchy. It is also an xml_node and a memory_pool through public inheritance. Use parse() function to build a DOM tree from a zero-terminated XML text string. parse() function allocates memory for nodes and attributes by using functions of xml_document, which are inherited from memory_pool. To access root node of the document, use the document itself, as if it was an xml_node.

参数

Ch	Character type to use.

6.430.2 成函数明

6.430.2.1 clear()

```
template<class Ch = char>
void rapidxml::xml_document< Ch >::clear ( ) [inline]
```

Clears the document by deleting all nodes and clearing the memory pool. All nodes owned by document pool are destroyed.

6.430.2.2 fixup()

```
template<class Ch = char>
template<int Flags>
void rapidxml::xml_document< Ch >::fixup (
            xml_node< Ch > * element,
            bool recurse ) [inline]
```

Terminates and/or decodes existing parsed tree, optionally recursively.

6.430.2.3 parse()

```
template<class Ch = char>
template<int Flags>
Ch* rapidxml::xml_document< Ch >::parse (
            Ch * text,
            xml_document< Ch > * parent = 0 ) [inline]
```

Parses zero-terminated XML string according to given flags. Passed string will be modified by the parser, unless rapidxml::parse_non_destructive flag is used. The string must persist for the lifetime of the document. In case of error, rapidxml::parse_error exception will be thrown.

If you want to parse contents of a file, you must first load the file into the memory, and pass pointer to its beginning. Make sure that data is zero-terminated.

Document can be parsed into multiple times. Each new call to parse removes previous nodes and attributes (if any), but does not clear memory pool.

参数

text	XML data to parse; pointer is non-const to denote fact that this data may be modified by the parser.

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml.hpp

6.431 rapidxml::xml_node< Ch > 模板 参考

```
#include <rapidxml.hpp>
```

Public 成函数

- xml_node (node_type type)
- node_type type () const
- void **prefix** (const Ch *prefix, std::size_t size)
- void **prefix** (const Ch *prefix)
- Ch * **prefix** () const
- std::size_t **prefix_size** () const
- void **contents** (Ch const *contents, std::size_t contents_size)
- const Ch * **contents** () const
- std::size_t **contents_size** () const
- Ch * **xmlns** () const
- void **xmlns_lookup** (Ch *&xmlns, size_t &xmlns_size, Ch *prefix, size_t prefix_size) const
- std::size_t **xmlns_size** () const
- xml_document< Ch > * document () const
- xml_node< Ch > * first_node (const Ch *name=0, const Ch *xmlns=0, std::size_t name_size=0, std::size_t xmlns_size=0, bool case_sensitive=true) const
- xml_node< Ch > * last_node (const Ch *name=0, const Ch *xmlns=0, std::size_t name_size=0, std::size_t xmlns_size=0, bool case_sensitive=true) const
- xml_node< Ch > * previous_sibling (const Ch *name=0, const Ch *xmlns=0, std::size_t name_size=0, std↩::size_t xmlns_size=0, bool case_sensitive=true) const
- xml_node< Ch > * next_sibling (const Ch *name=0, const Ch *xmlns=0, std::size_t name_size=0, std::size_t xmlns_size=0, bool case_sensitive=true) const
- xml_attribute< Ch > * first_attribute (const Ch *name=0, std::size_t name_size=0, bool case_sensitive=true) const
- xml_attribute< Ch > * last_attribute (const Ch *name=0, std::size_t name_size=0, bool case_sensitive=true) const
- void type (node_type type)
- void prepend_node (xml_node< Ch > *child)
- void append_node (xml_node< Ch > *child)
- void insert_node (xml_node< Ch > *where, xml_node< Ch > *child)
- void remove_first_node ()
- void remove_last_node ()
- void remove_node (xml_node< Ch > *where)

  Removes specified child from the node.
- void remove_all_nodes ()

  Removes all child nodes (but not attributes).
- void prepend_attribute (xml_attribute< Ch > *attribute)
- void append_attribute (xml_attribute< Ch > *attribute)
- void insert_attribute (xml_attribute< Ch > *where, xml_attribute< Ch > *attribute)
- void remove_first_attribute ()
- void remove_last_attribute ()
- void remove_attribute (xml_attribute< Ch > *where)
- void remove_all_attributes ()

  Removes all attributes of node.
- void **validate** () const

6.431.1 描述

template<class Ch = char>
class rapidxml::xml_node< Ch >

Class representing a node of XML document. Each node may have associated name and value strings, which are available through name() and value() functions. Interpretation of name and value depends on type of the node. Type of node can be determined by using type() function.

Note that after parse, both name and value of node, if any, will point interior of source text used for parsing. Thus, this text must persist in the memory for the lifetime of node.

参数

Ch	Character type to use.

6.431.2 构造及析构函数明

6.431.2.1 xml_node()

```
template<class Ch = char>
rapidxml::xml_node< Ch >::xml_node (
            node_type type )  [inline]
```

Constructs an empty node with the specified type. Consider using memory_pool of appropriate document to allocate nodes manually.

参数

type	Type of node to construct.

6.431.3 成函数明

6.431.3.1 append_attribute()

```
template<class Ch = char>
void rapidxml::xml_node< Ch >::append_attribute (
            xml_attribute< Ch > * attribute )  [inline]
```

Appends a new attribute to the node.

参数

attribute	Attribute to append.

6.431.3.2 append_node()

```
template<class Ch = char>
void rapidxml::xml_node< Ch >::append_node (
            xml_node< Ch > * child )  [inline]
```

Appends a new child node. The appended child becomes the last child.

参数

child	Node to append.

6.431.3.3 document()

```
template<class Ch = char>
xml_document<Ch>* rapidxml::xml_node< Ch >::document ( ) const  [inline]
```

Gets document of which node is a child.

返回

Pointer to document that contains this node, or 0 if there is no parent document.

6.431.3.4 first_attribute()

```
template<class Ch = char>
xml_attribute<Ch>* rapidxml::xml_node< Ch >::first_attribute (
            const Ch * name = 0,
            std::size_t name_size = 0,
            bool case_sensitive = true ) const  [inline]
```

Gets first attribute of node, optionally matching attribute name.

参数

name	Name of attribute to find, or 0 to return first attribute regardless of its name; this string doesn't have to be zero-terminated if name_size is non-zero
name_size	Size of name, in characters, or 0 to have size calculated automatically from string
case_sensitive	Should name comparison be case-sensitive; non case-sensitive comparison works properly only for ASCII characters

返回

Pointer to found attribute, or 0 if not found.

6.431.3.5 first_node()

```
template<class Ch = char>
xml_node<Ch>* rapidxml::xml_node< Ch >::first_node (
            const Ch * name = 0,
            const Ch * xmlns = 0,
            std::size_t name_size = 0,
            std::size_t xmlns_size = 0,
            bool case_sensitive = true ) const  [inline]
```

Gets first child node, optionally matching node name.

参数

name	Name of child to find, or 0 to return first child regardless of its name; this string doesn't have to be zero-terminated if name_size is non-zero
name_size	Size of name, in characters, or 0 to have size calculated automatically from string
case_sensitive	Should name comparison be case-sensitive; non case-sensitive comparison works properly only for ASCII characters

返回

Pointer to found child, or 0 if not found.

6.431.3.6 insert_attribute()

```
template<class Ch = char>
void rapidxml::xml_node< Ch >::insert_attribute (
            xml_attribute< Ch > * where,
            xml_attribute< Ch > * attribute )  [inline]
```

Inserts a new attribute at specified place inside the node. All attributes after and including the specified attribute are moved one position back.

参数

where	Place where to insert the attribute, or 0 to insert at the back.
attribute	Attribute to insert.

6.431.3.7 insert_node()

```
template<class Ch = char>
void rapidxml::xml_node< Ch >::insert_node (
            xml_node< Ch > * where,
            xml_node< Ch > * child )  [inline]
```

Inserts a new child node at specified place inside the node. All children after and including the specified node are moved one position back.

参数

where	Place where to insert the child, or 0 to insert at the back.
child	Node to insert.

6.431.3.8 last_attribute()

```
template<class Ch = char>
xml_attribute<Ch>* rapidxml::xml_node< Ch >::last_attribute (
            const Ch * name = 0,
            std::size_t name_size = 0,
            bool case_sensitive = true ) const  [inline]
```

Gets last attribute of node, optionally matching attribute name.

参数

name	Name of attribute to find, or 0 to return last attribute regardless of its name; this string doesn't have to be zero-terminated if name_size is non-zero
name_size	Size of name, in characters, or 0 to have size calculated automatically from string
case_sensitive	Should name comparison be case-sensitive; non case-sensitive comparison works properly only for ASCII characters

返回

Pointer to found attribute, or 0 if not found.

6.431.3.9 last_node()

```
template<class Ch = char>
xml_node<Ch>* rapidxml::xml_node< Ch >::last_node (
            const Ch * name = 0,
            const Ch * xmlns = 0,
            std::size_t name_size = 0,
            std::size_t xmlns_size = 0,
            bool case_sensitive = true ) const  [inline]
```

Gets last child node, optionally matching node name. Behaviour is undefined if node has no children. Use first_node() to test if node has children.

参数

name	Name of child to find, or 0 to return last child regardless of its name; this string doesn't have to be zero-terminated if name_size is non-zero
name_size	Size of name, in characters, or 0 to have size calculated automatically from string
case_sensitive	Should name comparison be case-sensitive; non case-sensitive comparison works properly only for ASCII characters

返回

Pointer to found child, or 0 if not found.

6.431.3.10 next_sibling()

```
template<class Ch = char>
xml_node<Ch>* rapidxml::xml_node< Ch >::next_sibling (
            const Ch * name = 0,
            const Ch * xmlns = 0,
            std::size_t name_size = 0,
            std::size_t xmlns_size = 0,
            bool case_sensitive = true ) const  [inline]
```

Gets next sibling node, optionally matching node name. Behaviour is undefined if node has no parent. Use parent() to test if node has a parent.

参数

name	Name of sibling to find, or 0 to return next sibling regardless of its name; this string doesn't have to be zero-terminated if name_size is non-zero
name_size	Size of name, in characters, or 0 to have size calculated automatically from string
case_sensitive	Should name comparison be case-sensitive; non case-sensitive comparison works properly only for ASCII characters

返回

Pointer to found sibling, or 0 if not found.

6.431.3.11 prepend_attribute()

```
template<class Ch = char>
void rapidxml::xml_node< Ch >::prepend_attribute (
            xml_attribute< Ch > * attribute )  [inline]
```

Prepends a new attribute to the node.

参数

attribute	Attribute to prepend.

6.431.3.12 prepend_node()

```
template<class Ch = char>
void rapidxml::xml_node< Ch >::prepend_node (
            xml_node< Ch > * child )  [inline]
```

Prepends a new child node. The prepended child becomes the first child, and all existing children are moved one position back.

参数

child	Node to prepend.

6.431.3.13 previous_sibling()

```
template<class Ch = char>
xml_node<Ch>* rapidxml::xml_node< Ch >::previous_sibling (
            const Ch * name = 0,
            const Ch * xmlns = 0,
            std::size_t name_size = 0,
            std::size_t xmlns_size = 0,
            bool case_sensitive = true ) const  [inline]
```

Gets previous sibling node, optionally matching node name. Behaviour is undefined if node has no parent. Use parent() to test if node has a parent.

参数

name	Name of sibling to find, or 0 to return previous sibling regardless of its name; this string doesn't have to be zero-terminated if name_size is non-zero
name_size	Size of name, in characters, or 0 to have size calculated automatically from string
case_sensitive	Should name comparison be case-sensitive; non case-sensitive comparison works properly only for ASCII characters

返回

Pointer to found sibling, or 0 if not found.

6.431.3.14 remove_attribute()

```
template<class Ch = char>
void rapidxml::xml_node< Ch >::remove_attribute (
            xml_attribute< Ch > * where )  [inline]
```

Removes specified attribute from node.

参数

where	Pointer to attribute to be removed.

6.431.3.15 remove_first_attribute()

```
template<class Ch = char>
void rapidxml::xml_node< Ch >::remove_first_attribute ( )  [inline]
```

Removes first attribute of the node. If node has no attributes, behaviour is undefined. Use first_attribute() to test if node has attributes.

6.431.3.16 remove_first_node()

```
template<class Ch = char>
void rapidxml::xml_node< Ch >::remove_first_node ( )  [inline]
```

Removes first child node. If node has no children, behaviour is undefined. Use first_node() to test if node has children.

6.431.3.17 remove_last_attribute()

```
template<class Ch = char>
void rapidxml::xml_node< Ch >::remove_last_attribute ( )  [inline]
```

Removes last attribute of the node. If node has no attributes, behaviour is undefined. Use first_attribute() to test if node has attributes.

6.431.3.18 remove_last_node()

```
template<class Ch = char>
void rapidxml::xml_node< Ch >::remove_last_node ( )  [inline]
```

Removes last child of the node. If node has no children, behaviour is undefined. Use first_node() to test if node has children.

6.431.3.19 type() [1/2]

```
template<class Ch = char>
node_type rapidxml::xml_node< Ch >::type ( ) const  [inline]
```

Gets type of node.

返回

Type of node.

6.431.3.20 type() [2/2]

```
template<class Ch = char>
void rapidxml::xml_node< Ch >::type (
            node_type type )  [inline]
```

Sets type of node.

参数

type	Type of node to set.

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml.hpp

6.432 ZtGeographyCoordinateTransform::XY2BLInput构体 参考

Public 属性

- double **x**
- double **y**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform↩_m.h

6.433 ZtGeographyCoordinateTransform::XY2BLOutput构体 参考

Public 属性

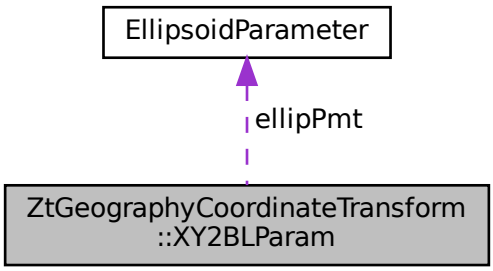
- double **lat**
- double **lon**
- bool **isValidCoord**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform↩_m.h

6.434 ZtGeographyCoordinateTransform::XY2BLParam构体 参考

ZtGeographyCoordinateTransform::XY2BLParam 的作:



Public 属性

- EllipsoidParameter **ellipPmt**
- double **meridianLine** = -360
- char **projType** = 'g'

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform↩_m.h

6.435 ZtGeographyCoordinateTransform::XYZ2BLHInput构体 参考

Public 属性

- double **x**
- double **y**
- double **z**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform↩_m.h

6.436 ZtGeographyCoordinateTransform::XYZ2BLHOutput构体 参考

Public 属性

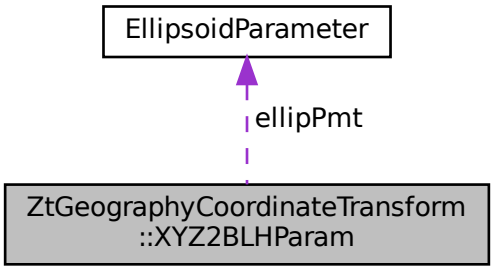
- double **lat**
- double **lon**
- double **ht**
- bool **isValidCoord**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform↩_m.h

6.437 ZtGeographyCoordinateTransform::XYZ2BLHParam构体 参考

ZtGeographyCoordinateTransform::XYZ2BLHParam 的作:



Public 属性

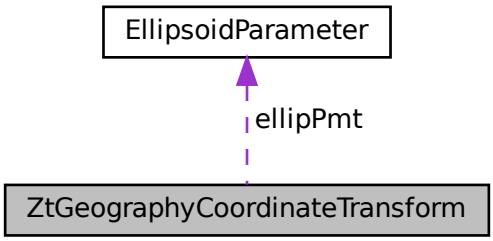
- EllipsoidParameter **ellipPmt**
- double **meridianLine** = -360
- char **projType** = 'g'

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform↩_m.h

6.438 ZtGeographyCoordinateTransform 参考

ZtGeographyCoordinateTransform 的作:



- struct BL2XYInput
- struct BL2XYOutput
- struct BL2XYParam
- struct BLH2XYZInput
- struct BLH2XYZOutput
- struct BLH2XYZParam
- struct XY2BLInput
- struct XY2BLOutput
- struct XY2BLParam
- struct XYZ2BLHInput
- struct XYZ2BLHOutput
- struct XYZ2BLHParam

Public 成函数

- bool **XY2BL** (double x, double y, double &lat, double &lon)
- bool **BL2XY** (double lat, double lon, double &x, double &y)
- bool **XYZ2BLH** (double x, double y, double z, double &lat, double &lon, double &ht)
- bool **BLH2XYZ** (double lat, double lon, double ht, double &x, double &y, double &z)

Public 属性

- EllipsoidParameter **ellipPmt**
- double **meridianLine**
- char **projType**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform.↩ h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/ztgeographycoordinatetransform.↩ cpp

Chapter 7

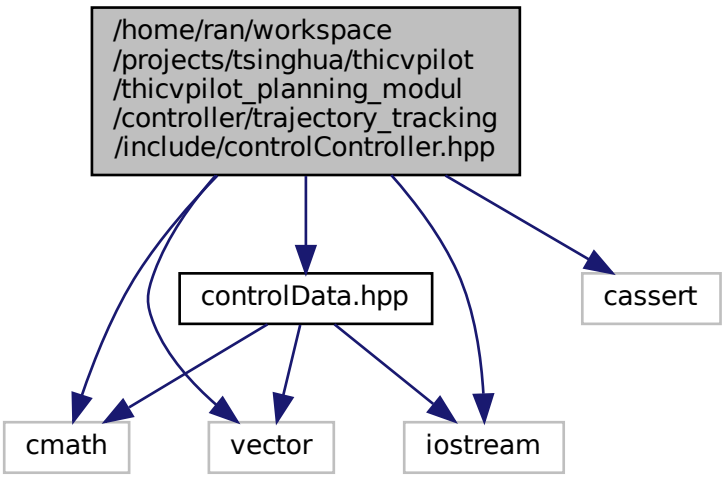
文件明

7.1 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlController.hpp 文件参考

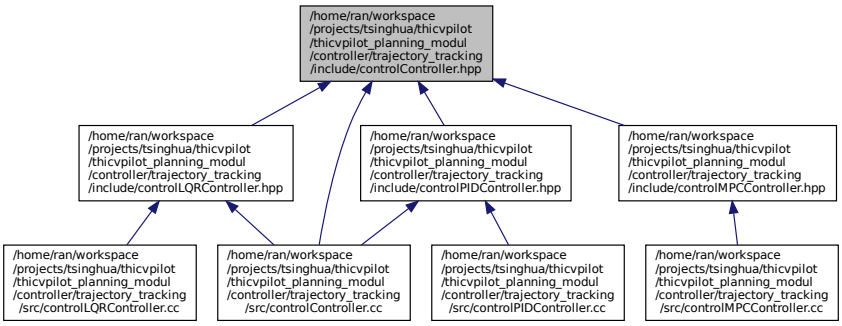
控制器模，横向控制基，向控制定

```
#include <cmath>
#include <vector>
#include <iostream>
#include "controlData.hpp"
#include <cassert>
```

controlController.hpp 的引用(Include)系:



此展示文件直接或接的被哪些文件引用了:



- class Control::LatControllerBase

  横向控制器基。
- class Control::LonController

  longitude 向控制器
- class Control::Controller

  横向控制器基 控制模主要模。内部分横向控制和向控制

函数

- int Control::getNearestIndex (State state, const Traj &traj)

  找最近点*ID*. 比曲中个点和当前点的距离

7.1.1 描述

控制器模，横向控制基，向控制定

作者

yibin, yining (852488062@qq.com)

版本

1.0

日期

2022-08-18

版所有

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7.1.2 函数明

7.1.2.1 getNearestIndex()

```
int Control::getNearestIndex (
            State state,
            const Traj & traj )
```

找最近点ID. 比曲中个点和当前点的距离

参数

state	状
traj	待跟踪迹

返回

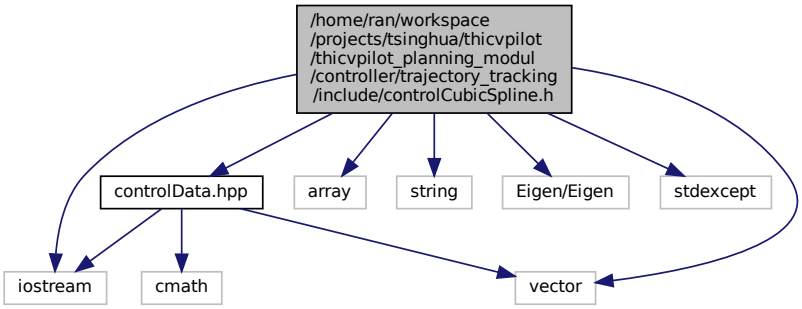
int nID: nearset ID 最近点ID

7.2 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_↩planning_modul/controller/trajectory_tracking/include/controlCubic↩Spline.h 文件参考

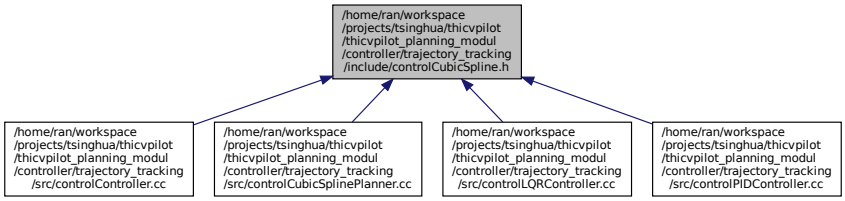
ref. "Cpp Robotics. Python Robotics."

```
#include <iostream>
#include <vector>
#include <array>
#include <string>
#include <Eigen/Eigen>
#include <stdexcept>
#include "controlData.hpp"
```

controlCubicSpline.h 的引用(Include)系:



此展示文件直接或接的被哪些文件引用了:



- class Control::Spline
- class Control::Spline2D
- class Control::Spline2DWithV

  TrajRaw -> Traj。插所需

型定

- using **Control::Vec_d** = std::vector< double >
- using **Control::Poi_d** = std::array< double, 2 >
- using **Control::Vec_Poi** = std::vector< Poi_d >

函数

- Vec_d **Control::vecDiff** (Vec_d input)
- Vec_d **Control::cumSum** (Vec_d input)
- void Control::calcSplineCourse (Vec_d x, Vec_d y, Vec_d &rX, Vec_d &rY, Vec_d &rYaw, Vec_d &rCurvature, Vec_d &rS, double ds)

  条曲插迹，并算相的切向角度和曲率
- Traj Control::calcSplineWithVCourse (const Traj &trajRaw, double ds)

  理决策入的原始迹*(Traj: x y yaw)*，出插并具有信息的迹*(Traj: xy yaw v k s).*
- Traj **Control::removeDuplicate** (const Traj &trajRaw)

7.2.1 描述

ref. ”Cpp Robotics. Python Robotics.”

作者

yibin, yining (852488062@qq.com)

版本

1.0

日期

2022-08-16

版所有

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7.2.2 函数明

7.2.2.1 calcSplineCourse()

```
void Control::calcSplineCourse (
            Vec_d x,
            Vec_d y,
            Vec_d & rX,
            Vec_d & rY,
            Vec_d & rYaw,
            Vec_d & rCurvature,
            Vec_d & rS,
            double ds )
```

条曲插迹，并算相的切向角度和曲率

参数

x	原始迹点
y	
rX	
rY	
rYaw	
rCurvature	
rS	
ds	

7.2.2.2 calcSplineWithVCourse()

```
Traj Control::calcSplineWithVCourse (
            const Traj & trajRaw,
            double ds )
```

理决策入的原始迹(Traj: x y yaw)，出插并具有信息的迹(Traj: xy yaw v k s).

参数

trajRaw	原始迹
ds	采隔

返回

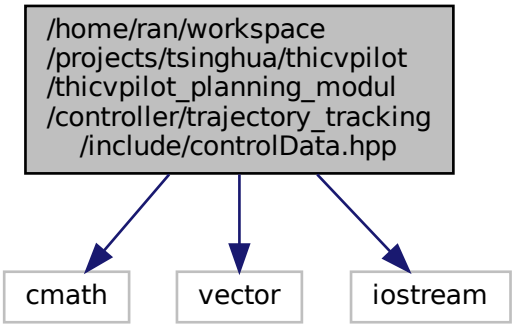
Traj 出迹

7.3 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlData.hpp 文件参考

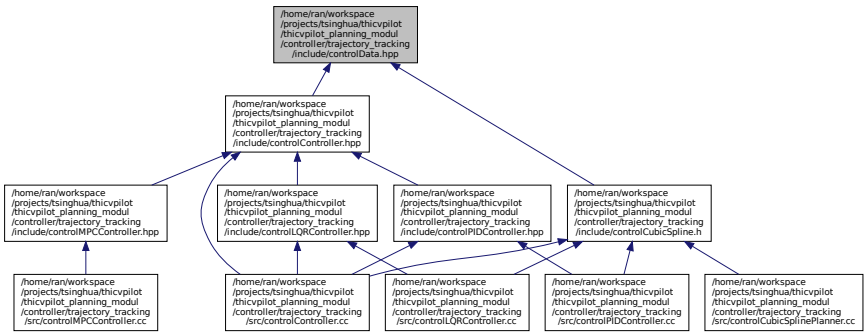
模使用数据型

```
#include <cmath>
#include <vector>
#include <iostream>
```

controlData.hpp 的引用(Include)系:



此展示文件直接或接的被哪些文件引用了:



- struct Control::State
  入位置信息
- struct Control::TrajPoint
- struct Control::ControlCMD

型定

- using **Control::Traj** = std::vector< TrajPoint >

7.3.1 描述

模使用数据型

作者

yibin, yining (852488062@qq.com)

版本

1.0

日期

2022-08-16

版所有

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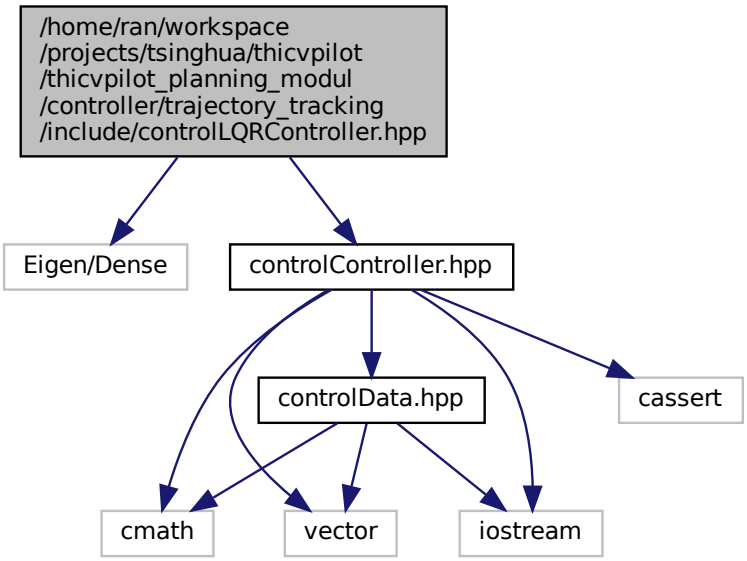
7.4 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/include/controlLQRController.hpp 文件参考

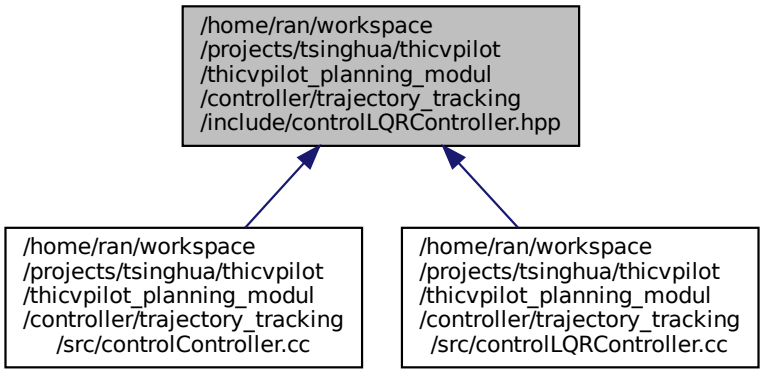
LQR横向控制模定

```
#include <Eigen/Dense>
#include "controlController.hpp"
```

controlLQRController.hpp 的引用(Include)系:



此展示文件直接或接的被哪些文件引用了:



- class Control::LQR
- class Control::LQRController

  *LQR*横向控制

型定

- using **Mat** = Eigen::MatrixXd
- using **Vec** = Eigen::VectorXd
- using **Matrix5d** = Eigen::Matrix< double, 5, 5 >
- using **Matrix52d** = Eigen::Matrix< double, 5, 2 >
- using **Matrix25d** = Eigen::Matrix< double, 2, 5 >
- using **RowVector5d** = Eigen::Matrix< double, 1, 5 >
- using **Vector5d** = Eigen::Matrix< double, 5, 1 >

7.4.1 描述

LQR横向控制模定

作者

yibin, yining (852488062@qq.com)

版本

0.1

日期

2022-08-18

版所有

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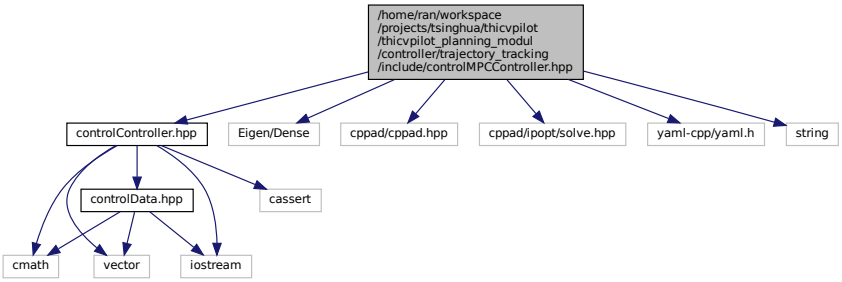
7.5 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_↩ planning_modul/controller/trajectory_tracking/include/controlMPC↩ Controller.hpp 文件参考

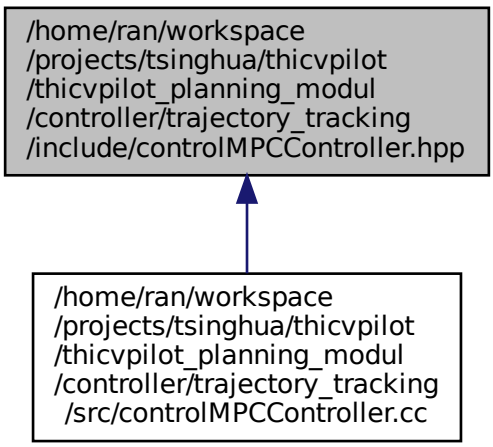
```
#include "controlController.hpp"
#include <Eigen/Dense>
#include <cppad/cppad.hpp>
#include <cppad/ipopt/solve.hpp>
#include <yaml-cpp/yaml.h>
```

```
#include <string>
```

controlMPCController.hpp 的引用(Include)系:



此展示文件直接或接的被哪些文件引用了:



- class Control::FG_EVAL
- class Control::MPC
- class Control::MPCController

型定

- using **Matf** = Eigen::MatrixXf
- using **Vec_f** = std::vector< float >
- using **Poi_f** = std::array< float, 2 >
- using **Vec_Poi** = std::vector< Poi_f >

函数

- typedef **Control::CPPAD_TESTVECTOR** (double) Dvector

7.5.1 描述

作者

your name (you@domain.com)

版本

0.1

日期

2022-09-06

版所有

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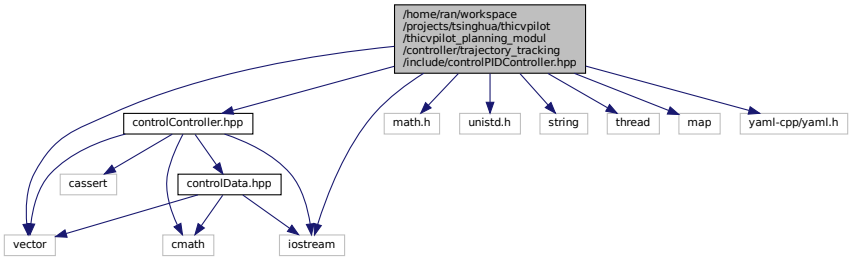
7.6 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_↩ planning_modul/controller/trajectory_tracking/include/controlPID↩ Controller.hpp 文件参 考

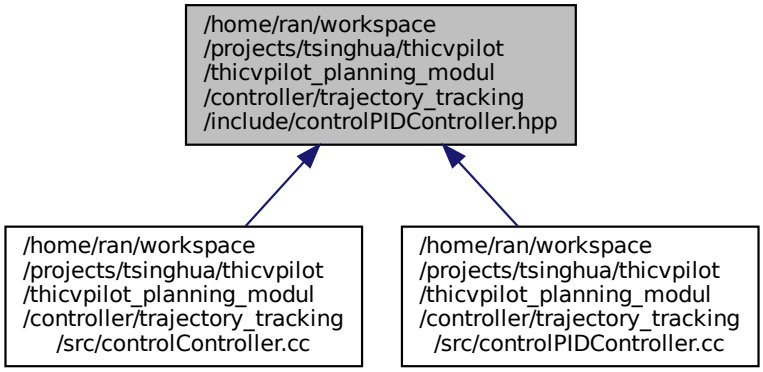
PID控制器 和 PID横向控制器

```
#include "controlController.hpp"
#include <math.h>
#include <unistd.h>
#include <string>
#include <thread>
#include <iostream>
#include <vector>
#include <map>
#include <yaml-cpp/yaml.h>
```

controlPIDController.hpp 的引用(Include)系:



此展示文件直接或接的被哪些文件引用了:



- class Control::PID

  PID 控制器
- class Control::PIDController

  *PID*横向运控制器

型定

- using **Control::PIDLibs** = std::map< std::string, PID >

7.6.1 描述

PID控制器 和 PID横向控制器

作者

yibin, yining (852488062@qq.com)

版本

1.0

日期

2022-08-16

版所有

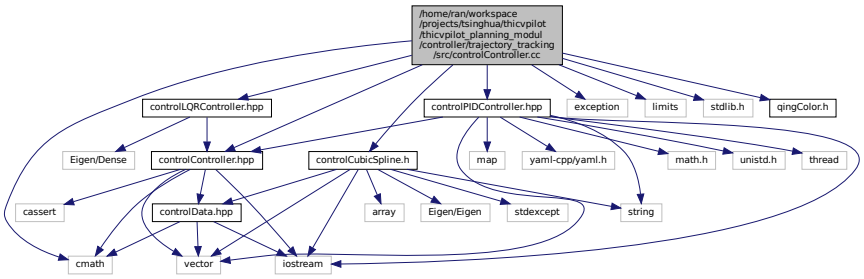
Copyright (c) 2022

7.7 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_↩ planning_modul/controller/trajectory_tracking/src/control↩ Controller.cc 文件参 考

横向控制器基

```
#include <cmath>
#include <exception>
#include <limits>
#include <stdlib.h>
#include "controlController.hpp"
#include "controlCubicSpline.h"
#include "controlPIDController.hpp"
#include "controlLQRController.hpp"
#include "qingColor.h"
```

controlController.cc 的引用(Include)系:



函数

- Traj **Control::transferGauss2RightHand** (const State &s, const Traj &trajRaw, State &s_)
- int Control::getNearestIndex (State state, const Traj &traj)

  找最近点*ID*. 比曲中个点和当前点的距离

量

- int **speedZeroFlag** =0

7.7.1 描述

横向控制器基

作者

yibin, yining (852488062@qq.com)

版本

1.0

日期

2022-08-16

版所有

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7.7.2 函数明

7.7.2.1 getNearestIndex()

```
int Control::getNearestIndex (
            State state,
            const Traj & traj )
```

找最近点ID. 比曲中个点和当前点的距离

参数

state	状
traj	待跟踪迹

返回

int nID: nearset ID 最近点ID

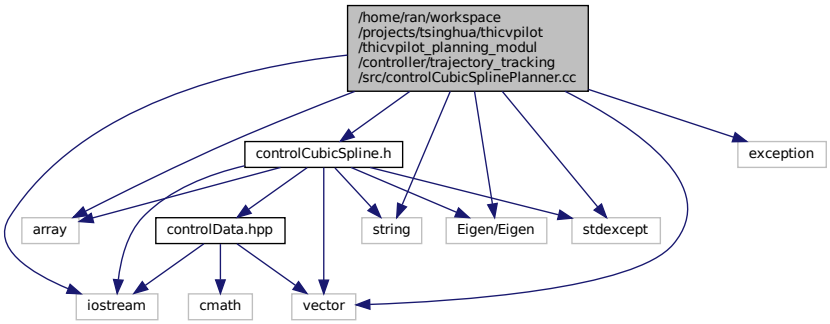
7.8 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/controlCubicSplinePlanner.cc 文件参考

理决策入的原始迹(Traj: x y yaw)，出插并具有信息的迹(Traj: xy yaw v k s).

```
#include "controlCubicSpline.h"
#include <Eigen/Eigen>
#include <array>
#include <exception>
#include <iostream>
#include <stdexcept>
```

```
#include <string>
#include <vector>
```

controlCubicSplinePlanner.cc 的引用(Include)系:



函数

- Vec_d **Control::vecDiff** (Vec_d input)
- Vec_d **Control::cumSum** (Vec_d input)
- void Control::calcSplineCourse (Vec_d x, Vec_d y, Vec_d &rX, Vec_d &rY, Vec_d &rYaw, Vec_d &rCurvature, Vec_d &rS, double ds)

  条曲插迹，并算相的切向角度和曲率
- Traj **Control::removeDuplicate** (const Traj &trajRaw)
- Traj Control::calcSplineWithVCourse (const Traj &trajRaw, double ds)

  理决策入的原始迹*(Traj: x y yaw)*，出插并具有信息的迹*(Traj: xy yaw v k s).*

7.8.1 描述

理决策入的原始迹(Traj: x y yaw)，出插并具有信息的迹(Traj: xy yaw v k s).

作者

yibin, yining (852488062@qq.com) Cpp Robotics

版本

1.0

日期

2022-08-16

版所有

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7.8.2 函数明

7.8.2.1 calcSplineCourse()

```
void Control::calcSplineCourse (
            Vec_d x,
            Vec_d y,
            Vec_d & rX,
            Vec_d & rY,
            Vec_d & rYaw,
            Vec_d & rCurvature,
            Vec_d & rS,
            double ds )
```

条曲插迹，并算相的切向角度和曲率

参数

x	原始迹点
y	
rX	
rY	
rYaw	
rCurvature	
rS	
ds	

7.8.2.2 calcSplineWithVCourse()

```
Traj Control::calcSplineWithVCourse (
            const Traj & trajRaw,
            double ds )
```

理决策入的原始迹(Traj: x y yaw)，出插并具有信息的迹(Traj: xy yaw v k s).

参数

trajRaw	原始迹
ds	采隔

返回

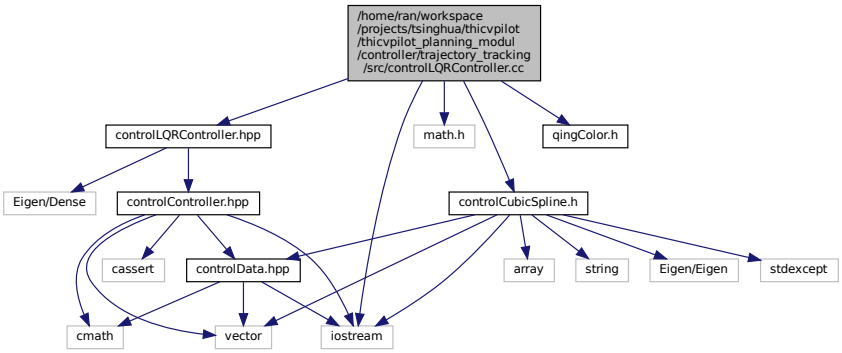
Traj 出迹

7.9 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/controlLQRController.cc 文件参考

LQR横向控制器

```
#include "controlLQRController.hpp"
#include <math.h>
#include <iostream>
#include "controlCubicSpline.h"
#include "qingColor.h"
```

controlLQRController.cc 的引用(Include)系:



7.9.1 描述

LQR横向控制器

作者

yibin, yining (852488062@qq.com)

版本

1.0

日期

2022-08-16

版所有

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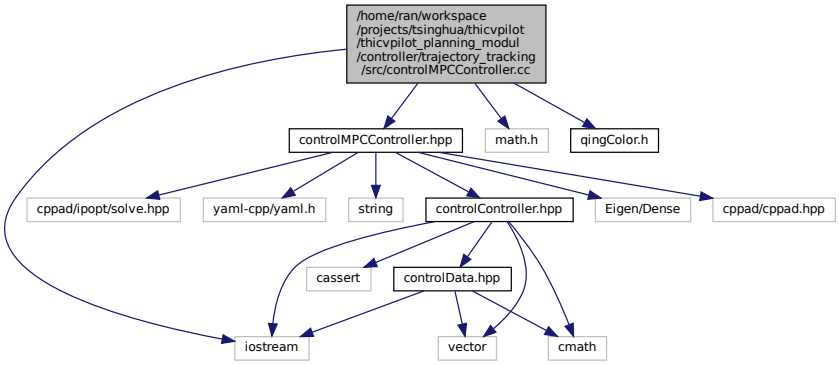
7.10 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/controlMPCController.cc 文件参考

```
#include "controlMPCController.hpp"
#include <math.h>
#include <iostream>
#include "qingColor.h"
```

controlMPCController.cc 的引用(Include)系:



7.10.1 描述

作者

your name (you@domain.com)

版本

0.1

日期

2022-09-06

版所有

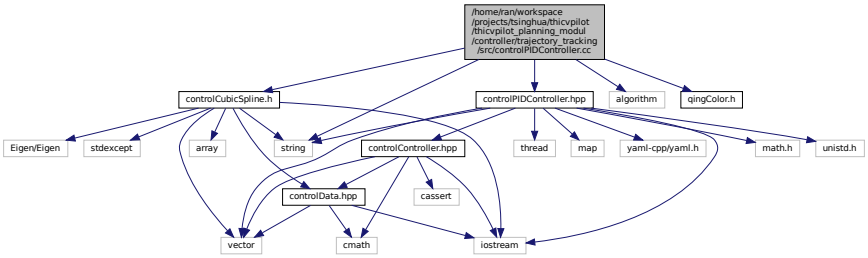
Copyright (c) 2022

7.11 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/controller/trajectory_tracking/src/controlPIDController.cc 文件参考

PID控制器，PID横向运控制器

```
#include "controlPIDController.hpp"
#include "controlCubicSpline.h"
#include <algorithm>
#include <string>
#include "qingColor.h"
```

controlPIDController.cc 的引用(Include)系:



函数

- PID Control::loadPIDYamlConfig (YAML::Node config, std::string name, double dt)

  取 *yaml* 文件中*pid_conf*相字段内容，并返回控制器
- bool Control::compareTrajDistance (const TrajPoint &p1, const TrajPoint &p2)

  找瞄点，根据路程大小，比两个路点的先后。用于后面的*getPreviewID*中

量

- double **Control::angleErrorWJ** = 0

7.11.1 描述

PID控制器，PID横向运控制器

作者

yibin, yining (852488062@qq.com)

版本

1.0

日期

2022-08-16

版所有

Copyright (c) 2022

7.11.2 函数明

7.11.2.1 compareTrajDistance()

```
bool Control::compareTrajDistance (
            const TrajPoint & p1,
            const TrajPoint & p2 )
```

找瞄点，根据路程大小，比两个路点的先后。用于后面的getPreviewID中

参数

p1	路点1
p2	路点2

返回

true 路点1在前
false 路点2在前

7.11.2.2 loadPIDYamlConfig()

```
PID Control::loadPIDYamlConfig (
            YAML::Node config,
            std::string name,
            double dt )
```

取 yaml 文件中pid_conf相字段内容，并返回控制器

参数

config	yaml node配置文件信息
name	控制差名称
dt	控制器率

返回

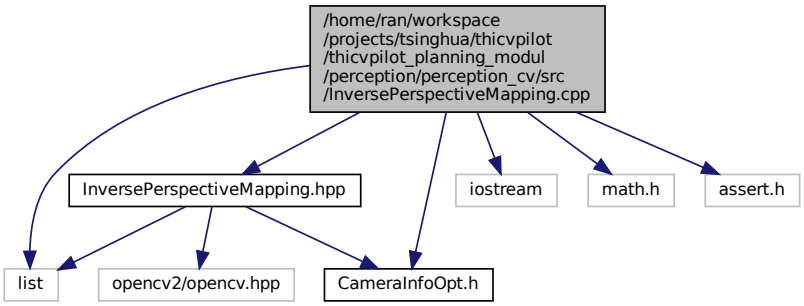
PID

7.12 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/perception/perception_cv/src/InversePerspectiveMapping.cpp 文件参考

Implements inverse perspective mapping functionality for lane detection.

```
#include "InversePerspectiveMapping.hpp"
#include "CameraInfoOpt.h"
#include <iostream>
#include <math.h>
#include <assert.h>
#include <list>
```

InversePerspectiveMapping.cpp 的引用(Include)系:



宏定

- #define **VP_PORTION** 0.05

函数

- void LaneDetector::imGetIPM (const cv::Mat &inImage, cv::Mat &outImage, IPMInfo *ipmInfo, const CameraInfo *cameraInfo, std::list< cv::Point > *outPoints)

  Coordinate system description.
- void LaneDetector::imTransformImage2Ground (const cv::Mat &inPoints, cv::Mat &outPoints, const CameraInfo *cameraInfo)

  Transforms points from image frame to ground plane.
- void LaneDetector::imTransformGround2Image (const cv::Mat &inPoints, cv::Mat &outPoints, const CameraInfo *cameraInfo)

  Transforms points from ground plane to image frame.
- cv::Point2f LaneDetector::imGetVanishingPoint (const CameraInfo *cameraInfo)

Computes vanishing point in image plane.

- void LaneDetector::imPointImIPM2World (cv::Point2f &point, const IPMInfo ∗ipmInfo)

  Converts point from IPM pixel coordinates to world coordinates.
- void LaneDetector::imTransformImIPM2Ground (const cv::Mat &inMat, cv::Mat &outMat, const IPMInfo ∗ipmInfo)

  Converts points from IPM pixel coordinates to world coordinates.
- void LaneDetector::imTransformImIPM2Im (const cv::Mat &inMat, cv::Mat &outMat, const IPMInfo ∗ipmInfo, const CameraInfo ∗cameraInfo)

  Converts points from IPM pixel coordinates to image coordinates.
- void LaneDetector::imInitCameraInfo (char ∗const fileName, CameraInfo ∗cameraInfo)

  Initializes camera info structure from config file.
- void LaneDetector::imScaleCameraInfo (CameraInfo ∗cameraInfo, cv::Size size)

  Scales camera parameters according to input image size.
- void LaneDetector::imGetIPMExtent (const CameraInfo ∗cameraInfo, IPMInfo ∗ipmInfo)

  Calculates extent of image on ground plane.

7.12.1 描述

Implements inverse perspective mapping functionality for lane detection.

作者

Mohamed Aly malaa@caltech.edu

日期

11/29/2006

7.12.2 函数明

7.12.2.1 imGetIPM()

```
void LaneDetector::imGetIPM (
            const cv::Mat & inImage,
            cv::Mat & outImage,
            IPMInfo * ipmInfo,
            const CameraInfo * cameraInfo,
            std::list< cv::Point > * outPoints )
```

Coordinate system description.

We are assuming the world coordinate frame center is at the camera, the ground plane is at height -h, the X-axis is going right, the Y-axis is going forward, the Z-axis is going up. The camera is looking forward with optical axis in direction of Y-axis, with possible pitch angle (above or below the Y-axis) and yaw angle (left or right). The camera coordinates have the same center as the world, but the Xc-axis goes right, the Yc-axis goes down, and the Zc-axis (optical cxis) goes forward. The uv-plane of the image is such that u is horizontal going right, v is vertical going down. The image coordinates uv are such that the pixels are at half coordinates i.e. first pixel is (.5,.5) ...etc where the top-left point is (0,0) i.e. the tip of the first pixel is (0,0)

Generates Inverse Perspective Mapping of input image

参数

inImage	Input image to transform
outImage	Output IPM transformed image
ipmInfo	IPM parameters and info for the transformation
cameraInfo	Camera calibration parameters
outPoints	Optional list to store indices of points outside image bounds

Performs IPM transform assuming flat ground plane using camera parameters

7.12.2.2 imGetIPMExtent()

```
void LaneDetector::imGetIPMExtent (
            const CameraInfo * cameraInfo,
            IPMInfo * ipmInfo )
```

Calculates extent of image on ground plane.

参数

cameraInfo	Camera calibration parameters
ipmInfo	IPM parameters structure (xLimits and yLimits are updated)

Computes the boundaries of the image when projected onto ground plane

Gets the extent of the image on the ground plane given the camera parameters

参数

cameraInfo	the input camera info
ipmInfo	the IPM info containing the extent on ground plane: xLimits & yLimits only are changed

7.12.2.3 imGetVanishingPoint()

```
cv::Point2f LaneDetector::imGetVanishingPoint (
            const CameraInfo * cameraInfo )
```

Computes vanishing point in image plane.

参数

cameraInfo	Camera calibration parameters

返回

Vanishing point coordinates in image frame

Finds intersection of image plane with line in XY-plane making yaw angle with Y-axis

Computes the vanishing point in the image plane uv. It is the point of intersection of the image plane with the line in the XY-plane in the world coordinates that makes an angle yaw clockwise (form Y-axis) with Y-axis

参数

cameraInfo	the input camera parameter

返回

the computed vanishing point in image frame

7.12.2.4 imInitCameraInfo()

```
void LaneDetector::imInitCameraInfo (
            char *const fileName,
            CameraInfo * cameraInfo )
```

Initializes camera info structure from config file.

参数

fileName	Path to camera configuration file
cameraInfo	Camera parameters structure to initialize

Initializes the cameraInfo structure with data read from the conf file

参数

fileName	the input camera conf file name
cameraInfo	the returned camera parametrs struct

7.12.2.5 imPointImIPM2World()

```
void LaneDetector::imPointImIPM2World (
            cv::Point2f & point,
            const IPMInfo * ipmInfo )
```

Converts point from IPM pixel coordinates to world coordinates.

参数

point	Point to transform (in/out parameter)
ipmInfo	IPM parameters from imGetIPM

Converts a point from IPM pixel coordinates into world coordinates

参数

point	in/out point
ipmInfo	the ipm info from imGetIPM

7.12.2.6 imScaleCameraInfo()

```
void LaneDetector::imScaleCameraInfo (
            CameraInfo * cameraInfo,
            cv::Size size )
```

Scales camera parameters according to input image size.

参数

cameraInfo	Camera parameters to scale (in/out)
size	Target image dimensions

Scales the cameraInfo according to the input image size

参数

cameraInfo	the input/return structure
size	the input image size

7.12.2.7 imTransformGround2Image()

```
void LaneDetector::imTransformGround2Image (
            const cv::Mat & inPoints,
            cv::Mat & outPoints,
            const CameraInfo * cameraInfo )
```

Transforms points from ground plane to image frame.

参数

inPoints	Input points in world coordinates on ground plane (2xN matrix)
outPoints	Output points in image coordinates (2xN matrix)
cameraInfo	Camera calibration parameters

Projects points from XY coordinates on ground plane Z=-height to image UV coordinates

Transforms points from the ground plane (z=-h) in the world frame into points on the image in image frame (uv-coordinates)

参数

inPoints	2xN array of input points on the ground in world coordinates
outPoints	2xN output points in on the image in image coordinates
cameraInfo	the camera parameters

7.12.2.8 imTransformImage2Ground()

```
void LaneDetector::imTransformImage2Ground (
            const cv::Mat & inPoints,
            cv::Mat & outPoints,
            const CameraInfo * cameraInfo )
```

Transforms points from image frame to ground plane.

参数

inPoints	Input points in image coordinates (2xN matrix)
outPoints	Output points in world coordinates on ground plane (2xN matrix)
cameraInfo	Camera calibration parameters

Projects points from image UV coordinates to XY coordinates on ground plane Z=-height

Transforms points from the image frame (uv-coordinates) into the real world frame on the ground plane (z=-height)

参数

inPoints	input points in the image frame (2xN matrix)
outPoints	output points in the world frame on the ground (z=-height) (2xN matrix with xw, yw and implicit z=-height)
cemaraInfo	the input camera parameters

7.12.2.9 imTransformImIPM2Ground()

```
void LaneDetector::imTransformImIPM2Ground (
            const cv::Mat & inMat,
            cv::Mat & outMat,
            const IPMInfo * ipmInfo )
```

Converts points from IPM pixel coordinates to world coordinates.

参数

inMat	Input matrix of points (2xN)
outMat	Output matrix of transformed points (2xN)
ipmInfo	IPM parameters from imGetIPM

Converts from IPM pixel coordinates into world coordinates

参数

inMat	input matrix 2xN
outMat	output matrix 2xN
ipmInfo	the ipm info from imGetIPM

7.12.2.10 imTransformImIPM2Im()

```
void LaneDetector::imTransformImIPM2Im (
            const cv::Mat & inMat,
            cv::Mat & outMat,
            const IPMInfo * ipmInfo,
            const CameraInfo * cameraInfo )
```

Converts points from IPM pixel coordinates to image coordinates.

参数

inMat	Input matrix of points (2xN)
outMat	Output matrix of transformed points (2xN)
ipmInfo	IPM parameters from imGetIPM
cameraInfo	Camera calibration parameters

Converts from IPM pixel coordinates into Image coordinates

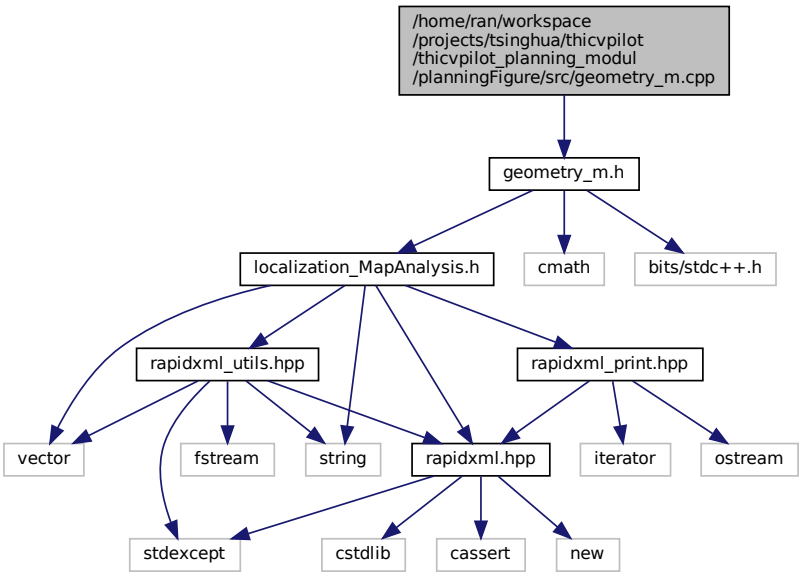
参数

inMat	input matrix 2xN
outMat	output matrix 2xN
ipmInfo	the ipm info from imGetIPM
cameraInfo	the camera info

7.13 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/geometry_m.cpp 文件参考

geometry

`#include "geometry_m.h"`
geometry_m.cpp 的引用(Include)系:



命名空

- geometry
  是一个算欧式距离的*namespace*。

函数

- void geometry::pointToPolygon (const pointToPolygonParam &param, const pointToPolygonInput &input, pointToPolygonOutput &output)
  算点到凸多形的距离
- void geometry::cross (const crossParam &param, const crossInput &input, crossOutput &output)
  点乘
- void geometry::pointToLine (const pointToLineParam &param, const pointToLineInput &input, pointToLine↩Output &output)
  算点到段的距离
- void geometry::isOnline (const isOnlineParam &param, const isOnlineInput &input, isOnlineOutput &output)
  判断点是否在段上
- void geometry::isInSimple (const isInSimpleParam &param, const isInSimpleInput &input, isInSimpleOutput &output)

判断点是否在横平直的多形内部

- void geometry::isLineIntersection (const isLineIntersectionParam &param, const isLineIntersectionInput &input, isLineIntersectionOutput &output)

  判断两段是否相交

- void geometry::isPolygonsIntersection (const isPolygonsIntersectionParam &param, const isPolygons↩ IntersectionInput &input, isPolygonsIntersectionOutput &output)

  判断两多形是否相交

- void geometry::isPointInPolygon (const isPointInPolygonParam &param, const isPointInPolygonInput &input, isPointInPolygonOutput &output)

  判断点是否在多形内部

- void geometry::inOrOutPolygon (const inOrOutPolygonParam &param, const inOrOutPolygonInput &input, inOrOutPolygonOutput &output)

  判断点是否在多形内部

7.13.1 描述

geometry

作者

版本

1.0

日期

2024-12-04

版所有

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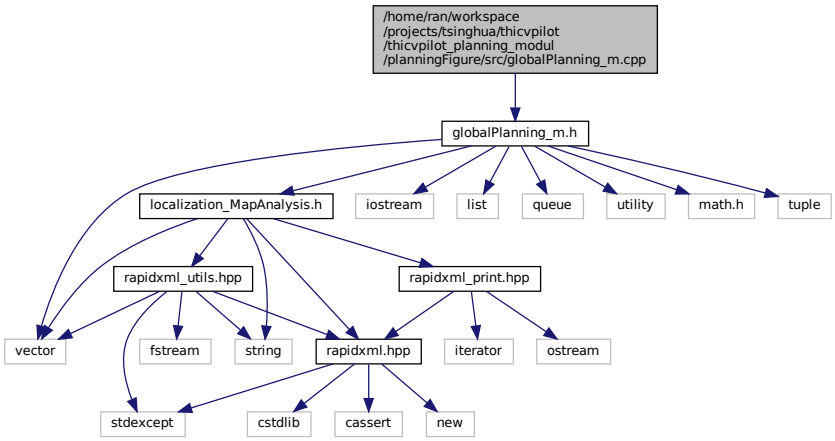
7.14 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_↩ planning_modul/planningFigure/src/globalPlanning_m.cpp 文件参考

globalPlanning

```
#include "globalPlanning_m.h"
```

globalPlanning_m.cpp 的引用(Include)系:



命名空

- GolbalPlanning

  是一个名*GolbalPlanning*的*namespace*。

函数

- void GolbalPlanning::AstarMapToAstar (const AstarMapToAstarParam &param, const AstarMapToAstarInput &input, AstarMapToAstarOutput &output)

  使用*Map*构建*Astar*
- void GolbalPlanning::AstarGetPath (const AstarGetPathParam &param, const AstarGetPathInput &input, AstarGetPathOutput &output)

  使用*Astar*算法取路径，包括理、后理等部分
- void GolbalPlanning::AstarInitRoad (const AstarInitRoadParam &param, const AstarInitRoadInput &input, AstarInitRoadOutput &output)

  *Astar*道路初始化
- void GolbalPlanning::AstarInitLink (const AstarInitLinkParam &param, const AstarInitLinkInput &input, AstarInitLinkOutput &output)

  *Astar*道路接系初始化
- void GolbalPlanning::AstarReset (const AstarResetParam &param, const AstarResetInput &input, Astar↩ ResetOutput &output)

  在保留地的情况下，初始化*Astar*参数
- void GolbalPlanning::AstarModuleSelfCheck (const AstarModuleSelfCheckParam &param, const Astar↩ ModuleSelfCheckInput &input, AstarModuleSelfCheckOutput &output)

  *Astar*道路*Id*
- void GolbalPlanning::AstarModuleSelfCheckPrint (const AstarModuleSelfCheckPrintParam &param, const AstarModuleSelfCheckPrintInput &input, AstarModuleSelfCheckPrintOutput &output)

  *Astar*道路*Id*打印
- void GolbalPlanning::AstarFindLane (const AstarFindLaneParam &param, const AstarFindLaneInput &input, AstarFindLaneOutput &output)

  根据*Astar*道路全局划果，得道全局划果

- void GolbalPlanning::AstarFindLaneR (const AstarFindLaneRParam &param, const AstarFindLaneRInput &input, AstarFindLaneROutput &output)

  根据*Astar*道路全局划果，使用的方法得道全局划果
- void GolbalPlanning::AstarRecursiveFindLane (const AstarRecursiveFindLaneParam &param, const AstarRecursiveFindLaneInput &input, AstarRecursiveFindLaneOutput &output)

  *Astar*得道全局划果主体函数

7.14.1 描述

globalPlanning

作者

版本

1.0

日期

2024-12-04

版所有

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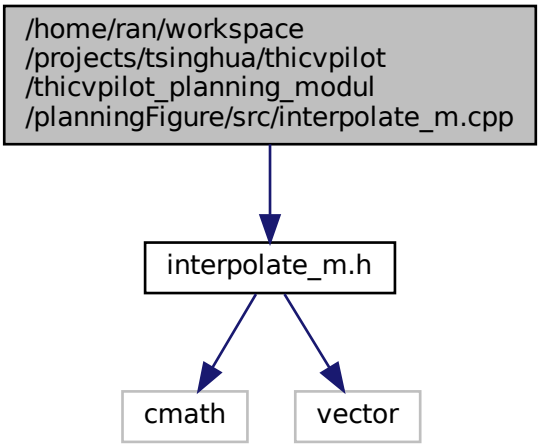
7.15 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/interpolate_m.cpp 文件参考

Interpolate speed along the path.

`#include "interpolate_m.h"`
interpolate_m.cpp 的引用(Include)系:



命名空

- interp_m

  是一个名*interp_m*的*namespace*。

函数

- void interp_m::linear (const interpLinearParam &param, const interpLinearInput &input, interpLinearOutput &output)

  根据初、和数，行性插
- void interp_m::constant_acceleration (const interpConstAccParam &param, const interpConstAccInput &input, interpConstAccOutput &output)

  在等隔的情况下，行匀加速度插

7.15.1 描述

Interpolate speed along the path.

作者

版本

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日期

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7.16 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/localPlanning_m.cpp 文件参考

local planning

```
#include "localPlanning_m.hpp"
```

localPlanning_m.cpp 的引用(Include)系:

命名空

- LocalPlanning

  是一个名*LocalPlanning*的*namespace*。

宏定

- #define **PLANNING_VELOCITY_MULTIPLIER** 2.6
- #define **DEFAULT_PLANNING_DISTANCE** 5
- #define **MAX_PLANNING_DISTANCE** 6
- #define **MIN_PLANNING_DISTANCE** 5
- #define **DISTANCE_RANGE** 3
- #define **DEFAULT_SECOND_PLANNING_DISTANCE_LONGER** 5
- #define **POINT_DIS_THRESHOLD_ON_CURRENT_LANE** 3
- #define **POINT_DIS_THRESHOLD_ON_OTHER_LANE** 2
- #define **POINT_ANGLE_THRESHOLD** 120
- #define **END_SPEED_NUM** 1
- #define **DESIRED_SPEED** 12 / 3.6
- #define **SPEED_RANGE** 3 / 3.6
- #define **CURVATURE_SPEED** 10 / 3.6
- #define **CURVATURE_THRESHOLS** 0.1
- #define **VEHICLE_DIAGONAL** 2.7951
- #define **VEHICLE_DIAGONAL_DEGREE** 26.5651
- #define **PREDICT_FREQUENCY** 0.1
- #define **STOP_JUDGE_DISTANCE** 6
- #define **MAX_FRENET_S** 8
- #define **ACC_RELA_DIST** 15.0
- #define **ACC_BEGIN_DIST** (ACC_RELA_DIST + 30)
- #define **ACC_FOLLOW_OBJECT_ID** 2
- #define **doNothing**()
- #define **SAFE_DISTANCE** 1.0

函数

- void LocalPlanning::gaussConvert (const gaussConvertParam &param, const gaussConvertInput &input, gaussConvertOutput &output)

  度全局坐
- void LocalPlanning::coordTran2D (const coordTran2DParam &param, const coordTran2DInput &input, coordTran2DOutput &output)

  全局坐局部坐
- void LocalPlanning::inArea (const inAreaParam &param, const inAreaInput &input, inAreaOutput &output)

  判断点是否在区域内
- void LocalPlanning::pointDistance (const pointDistanceParam &param, const pointDistanceInput &input, pointDistanceOutput &output)

  算两点之的距离
- void LocalPlanning::courseAngleRevise (const courseAngleReviseParam &param, const courseAngle↩ReviseInput &input, courseAngleReviseOutput &output)

  逆旋坐系*90*度，向*0*度成向北*0*度
- void LocalPlanning::loadStopPoints (const loadStopPointsParam &param, const loadStopPointsInput &input, loadStopPointsOutput &output)

  从文件中加停止点

- void LocalPlanning::loadStopPointsFromYaml (const loadStopPointsFromYamlParam &param, const load↩ StopPointsFromYamlInput &input, loadStopPointsFromYamlOutput &output)

  从文件中加停止点
- void LocalPlanning::getNextRoadLaneId (const getNextRoadLaneIdParam &param, const getNextRoad↩ LaneIdInput &input, getNextRoadLaneIdOutput &output)

  从全局划中，取后道和道路*Id*
- void LocalPlanning::getPlanningDistance (const getPlanningDistanceParam &param, const getPlanning↩ DistanceInput &input, getPlanningDistanceOutput &output)

  根据速度、曲率取划距离
- void LocalPlanning::getPlanningPoint (const getPlanningPointParam &param, const getPlanningPointInput &input, getPlanningPointOutput &output)

  根据信息取划瞄点
- void LocalPlanning::limitPlanningDistance (const limitPlanningDistanceParam &param, const limitPlanning↩ DistanceInput &input, limitPlanningDistanceOutput &output)

  限制划距离，防止突
- void LocalPlanning::getAngle (const getAngleParam &param, const getAngleInput &input, getAngleOutput &output)

  算坐系中两点角度
- void LocalPlanning::getAngleDiff (const getAngleDiffParam &param, const getAngleDiffInput &input, get↩ AngleDiffOutput &output)

  算两角度之差，并正化
- void **LocalPlanning::localPlanning** (const pc::Imu &imu, double velocity, const RoadMap &map, Decision↩ Data &decisionData, const prediction::ObjectList &predictionMsg, const std::vector< std::tuple< int32_↩ t, int32_t >> &routingList, const std::vector< GaussRoadPoint > stopPoints, const infopack::TrafficLight &trafficLight, std::vector< infopack::ObjectsProto > objectsCmd)
- void LocalPlanning::restart (const restartParam &param, const restartInput &input, restartOutput &output)

  停后，判断能否重新后
- void LocalPlanning::stop (const stopParam &param, const stopInput &input, stopOutput &output)

  停修改速度曲
- void LocalPlanning::stopPointJudge (const stopPointJudgeParam &param, const stopPointJudgeInput &input, stopPointJudgeOutput &output)

  判断是否在停区域
- void LocalPlanning::pointOnCubicBezier (const pointOnCubicBezierParam &param, const pointOnCubic↩ BezierInput &input, pointOnCubicBezierOutput &output)

  算塞曲上的点
- void LocalPlanning::generateBezierPathInFrenet (const generateBezierPathInFrenetParam &param, const generateBezierPathInFrenetInput &input, generateBezierPathInFrenetOutput &output)

  在*frenet*中生成塞曲
- void LocalPlanning::clearPathList (const clearPathListParam &param, const clearPathListInput &input, clearPathListOutput &output)

  清除上次生成的迹
- void LocalPlanning::getTrajectoryPlanningResult (const getTrajectoryPlanningResultParam &param, const getTrajectoryPlanningResultInput &input, getTrajectoryPlanningResultOutput &output)

  完成了多条迹和速度曲的生成，并化到笛卡坐系中，作出了迹的
- void LocalPlanning::getDistance (const getDistanceParam &param, const getDistanceInput &input, get↩ DistanceOutput &output)

  算两点之的距离
- void LocalPlanning::pointToLineDistance (const pointToLineDistanceParam &param, const pointToLine↩ DistanceInput &input, pointToLineDistanceOutput &output)

  算点到段的距离
- void LocalPlanning::angleRegulate (const angleRegulateParam &param, const angleRegulateInput &input, angleRegulateOutput &output)

  化角度到*0-180*中，便于比

- void LocalPlanning::getCurrentPosition (const getCurrentPositionParam &param, const getCurrentPosition↩ Input &input, getCurrentPositionOutput &output)

  找距离位置最近的路点
- void LocalPlanning::generateSpeedList (const generateSpeedListParam &param, const generateSpeed↩ ListInput &input, generateSpeedListOutput &output)

  根据道路信息等，采生成速度曲
- void LocalPlanning::generateTrajectoryList (const generateTrajectoryListParam &param, const generate↩ TrajectoryListInput &input, generateTrajectoryListOutput &output)

  生成采迹，供后
- void LocalPlanning::processTrafficLight (const processTrafficLightParam &param, const processTrafficLight↩ Input &input, processTrafficLightOutput &output)

  根据灯信息更新决策数据
- void LocalPlanning::getOptimalTrajectoryIndex (const getOptimalTrajectoryIndexParam &param, const get↩ OptimalTrajectoryIndexInput &input, getOptimalTrajectoryIndexOutput &output)

  估迹，并取最迹索引
- void LocalPlanning::trajectoryCollisionCheck (const trajectoryCollisionCheckParam &param, const trajectoryCollisionCheckInput &input, trajectoryCollisionCheckOutput &output)

  迹碰撞，*true*表示可行，*false*表示碰撞
- void LocalPlanning::ableToPassYellowLight (const ableToPassYellowLightParam &param, const ableTo↩ PassYellowLightInput &input, ableToPassYellowLightOutput &output)

  判断是否能通黄灯
- void LocalPlanning::pointCollisionCheck (const pointCollisionCheckParam &param, const pointCollision↩ CheckInput &input, pointCollisionCheckOutput &output)

  迹点碰撞
- void LocalPlanning::boundaryCollisionCheck (const boundaryCollisionCheckParam &param, const boundaryCollisionCheckInput &input, boundaryCollisionCheckOutput &output)

  碰撞
- void LocalPlanning::innerCollisionCheck (const innerCollisionCheckParam &param, const innerCollision↩ CheckInput &input, innerCollisionCheckOutput &output)

  内部包含
- void LocalPlanning::changeCurrentPoint (const changeCurrentPointParam &param, const changeCurrent↩ PointInput &input, changeCurrentPointOutput &output)

  判断是否可以道
- void LocalPlanning::changeLocalToGlobal (const changeLocalToGlobalParam &param, const changeLocal↩ ToGlobalInput &input, changeLocalToGlobalOutput &output)

  局部坐迹全局迹
- void LocalPlanning::initLocalPlanning (const initLocalPlanningParam &param, const initLocalPlanningInput &input, initLocalPlanningOutput &output)

  局部划决策数据初始化
- void LocalPlanning::fixRoadLaneIndex (const fixRoadLaneIndexParam &param, const fixRoadLaneIndex↩ Input &input, fixRoadLaneIndexOutput &output)

  返回*decisionDate*中*roadID* 和*laneID*的索引
- void LocalPlanning::calculateCurvature (const calculateCurvatureParam &param, const calculateCurvature↩ Input &input, calculateCurvatureOutput &output)

  由相三点近似算曲率
- void LocalPlanning::assessTrajectory (const assessTrajectoryParam &param, const assessTrajectoryInput &input, assessTrajectoryOutput &output)

  返回距离障碍物最近距离，行迹估
- void LocalPlanning::getMinDistanceOfPoint (const getMinDistanceOfPointParam &param, const getMin↩ DistanceOfPointInput &input, getMinDistanceOfPointOutput &output)

  返回点距离障碍物最近距离
- void LocalPlanning::findMin (const findMinParam &param, const findMinInput &input, findMinOutput &output)

  返回*double*数最小

- void LocalPlanning::findMax (const findMaxParam &param, const findMaxInput &input, findMaxOutput &output)

  返回*double*数最大
- void LocalPlanning::getReferenceLine (const getReferenceLineParam &param, const getReferenceLineInput &input, getReferenceLineOutput &output)

  取局部划参考迹
- void LocalPlanning::getFrenetLocationPoint (const getFrenetLocationPointParam &param, const getFrenet↩ LocationPointInput &input, getFrenetLocationPointOutput &output)

  取当前位置在*frenet*坐系中坐
- void LocalPlanning::frenet2Cartesian (const frenet2CartesianParam &param, const frenet2CartesianInput &input, frenet2CartesianOutput &output)

  将*frenet*坐系坐自坐系坐，*xy*右前
- void LocalPlanning::generateBezierPathListInFrenet (const generateBezierPathListInFrenetParam &param, const generateBezierPathListInFrenetInput &input, generateBezierPathListInFrenetOutput &output)

  在*frenet*坐系中生成多条塞曲
- void LocalPlanning::hasPrecedingVehicle (const hasPrecedingVehicleParam &param, const hasPreceding↩ VehicleInput &input, hasPrecedingVehicleOutput &output)

  判断前方是否有需要跟随的
- void LocalPlanning::cruiseController (const cruiseControllerParam &param, const cruiseControllerInput &input, cruiseControllerOutput &output)

  *acc*景下，目速度算

7.16.1 描述

local planning

作者

版本

1.0

日期

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版所有

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7.17 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_↩ planning_modul/planningFigure/src/planningFigure_m.cpp 文件参考

planning

```
#include "PlanningMsg.pb.h"
#include "defineColor.h"
#include "globalPlanning_m.h"
#include "localPlanning_m.hpp"
#include "opencv2/highgui/highgui.hpp"
#include "opencv2/imgproc/imgproc.hpp"
#include "planning.hpp"
#include "sys/time.h"
#include "unistd.h"
#include <chrono>
#include <thread>
```

planningFigure_m.cpp 的引用(Include)系:

宏定

- #define **STOP_POINT_FILE_NAME** "../mapMsg/stopPoint.txt"
- #define **STOP_POINT_FILE_NAME_YAML** "../mapMsg/stopPoint.yaml"
- #define **RECV_PERIOD** 40
- #define **SEND_PERIOD** 100
- #define **GLOBAL_PLANNING_PERIOD** 100
- #define **LOCAL_PLANNING_PERIOD** 100
- #define **MAP_PATH** "../mapMsg/roadMap_KCYTemp.xodr"
- #define **gridInOneMeter** 5
- #define **mapSizeXMeter** 120
- #define **mapSizeYMeter** 120
- #define **mapYZero** 200
- #define **HEARTBEAT_PERIOD** 500
- #define **myselfVehicleID** 15
- #define **myselfVehicleeType** 0
- #define **myselfVCType** 1

函数

- cv::Point **drawRegulate** (cv::Point input)
- long int **cvtThreadId2Long** (thread::id id)
- void **checkThreadStatus** (int tStatus, thread::id id)
- int **main** ()

量

- auto **lastTheadTime** = std::chrono::steady_clock::now()
- auto **lastTheadTimeofSub** = std::chrono::steady_clock::now()
- auto **durationFroShowofSub** = std::chrono::duration_cast<std::chrono::milliseconds>(std::chrono↩ ::steady_clock::now() - lastTheadTimeofSub)

7.17.1 描述

planning

作者

版本

1.0

日期

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版所有

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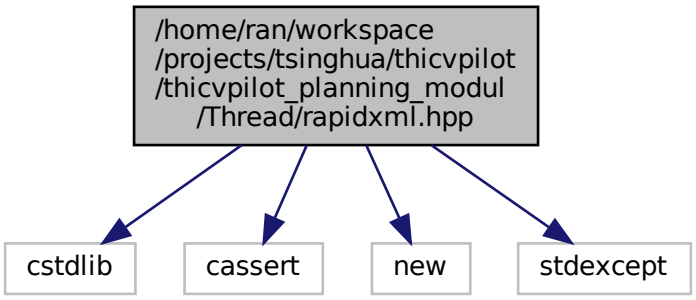
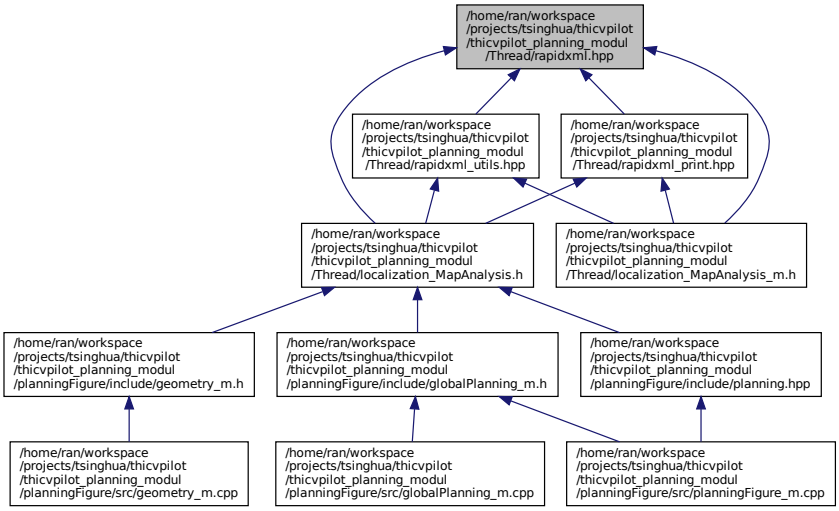
7.18 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_↩ planning_modul/Thread/rapidxml.hpp 文件参考

```
#include <cstdlib>
#include <cassert>
#include <new>
#include <stdexcept>
```

rapidxml.hpp 的引用(Include)系:



此展示文件直接或接的被哪些文件引用了:



- class rapidxml::parse_error
- class rapidxml::eof_error
- class rapidxml::validation_error
- class rapidxml::xml_node< Ch >
- class rapidxml::xml_attribute< Ch >
- class rapidxml::xml_document< Ch >
- class rapidxml::memory_pool< Ch >
- class rapidxml::xml_base< Ch >
- class rapidxml::xml_attribute< Ch >
- class rapidxml::xml_node< Ch >
- class rapidxml::xml_document< Ch >

宏定

- #define **RAPIDXML_PARSE_ERROR**(what, where) {if (∗where == Ch(0)) throw eof_error(what, where); else throw parse_error(what, where);} (void)0
- #define **RAPIDXML_EOF_ERROR**(what, where) throw eof_error(what, where)
- #define **RAPIDXML_STATIC_POOL_SIZE** (64 ∗ 1024)
- #define **RAPIDXML_DYNAMIC_POOL_SIZE** (64 ∗ 1024)
- #define **RAPIDXML_ALIGNMENT** sizeof(void ∗)

枚

- enum rapidxml::node_type {
  rapidxml::node_document, rapidxml::node_element, rapidxml::node_data, rapidxml::node_cdata, rapidxml::node_comment, rapidxml::node_declaration, rapidxml::node_doctype, rapidxml::node_pi, rapidxml::node_literal }

量

- const int rapidxml::parse_no_data_nodes = 0x1
- const int rapidxml::parse_no_element_values = 0x2
- const int rapidxml::parse_no_string_terminators = 0x4
- const int rapidxml::parse_no_entity_translation = 0x8
- const int rapidxml::parse_no_utf8 = 0x10
- const int rapidxml::parse_declaration_node = 0x20
- const int rapidxml::parse_comment_nodes = 0x40
- const int rapidxml::parse_doctype_node = 0x80
- const int rapidxml::parse_pi_nodes = 0x100
- const int rapidxml::parse_validate_closing_tags = 0x200
- const int rapidxml::parse_trim_whitespace = 0x400
- const int rapidxml::parse_normalize_whitespace = 0x800
- const int rapidxml::parse_open_only = 0x1000
- const int rapidxml::parse_parse_one = 0x2000
- const int rapidxml::parse_validate_xmlns = 0x4000
- const int rapidxml::parse_default = 0
- const int rapidxml::parse_non_destructive = parse_no_string_terminators | parse_no_entity_translation
- const int rapidxml::parse_fastest = parse_non_destructive | parse_no_data_nodes
- const int rapidxml::parse_full = parse_declaration_node | parse_comment_nodes | parse_doctype_node | parse_pi_nodes | parse_validate_closing_tags

7.18.1 描述

This file contains rapidxml parser and DOM implementation

7.18.2 枚型明

7.18.2.1 node_type

`enum rapidxml::node_type`

Enumeration listing all node types produced by the parser. Use xml_node::type() function to query node type.

枚

node_document	A document node. Name and value are empty.
node_element	An element node. Name contains element name. Value contains text of first data node.
node_data	A data node. Name is empty. Value contains data text.
node_cdata	A CDATA node. Name is empty. Value contains data text.
node_comment	A comment node. Name is empty. Value contains comment text.
node_declaration	A declaration node. Name and value are empty. Declaration parameters (version, encoding and standalone) are in node attributes.
node_doctype	A DOCTYPE node. Name is empty. Value contains DOCTYPE text.
node_pi	A PI node. Name contains target. Value contains instructions.
node_literal	Value is unencoded text (used for inserting pre-rendered XML).

7.18.3 量明

7.18.3.1 parse_comment_nodes

```
const int rapidxml::parse_comment_nodes = 0x40
```

Parse flag instructing the parser to create comments nodes. By default, comment nodes are not created. Can be combined with other flags by use of | operator.

See xml_document::parse() function.

7.18.3.2 parse_declaration_node

```
const int rapidxml::parse_declaration_node = 0x20
```

Parse flag instructing the parser to create XML declaration node. By default, declaration node is not created. Can be combined with other flags by use of | operator.

See xml_document::parse() function.

7.18.3.3 parse_default

```
const int rapidxml::parse_default = 0
```

Parse flags which represent default behaviour of the parser. This is always equal to 0, so that all other flags can be simply ored together. Normally there is no need to inconveniently disable flags by anding with their negated (~) values. This also means that meaning of each flag is a *negation* of the default setting. For example, if flag name is rapidxml::parse_no_utf8, it means that utf-8 is *enabled* by default, and using the flag will disable it.

See xml_document::parse() function.

7.18.3.4 parse_doctype_node

```
const int rapidxml::parse_doctype_node = 0x80
```

Parse flag instructing the parser to create DOCTYPE node. By default, doctype node is not created. Although W3C specification allows at most one DOCTYPE node, RapidXml will silently accept documents with more than one. Can be combined with other flags by use of | operator.

See xml_document::parse() function.

7.18.3.5 parse_fastest

```
const int rapidxml::parse_fastest = parse_non_destructive | parse_no_data_nodes
```

A combination of parse flags resulting in fastest possible parsing, without sacrificing important data.

See xml_document::parse() function.

7.18.3.6 parse_full

```
const int rapidxml::parse_full = parse_declaration_node | parse_comment_nodes | parse_doctype_node
| parse_pi_nodes | parse_validate_closing_tags
```

A combination of parse flags resulting in largest amount of data being extracted. This usually results in slowest parsing.

See xml_document::parse() function.

7.18.3.7 parse_no_data_nodes

```
const int rapidxml::parse_no_data_nodes = 0x1
```

Parse flag instructing the parser to not create data nodes. Text of first data node will still be placed in value of parent element, unless rapidxml::parse_no_element_values flag is also specified. Can be combined with other flags by use of | operator.

See xml_document::parse() function.

7.18.3.8 parse_no_element_values

```
const int rapidxml::parse_no_element_values = 0x2
```

Parse flag instructing the parser to not use text of first data node as a value of parent element. Can be combined with other flags by use of | operator. Note that child data nodes of element node take precendence over its value when printing. That is, if element has one or more child data nodes *and* a value, the value will be ignored. Use rapidxml::parse_no_data_nodes flag to prevent creation of data nodes if you want to manipulate data using values of elements.

See xml_document::parse() function.

7.18.3.9 parse_no_entity_translation

```
const int rapidxml::parse_no_entity_translation = 0x8
```

Parse flag instructing the parser to not translate entities in the source text. By default entities are translated, modifying source text. Can be combined with other flags by use of | operator.

See xml_document::parse() function.

7.18.3.10 parse_no_string_terminators

```
const int rapidxml::parse_no_string_terminators = 0x4
```

Parse flag instructing the parser to not place zero terminators after strings in the source text. By default zero terminators are placed, modifying source text. Can be combined with other flags by use of | operator.

See xml_document::parse() function.

7.18.3.11 parse_no_utf8

```
const int rapidxml::parse_no_utf8 = 0x10
```

Parse flag instructing the parser to disable UTF-8 handling and assume plain 8 bit characters. By default, UTF-8 handling is enabled. Can be combined with other flags by use of | operator.

See xml_document::parse() function.

7.18.3.12 parse_non_destructive

```
const int rapidxml::parse_non_destructive = parse_no_string_terminators | parse_no_entity_translation
```

A combination of parse flags that forbids any modifications of the source text. This also results in faster parsing. However, note that the following will occur:

- names and values of nodes will not be zero terminated, you have to use xml_base::name_size() and xml_base::value_size() functions to determine where name and value ends
- entities will not be translated
- whitespace will not be normalized

See xml_document::parse() function.

7.18.3.13 parse_normalize_whitespace

```
const int rapidxml::parse_normalize_whitespace = 0x800
```

Parse flag instructing the parser to condense all whitespace runs of data nodes to a single space character. Trimming of leading and trailing whitespace of data is controlled by rapidxml::parse_trim_whitespace flag. By default, whitespace is not normalized. If this flag is specified, source text will be modified. Can be combined with other flags by use of | operator.

See xml_document::parse() function.

7.18.3.14 parse_open_only

```
const int rapidxml::parse_open_only = 0x1000
```

Parse flag to say "Parse only the initial element opening." Useful for XMLstreams used in XMPP.

7.18.3.15 parse_parse_one

```
const int rapidxml::parse_parse_one = 0x2000
```

Parse flag to say "Toss the children of the top node and parse off one element. Useful for parsing off XMPP top-level elements.

7.18.3.16 parse_pi_nodes

```
const int rapidxml::parse_pi_nodes = 0x100
```

Parse flag instructing the parser to create PI nodes. By default, PI nodes are not created. Can be combined with other flags by use of | operator.

See xml_document::parse() function.

7.18.3.17 parse_trim_whitespace

```
const int rapidxml::parse_trim_whitespace = 0x400
```

Parse flag instructing the parser to trim all leading and trailing whitespace of data nodes. By default, whitespace is not trimmed. This flag does not cause the parser to modify source text. Can be combined with other flags by use of | operator.

See xml_document::parse() function.

7.18.3.18 parse_validate_closing_tags

```
const int rapidxml::parse_validate_closing_tags = 0x200
```

Parse flag instructing the parser to validate closing tag names. If not set, name inside closing tag is irrelevant to the parser. By default, closing tags are not validated. Can be combined with other flags by use of | operator.

See xml_document::parse() function.

7.18.3.19 parse_validate_xmlns

```
const int rapidxml::parse_validate_xmlns = 0x4000
```

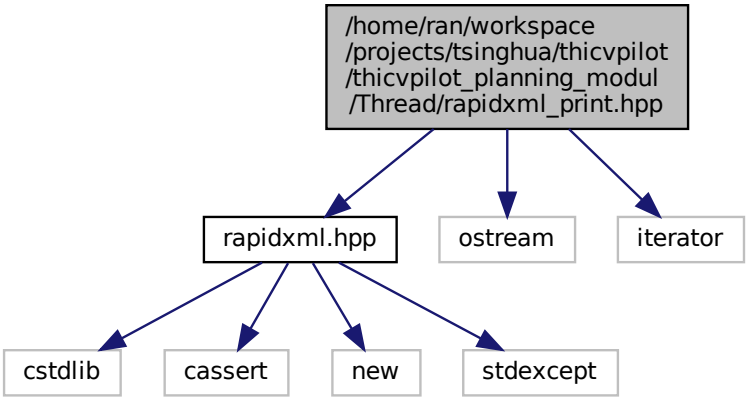
Parse flag to say "Validate XML namespaces fully." This will generate additional errors, including unbound prefixes and duplicate attributes (with different prefices)

7.19 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml_print.hpp 文件参考

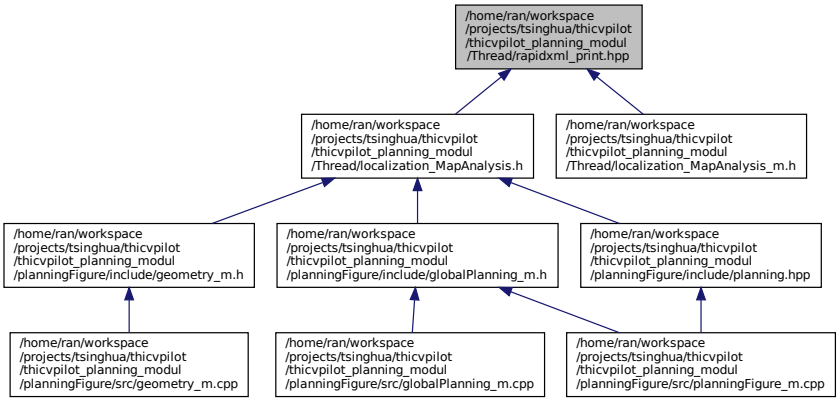
```
#include "rapidxml.hpp"
#include <ostream>
```

```
#include <iterator>
```

rapidxml_print.hpp 的引用(Include)系:



此展示文件直接或接的被哪些文件引用了:



函数

- template<class OutIt , class Ch >
  OutIt rapidxml::print (OutIt out, const xml_node< Ch > &node, int flags=0)
- template<class Ch >
  std::basic_ostream< Ch > & rapidxml::print (std::basic_ostream< Ch > &out, const xml_node< Ch > &node, int flags=0)
- template<class Ch >
  std::basic_ostream< Ch > & rapidxml::operator<< (std::basic_ostream< Ch > &out, const xml_node< Ch > &node)

量

- const int rapidxml::print_no_indenting = 0x1

  Printer flag instructing the printer to suppress indenting of XML. See print() function.

7.19.1 描述

This file contains rapidxml printer implementation

7.19.2 函数明

7.19.2.1 operator<<()

```
template<class Ch >
std::basic_ostream<Ch>& rapidxml::operator<< (
            std::basic_ostream< Ch > & out,
            const xml_node< Ch > & node )  [inline]
```

Prints formatted XML to given output stream. Uses default printing flags. Use print() function to customize printing process.

参数

out	Output stream to print to.
node	Node to be printed.

返回

Output stream.

7.19.2.2 print() [1/2]

```
template<class OutIt , class Ch >
OutIt rapidxml::print (
            OutIt out,
            const xml_node< Ch > & node,
            int flags = 0 )  [inline]
```

Prints XML to given output iterator.

参数

out	Output iterator to print to.
node	Node to be printed. Pass xml_document to print entire document.
flags	Flags controlling how XML is printed.

返回

Output iterator pointing to position immediately after last character of printed text.

7.19.2.3 print() [2/2]

```
template<class Ch >
std::basic_ostream<Ch>& rapidxml::print (
            std::basic_ostream< Ch > & out,
            const xml_node< Ch > & node,
            int flags = 0 )  [inline]
```

Prints XML to given output stream.

参数

out	Output stream to print to.
node	Node to be printed. Pass xml_document to print entire document.
flags	Flags controlling how XML is printed.

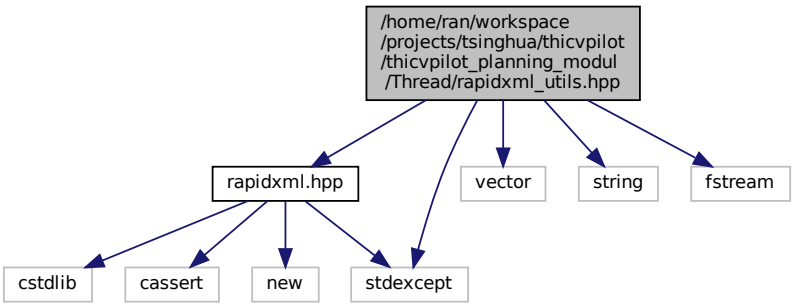
返回

Output stream.

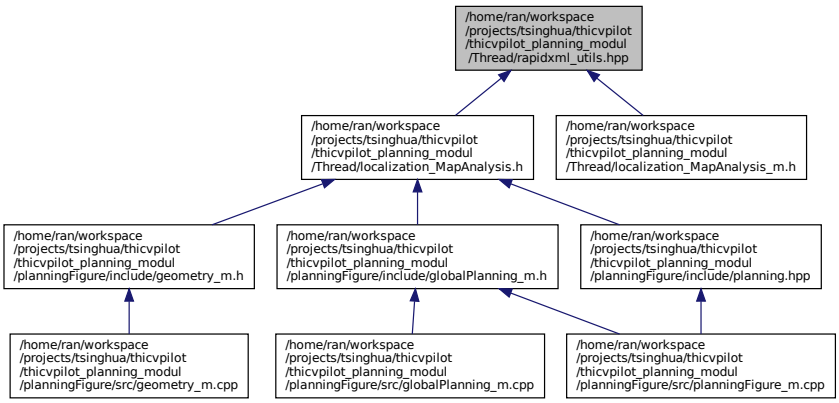
7.20 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/Thread/rapidxml_utils.hpp 文件参考

```
#include "rapidxml.hpp"
#include <vector>
#include <string>
#include <fstream>
#include <stdexcept>
```

rapidxml_utils.hpp 的引用(Include)系:



此展示文件直接或接的被哪些文件引用了:



- class rapidxml::file< Ch >

  Represents data loaded from a file.

函数

- template<class Ch >
  std::size_t rapidxml::count_children (xml_node< Ch > *node)
- template<class Ch >
  std::size_t rapidxml::count_attributes (xml_node< Ch > *node)

7.20.1 描述

This file contains high-level rapidxml utilities that can be useful in certain simple scenarios. They should probably not be used if maximizing performance is the main objective.

7.20.2 函数明

7.20.2.1 count_attributes()

```
template<class Ch >
std::size_t rapidxml::count_attributes (
            xml_node< Ch > * node )  [inline]
```

Counts attributes of node. Time complexity is O(n).

返回

Number of attributes of node

7.20.2.2 count_children()

```
template<class Ch >
std::size_t rapidxml::count_children (
            xml_node< Ch > * node )  [inline]
```

Counts children of node. Time complexity is O(n).

返回

Number of children of node

Chapter 8

示例明

8.1 controlMain.cc

几的构造用方法

8.2 controlMain4ros.cc

controlMain4ros.hpp ROS用示例

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